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Varicose Veins

Number: 0050

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Scope of Policy

This Clinical Policy Bulletin addresses treatment of varicose veins.

I. **Medically Necessary**

A. Aetna considers the following procedures medically necessary for treatment of varicose veins:

1. Great saphenous vein or small saphenous vein ligation / division / stripping;
2. Radiofrequency endovenous occlusion (VNUS procedure); *and*
3. Endovenous laser ablation of the saphenous vein (ELAS) (also known as endovenous laser treatment (EVLT))

When the following criteria are met:

1. Incompetence at the saphenofemoral junction or saphenopopliteal junction is documented by recent (performed within the past 6 months) Doppler or duplex ultrasound scanning, and *all* of the following criteria are met:

   a. Ultrasound documented junctional reflux duration of 500 milliseconds (ms) or greater in the saphenofemoral or saphenopopliteal vein to be treated; *and*
   b. Vein size is 4.5 mm or greater in diameter measured by ultrasound below the saphenofemoral or saphenopopliteal junction (not valve diameter at junction); *and*
   c. Saphenous varicosities result in *any* of the following:

      i. Intractable ulceration secondary to venous stasis; *or*
      ii. More than 1 episode of minor hemorrhage from a ruptured superficial varicosity; or a single significant hemorrhage from a ruptured superficial varicosity, especially if transfusion of blood is required; *or*
      iii. Saphenous varicosities result in *either* of the following, and symptoms persist despite a 3-month trial of conservative managementFootnote1* (including medical grade (20 mmHg or greater) gradient support compression

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stockings).

a. Recurrent superficial thrombophlebitis; *or*
b. Severe and persistent pain and swelling interfering with activities of daily living.

2. Surgical ligation (including subfascial endoscopic perforator vein surgery (SEPS)) or endovenous ablation procedures are considered medically necessary for the treatment of incompetent perforating veins when *all* of the following are met:

   a. Perforating vein diameter measured by recent ultrasound of 3.5 mm or greater; *and*
   b. Outward flow duration of 500 msec duration or more; *and*
   c. Perforating vein is located underneath an active or healed venous stasis ulcer (also known as CEAP C5 or C6). See Appendix.

3. Endovenous ablation procedures are considered medically necessary adjunctive treatment of symptomatic accessory saphenous veins

   a. who are being treated or have previously ;been treated by one of the procedures listed above for incompetence (i.e., reflux) at the saphenofemoral junction or saphenopopliteal junction; *and*
   b. anatomically related persistent junctional reflux is demonstrated after the great or small saphenous veins have been removed or ablated.

   Note: Initially, endovenous ablation therapy of the first vein and of the second and subsequent veins in each affected extremity is considered medically necessary when criteria are met. **Note**: Thus one primary code and one secondary code for each affected leg are considered medically necessary for initial endovenous ablation treatment.

   Note: Additional endovenous ablation therapy is considered medically necessary for persons with persistent or recurrent junctional reflux of the greater saphenous vein, lesser saphenous vein following initial endovenous ablation therapy. In order to authorize additional endovenous ablation, there should be documentation that the member continues to have symptoms and ultrasound showing persistent junctional reflux. Additional endovenous ablation therapy may also be necessary for treatment of accessory saphenous veins as noted above. These procedures are considered experimental, investigational, or unproven for treatment of varicose tributaries and accessory veins other than the accessory saphenous vein. These procedures are considered cosmetic for all other indications.

   Note: Doppler or duplex ultrasound studies are considered necessary prior to varicose vein treatment to assess the anatomy and to determine whether there is significant reflux at the saphenofemoral or saphenopopliteal junction requiring surgical repair, and after completion of the treatment to determine the success of the procedure and detect thrombosis. Ultrasound guidance is inclusive of the VNUS or ELAS procedures.

   Note: The term endovenous catheter ablation (EVCA) is a non-specific term that refers to the several catheter based minimally invasive alternatives to surgical stripping such as radiofrequency endovenous occlusion (VNUS procedure) and endovenous laser ablation of the saphenous vein (ELAS). In assessing the medical necessity of EVCA, reference should be made to the specific technique that is being employed.

B. Aetna considers liquid or foam sclerotherapy (endovenous chemical ablation) (e.g., Varithena) medically necessary treatment of symptomatic saphenous veins, varicose tributaries, accessory, and perforator veins when *all* of the following criteria are met:

   1. Vein size is 2.5 mm or greater in diameter, measured by recent ultrasound; *and*

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2. Varicosities result in *any* of the following:

    a. Intractable ulceration secondary to venous stasis; *or*
    b. More than 1 episode of minor hemorrhage from a ruptured superficial varicosity; or a single significant hemorrhage from a ruptured superficial varicosity, especially if transfusion of blood is required; *or*
    c. Varicosities result in *either* of the following, and symptoms persist despite a 3-month trial of conservative managementFootnote1* (including medical grade (20 mmHg or greater) gradient support compression stockings):

        i. Recurrent superficial thrombophlebitis; *or*
        ii. Severe and persistent pain and swelling interfering with activities of daily living; *and*

3. If member has incompetence (i.e., reflux) at the saphenofemoral junction, the junctional reflux is being treated by one or more of the endovenous ablation or ligation and division procedures noted in section A above to reduce the risk of varicose vein recurrence. **Note:** Varithena has not been proven to be more effective than other methods of foam sclerotherapy.

Note: Since ultrasound-monitored or duplex-guided techniques for sclerotherapy have not been shown to definitively increase the effectiveness or safety of this procedure, these tests are only considered medically necessary when initially performed to determine the extent and configuration of varicose veins.

Note: Ultrasound- or radiologically guided or monitoring techniques are of no proven value when performed solely to guide the needle or introduce the sclerosant into the varicose veins.

Note: The number of medically necessary sclerotherapy injection sessions varies with the number of anatomical areas that have to be injected, as well as the response to each injection. Usually 1 to 3 injections are necessary to obliterate any vessel, and 10 to 40 vessels, or a set of up to 20 injections in each leg, may be treated during one treatment session. Initially, up to two sets of injections of sclerosing solution in multiple veins in each affected leg (i.e., a total of four sets of injections if both legs are affected) are considered medically necessary when criteria are met. **Note**: A set of injections is defined as multiple sclerotherapy injections during a treatment session. Additional sets of injections of sclerosing solution are considered medically necessary for persons with persistent or recurrent symptoms.

C. Aetna considers ambulatory phlebectomy or transilluminated powered phlebectomy (TriVex System) medically necessary adjunctive treatment of symptomatic saphenous veins, varicose tributaries, accessory, and perforator veins when *all* of the following criteria are met:

1. Vein size is 2.5 mm or greater in diameter; *and*
2. Varicosities result in *any* of the following:

    a. Intractable ulceration secondary to venous stasis; *or*
    b. More than 1 episode of minor hemorrhage from a ruptured superficial varicosity; or a single significant hemorrhage from a ruptured superficial varicosity, especially if transfusion of blood is required; *or*
    c. Varicosities result in *either* of the following, and symptoms persist despite a 3-month trial of conservative managementFootnote1* (including medical grade (20 mmHg or greater) gradient support compression stockings):

        i. Recurrent superficial thrombophlebitis; *or*
        ii. Severe and persistent pain and swelling interfering with activities of daily living; *and*

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3. If member has incompetence (i.e., reflux) at the saphenofemoral junction or saphenopopliteal junction, the junctional reflux is being treated by one or more of the procedures noted in section A above to reduce the risk of varicose vein recurrence.

Note: Initially, up to two multiple stab phlebectomy incisions in each affected extremity (i.e., a total of four multiple stab incisions if both legs are affected) are considered medically necessary when criteria are met. Additional multiple stab phlebectomy incisions are considered medically necessary for persons with persistent or recurrent symptoms. **Note**: A set of stab phlebectomy incisions is defined as multiple stab phlebectomy incisions during a treatment session.

Footnote1***Note**: A trial of conservative management is not required for persons with persistent or recurrent varicosities who have undergone prior endovenous catheter ablation procedures or stripping/division/ligation in the same leg because conservative management is unlikely to be successful in this situation.

D. Aetna considers valvular reconstruction medically necessary for chronic venous insufficiency.

II. Experimental, Investigational, or Unproven

A. The following procedures are considered experimental, investigational, or unproven because they have not been proven in direct comparative studies to be as effective as established alternatives or has not been proven for the listed indications:

1. Ambulatory phlebectomy or transilluminated powered phlebectomy for treatment of junctional reflux;
2. Ambulatory selective variceal ablation under local anesthetic (the ASVAL procedure) for the treatment of symptomatic great saphenous vein;
3. Endomechanical or mechanochemical ablation (MOCA) (e.g., ClariVein);
4. Endovenous ablation (laser or radiofrequency) for the treatment of reflux of the common femoral vein;
5. External valvuloplasty for the treatment of symptomatic great saphenous vein;
6. Measurements of plasma growth factors (e.g., angiopoietin-1 (ANG1), angiopoietin-2 (ANG2), epidermal growth factor (EGF), platelet-derived growth factor (PDGF), and vascular endothelial growth factor (VEGF)) for predicting adequacy of treatment and the possibility of recurrence of varicose veins before and after treatment with endovenous laser ablation;
7. Micronized purified flavonoid fraction;
8. Peri-procedural thromboprophylaxis (including pharmacotherapy) for the prevention of deep vein thrombosis in varicose vein interventions;
9. Polymorphism genotyping of matrix metalloproteinases genes (e.g., MMP1, MMP2, MMP3, and MMP7) as markers of predisposition to varicose veins;
10. Sclerotherapy for treatment of reflux of iliac veins, the saphenofemoral junction, or saphenopopliteal junction; sclerotherapy alone has not been shown to be effective for persons with reflux at the saphenofemoral or saphenopopliteal junctions; under established guidelines, individuals with reflux should also be treated with endovenous ablation, ligation or division of the junction to reduce the risk of varicose vein recurrence; **Note**: Gel sclerotherapy of ovarian veins with the internal iliac veins, is considered medically necessary for the treatment of pelvic congestion syndrome - see CPB 0441 - Pelvic Congestion Syndrome: Treatments;
11. Synthetic matrix metalloproteinases inhibitors;
12. Transdermal laser treatment of large varicose veins; **Note**: Although transdermal Nd:YAG laser has been shown to be effective for the treatment of telangiectasias and reticular veins, treatment of these small veins is considered cosmetic;
13. Use of medical adhesive (also referred to as cyanoacrylate superglue, n-butyl-cyanoacrylate) (e.g., VariClose Vein Sealing System, VenaSeal Closure System) for the treatment of varicose veins;

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14. VeinGogh Ohmic Thermolysis System;
15. VeinOPlus vascular device for the treatment of muscle atrophy due to varicose veins.

B. Endovenous ablation procedures are considered experimental, investigational, or unproven for treatment of varicose tributaries and accessory veins other than the accessory saphenous vein.

III. Cosmetic

A. The following procedures are considered cosmetic because they are used to treat small veins that do not cause pain bleeding, ulceration, or other medical problems:

1. Ambulatory phlebectomy and the TriVex system for veins less than 2.5 mm in diameter and all other indications; **Note**: Transilluminated powered phlebectomy has not been proven to be superior to other methods of varicose vein removal; therefore, the TriVex procedure should be billed as any other varicose vein removal procedure;
2. Asclera polidocanol injection; **Note**: Although Asclera has been approved by the Food and Drug Administration (FDA) for the treatment of telangiectasias and reticular veins less than 3 mm in diameter, treatment of these small veins is considered cosmetic;
3. Photothermal sclerosis (also referred to as an intense pulsed light source, e.g., the PhotoDerm VascuLight, VeinLase), used to treat small veins such as small varicose veins and spider veins;
4. Sclerotherapy for treatment of veins less than 2.5 mm in diameter and for all other indications.

B. Endovenous ablation procedures are considered cosmetic for all other indications than listed in Policy section I.

IV. Related Policies

- CPB 0100 - Cryoablation
- CPB 0441 - Pelvic Congestion Syndrome: Treatments

Table:

Applicable CPT / HCPCS / ICD-10 Codes

Code	Code Description
CPT codes covered if selection criteria are met:	
36465 - 36466	Injection of non-compounded foam sclerosant with ultrasound compression maneuvers to guide dispersion of the injectate, inclusive of all imaging guidance and monitoring
36470	Injection of sclerosant; single incompetent vein (other than telangiectasia)
36471	multiple incompetent veins (other than telangiectasia), same leg
36475	Endovenous ablation therapy of incompetent vein, extremity, inclusive of all imaging guidance and monitoring, percutaneous, radiofrequency; first vein treated
+ 36476	second and subsequent veins treated in a single extremity, each through separate access sites (List separately in addition to code for primary procedure)

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Code	Code Description
36478	Endovenous ablation therapy of incompetent vein, extremity, inclusive of all imaging guidance and monitoring, percutaneous, laser; first vein treated
+ 36479	second and subsequent veins treated in a single extremity, each through separate access sites (List separately in addition to code for primary procedure)
37500	Vascular endoscopy, surgical, with ligation of perforator veins, subfascial (SEPS)
37700	Ligation and division of long saphenous vein at saphenofemoral junction, or distal interruptions
37718	Ligation, division, and stripping, short saphenous vein
37722	Ligation, division, and stripping, long (greater) saphenous veins from saphenofemoral junction to knee or below
37735	Ligation and division and complete stripping of long or short saphenous veins with radical excision of ulcer and skin graft and/or interruption of communicating veins of lower leg, with excision of deep fascia
37760	Ligation of perforator veins, subfascial, radical (Linton type), including skin graft, when performed, open, 1 leg
37761	Ligation of perforator vein(s), subfascial, open, including ultrasound guidance, when performed, 1 leg
37765	Stab phlebectomy of varicose veins, one extremity; 10-20 stab incisions [ambulatory]
37766	more than 20 incisions [ambulatory]
37780	Ligation and division of short saphenous vein at saphenopopliteal junction (separate procedure)
37785	Ligation, division, and/or excision of varicose vein cluster(s), one leg
37799	Unlisted procedure, vascular surgery [reported for stab phlebectomy of varicose veins, 1-9 incisions, ambulatory]

CPT codes not covered for indications listed in the CPB:

***Polymorphism genotyping of matrix metalloproteinases genes, synthetic matrix metalloproteinases inhibitors, measurements of plasma growth factors (e.g., ANG1, ANG2, EGF, PDGF, and VEGF), Ambulatory selective variceal ablation under local anesthetic (the ASVAL procedure) and External valvuloplasty* - no specific code**

36011	Selective catheter placement, venous system; first order branch (e.g., renal vein, jugular vein)
36468	Single or multiple injections of sclerosing solutions, spider veins (telangiectasia); limb or trunk
36473	Endovenous ablation therapy of incompetent vein, extremity, inclusive of all imaging guidance and monitoring, percutaneous, mechanochemical; first vein treated
36474	Endovenous ablation therapy of incompetent vein, extremity, inclusive of all imaging guidance and monitoring, percutaneous, mechanochemical; subsequent vein(s) treated in a single extremity, each through separate access sites (List separately in addition to code for primary procedure)
36482 - 36483	Endovenous ablation therapy of incompetent vein, extremity, by transcatheter delivery of a chemical adhesive (eg, cyanoacrylate) remote from the access site, inclusive of all imaging guidance and monitoring, percutaneous
37204	Transcatheter occlusion or embolization (eg, for tumor destruction, to achieve hemostasis, to occlude a vascular malformation), percutaneous, any method, non-central nervous system, non-head or neck
37241	Vascular embolization or occlusion, inclusive of all radiological supervision and interpretation, intraprocedural roadmapping, and imaging guidance necessary to

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Code	Code Description
	complete the intervention; venous, other than hemorrhage (eg, congenital or acquired venous malformations, venous and capillary hemangiomas, varices, varicoceles)
37244	Vascular embolization or occlusion, inclusive of all radiological supervision and interpretation, intraprocedural roadmapping, and imaging guidance necessary to complete the intervention; for arterial or venous hemorrhage or lymphatic extravasation
75894	Transcatheter therapy, embolization, any method, radiological supervision and interpretation
76942	Ultrasonic guidance for needle placement (eg, biopsy, aspiration, injection, localization device), imaging supervision and interpretation [not covered when performed solely to guide the needle or introduce the sclerosant into the varicose veins]
76998	Ultrasonic guidance, intraoperative [not covered when performed solely to guide the needle or introduce the sclerosant into the varicose veins]

Other CPT codes related to the CPB:

37252	Intravascular ultrasound (noncoronary vessel) during diagnostic evaluation and/or therapeutic intervention, including radiological supervision and interpretation; initial noncoronary vessel (List separately in addition to code for primary procedure)
75820, 75822	Venography, extremity, unilateral or bilateral, radiological supervision and interpretation
93922	Limited bilateral non-invasive physiologic studies of upper or lower extremity arteries, (eg, for lower extremity: ankle/brachial indices at distal posterior tibial and anterior tibial/dorsalis pedis arteries plus bidirectional, Doppler waveform recording and analysis at 1-2 levels, or ankle/brachial indices at distal posterior tibial and anterior tibial/dorsalis pedis arteries plus volume plethysmography at 1-2 levels, or ankle/brachial indices at distal posterior tibial and anterior tibial/dorsalis pedis arteries with transcutaneous oxygen tension measurements at 1-2 levels)
93923	Complete bilateral non-invasive physiologic studies of upper or lower extremity arteries, 3 or more levels (eg, for lower extremity: ankle/brachial indices at distal posterior tibial and anterior tibial/dorsalis pedis arteries plus segmental blood pressure measurements with bidirectional Doppler waveform recording and analysis at 3 or more levels, or ankle/brachial indices at distal posterior tibial and anterior tibial/dorsalis pedis arteries plus segmental volume plethysmography at 3 or more levels, or ankle/brachial indices at distal posterior tibial and anterior tibial/dorsalis pedis arteries plus segmental transcutaneous oxygen tension measurements at 3 or more level(s), or single level study with provocative functional maneuvers (eg, measurements with postural provocative tests or measurements with reactive hyperemial)
93924	Non-invasive physiologic studies of lower extremity arteries, at rest and following treadmill stress testing, (ie, bidirectional Doppler waveform or volume plethysmography recording and analysis at rest with ankle/brachial indices immediately after and at timed intervals following performance of a standardized protocol on a motorized treadmill plus recording of time of onset of claudication or other symptoms, maximal walking time, and time to recovery) complete bilateral study
93970	Duplex scan of extremity veins including responses to compression and other maneuvers; complete bilateral study
93971	unilateral or limited study

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Code	Code Description

HCPCS codes covered if selection criteria are met:

S2202	Echosclerotherapy

HCPCS codes not covered for indications listed in the CPB:

***VeinOPlus vascular device* - no specific code**

Other HCPCS codes related to the CPB:

A6530 - A6549	Compression stockings

ICD-10 codes covered if selection criteria are met:

I80.00 - I80.03	Phlebitis and thrombophlebitis of superficial vessels of lower extremities
I82.401 - I82.499	Acute embolism and thrombosis of deep veins of lower extremity
I82.501 - I82.599	Chronic embolism and thrombosis of deep veins of lower extremity
I83.001 - I83.899	Varicose veins of lower extremities
I87.001 - I87.099	Postthrombotic syndrome
I87.2	Venous insufficiency (chronic) (peripheral) [not covered for saphenopopliteal reflux] [not covered for common femoral reflux][not covered for reflux of iliac veins]

ICD-10 codes not covered for indications listed in the CPB:

I83.90 - I83.93	Asymptomatic varicose veins of lower extremities
M62.50 - M62.59, M62.5A0, M62.5A1, M62.5A2, M62.5A9	Muscle wasting and atrophy, not elsewhere classified [due to varicose veins]
O22.00 - O22.03	Varicose veins of lower extremity in pregnancy
O22.20 - O22.23	Superficial thrombophlebitis in pregnancy
O22.90 - O22.93	Venous complication in pregnancy, unspecified
O87.0	Superficial thrombophlebitis in the puerperium
O87.4	Varicose veins of lower extremity in the puerperium
O87.9	Venous complication in the puerperium, unspecified

Background

Varicose veins are a common condition. In adult western populations visible varicose veins are present in 20 to 25 % of women and 10 to 15 % of men. In most persons, varicose veins do not cause symptoms other

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than poor cosmesis. Varicose vein surgery is one of the most commonly performed cosmetic procedures in the United States.

Most varicose veins do not require medical treatment (Tapley et al, 2003). In some cases, however, the circulation may be hindered enough to cause swelling of the foot and ankle, discomfort, a tingling sensation, or a feeling of heaviness. For most people with varicose veins, wearing specially fitted elastic stockings is all that is needed. The stockings should be carefully fitted to the individual, providing the most pressure in the lowest part of the leg. The stockings should be put on when first arising in the morning, preferably before getting out of bed. Exercise such as walking or cycling also helps promote better circulation from the lower part of the body. Resting with the legs elevated will help promote circulation; in contrast, sitting with the legs crossed can aggravate the condition. Authorities have recommended 6 or more months as a reasonable duration for a trial conservative management (NHS, 2005).

A substantial proportion of varicose vein symptoms respond to conservative management. A randomized controlled clinical trial compared surgery (n = 124) to conservative management (n = 122) of varicose veins (Michaels et al, 2006). Conservative management consisted of lifestyle advice relating to exercise, leg elevation, management of weight and diet, and the use of compression hosiery. In the surgical arm of the trial patients received the same lifestyle advice but also underwent surgical treatment, consisting of flush ligation of sites of reflux, stripping of the long saphenous vein and multiple phlebectomies, as appropriate. Although a greater proportion of patients assigned to surgery plus lifestyle advice at relieving symptoms at 1 year, approximately one-third of subjects assigned to conservative management reported some relief from conservative management with compression hosiery. At 2 years, there was no significant difference in symptom improvement between groups assigned to conservative management versus surgery. The authors posited that the lack of significant difference in symptomatology between groups at 2 years may have been due to cross-overs, with 7 patients in the conservative management group opting for surgery in year 1 and 37 patients opting for surgery in year 2. The study also found that persons assigned to surgery plus lifestyle advice had a greater improvement in cosmesis and quality of life than persons assigned to lifestyle advice alone, although it is not known whether improvements in quality of life were primarily related to improvements in cosmesis versus reductions in symptomatology. Weaknesses of the study included a substantial loss to follow-up in all groups. Fifteen of the 124 patients assigned to surgery either refused surgery in favor of conservative management or declined surgery due to fitness. Of the remaining 109 patients who underwent surgery, 43 were lost to follow up by the first year. Of subjects assigned to conservative treatment, 21 were lost to follow-up by the first year. The authors observed that, although surgery was more effective at improving symptomatology at 1 year, a substantial proportion of patients assigned to conservative treatment reported resolution or improvements in aching (26 %), heaviness (46 %), itching (56 %), and swelling (68 %). In addition, a substantial proportion of persons assigned to conservative management reported improvements in cosmesis. "Indeed, 22 % of the latter reported that they no longer had cosmetic concerns. These observations suggest a substantial benefit from surgery but perhaps support the case for careful evaluation of patients' symptoms and problems when considering surgical treatment."

An editorialist noted that the short follow-up of subjects assigned to surgery may result in an underestimate of the costs and an exaggeration of the benefits of surgery (van Rij, 2006). By the third year, only 40 % of subjects in the study by Michaels et al assigned to surgery were assessed. The editorialist noted, however, that most recurrences are diagnosed later than 3 years. Focusing on the short-term may lead to an under-estimate of cost and an over-estimate of benefit. The editorialist stated that prospective comparisons of durability up to 5 years and longer are infrequent and yet by this time the recurrence rate may be as high as 50 %.

In patients with varicose veins, leg pain may be associated with superficial thrombophlebitis or venous leg ulcers. In evaluating the role of varicose vein surgery in treatment of these conditions, the effectiveness of varicose vein surgery must be compared to conservative management.

If the patient is suffering from superficial thrombophlebitis, conservative management is indicated. According to available guidelines, uncomplicated superficial thrombophlebitis is usually treated symptomatically with heat, simple analgesia, non-steroidal anti-inflammatory drugs (NSAIDs), and compression stockings (SCHIN, 2002). Treatment should continue until symptoms have completely

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subsided (usually 2 to 6 weeks to subside but the thrombosed vein may be palpable and tender for months). More severe thrombophlebitis, as indicated by the degree of pain and redness and the extent of abnormality, should be treated by bed rest with elevation of the extremity and application of hot, wet compresses.

Leg ulcers arising from venous problems are called venous (varicose or stasis) ulcers. The main conservative treatment has been to apply a firm compression garment (bandage or stocking) to the lower leg in order to help the blood return back up the leg. Cullum et al (2002) conducted a meta-analysis of the literature on the effectiveness of compression bandaging and stockings in the treatment of varicose leg ulcers. The authors concluded that compression increases ulcer-healing rates compared with no compression. The authors also found that multi-layered systems are more effective than single-layered systems. High compression is more effective than low compression but there are no clear differences in the effectiveness of different types of high compression. In a meta-analysis, Nelson et al (2002) found circumstantial evidence of the benefit of compression in reducing recurrence of varicose ulcers. The authors also noted that recurrence rates may be lower in high compression hosiery than in medium compression hosiery and therefore patients should be offered the strongest compression with which they can comply.

In a trial conducted at 20 centers in the United Kingdom, funded by the National Institute for Health Research Health Technology Assessment Program, Gohel, et al. (2019) randomly assigned 450 patients with venous leg ulcers to receive compression therapy and undergo early endovenous ablation of superficial venous reflux within 2 weeks after randomization (early-intervention group) or to receive compression therapy alone, with consideration of endovenous ablation deferred until after the ulcer was healed or until 6 months after randomization if the ulcer was unhealed (deferred-intervention group). The primary outcome was the time to ulcer healing. Secondary outcomes were the rate of ulcer healing at 24 weeks, the rate of ulcer recurrence, the length of time free from ulcers (ulcer-free time) during the first year after randomization, and patient-reported health-related quality of life. Patient and clinical characteristics at baseline were similar in the two treatment groups. The time to ulcer healing was shorter in the early-intervention group than in the deferred-intervention group; more patients had healed ulcers with early intervention (hazard ratio for ulcer healing, 1.38; 95% confidence interval [CI], 1.13 to 1.68; P=0.001). The median time to ulcer healing was 56 days (95% CI, 49 to 66) in the early-intervention group and 82 days (95% CI, 69 to 92) in the deferred-intervention group. The rate of ulcer healing at 24 weeks was 85.6% in the early-intervention group and 76.3% in the deferred-intervention group. The median ulcer-free time during the first year after trial enrollment was 306 days (interquartile range, 240 to 328) in the early-intervention group and 278 days (interquartile range, 175 to 324) in the deferred-intervention group (P=0.002). The most common procedural complications of endovenous ablation were pain and deep-vein thrombosis. Early endovenous ablation of superficial venous reflux resulted in faster healing of venous leg ulcers and more time free from ulcers than deferred endovenous ablation.

A review (Weiß, et al., 2019) noted that weak points of the study by Gohel, et al. are

I. the relatively small (2–3 cm^2) ulcers, which are not representative of the ulcers commonly seen in inpatient/outpatient wound facilities,
II. the ulcer duration being very short (~3 months) compared with real-life situations, where patients present with long-standing ulcers,
III. the study population being almost exclusively white, and
IV. venous ulcers are included exclusively, whereas many ulcers are due to a combination of venous and arterial problems.

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The reviewers concluded: "Adequately powered, high-quality randomized controlled trials comparing the three available treatment options – compression, conventional venous surgery and endovenous treatment – are still required. These should measure and report outcomes that include time to ulcer healing, ulcer recurrence, adverse events, quality of life and cost-effectiveness."

Adequately powered, high-quality randomized controlled trials comparing the three available treatment options – compression, conventional venous surgery and endovenous treatment – are still required. These should measure and report outcomes that include time to ulcer healing, ulcer recurrence, adverse events, quality of life and cost-effectiveness.

According to a systematic review of the evidence, pentoxifilline has also been shown to be effective for treatment of venous leg ulcers (Nelson et al, 2002). According to the systematic evidence review, compression has been shown to prevent venous leg ulcers. The effectiveness of vein surgery for prevention or treatment of venous ulcers is "unknown" (Nelson et al, 2002).

Beyond conservative therapy, the treatment of varicose veins in the lower legs includes injection/compression sclerotherapy and surgical stripping or ligation or a combination of these approaches depending upon the severity of the condition. Despite many years of experience, there is still a disappointingly high recurrence rate of varices because many patients are inadequately investigated before treatment. As it has been shown that physical examination alone is unreliable, pre-treatment Doppler or Duplex ultrasound examination must be performed for localization of the sites of incompetence to allow the individualization of the treatment strategy for each patient. Photographs or office diagrams may be helpful in assessing the size and extent of the varices.

Under established guidelines, the basic tenet of successful treatment is to eliminate the primary and secondary sources of the reflux. These sources are usually a nearby perforator, or most often a major junction that causes redirected venous return through veins with intact valves.

Sclerotherapy has been found to be more effective in patients with dilated superficial or residual varicose veins, recurrent varicosities or incompetent perforating veins of small to moderate size (less than 6 mm) without vein reflux. Large varicosities do not respond as well as small varicosities to sclerotherapy (Rosenberg, 2006; MSAC, 2011; MAS, 2011). Inadvertent intra-arterial injection has been an untoward sequela of sclerotherapy. Almost all cases of painful varicosities are associated with junctional reflux. When reflux at the saphenofemoral and/or saphenopopliteal junctions is present, accepted guidelines provide that sclerotherapy should not be performed until surgical ligation and division of the junction has been done. The junctions themselves can not be adequately treated by sclerotherapy as junctional reflux must be addressed by endovenous ablation methods or surgical ligation or stripping (Jakobsen, 1979; MSAC, 2008; MSAC, 2011). Although varicosities can occasionally be present in the absence or reflux, there is a lack of evidence from reliable clinical studies of the effectiveness of sclerotherapy in relieving symptomatic varicosities not associated with junctional reflux. The sole randomized controlled clinical trial (n = 25) to address the efficacy of sclerotherapy in varicosities not associated with junctional reflux (Kalhe and Leng, 2004) evaluated sclerotherapy efficacy in obliterating varicosities, but did not address its effectiveness at relieving pain. Although sclerotherapy can be used to treat visible subcuticular veins (i.e., spider angiomas, and telangiectasias) less than 2.5 mm in size, these small veins do not cause symptoms and their treatment is purely cosmetic (MSAC, 2011).

Doppler ultrasound is often used in conjunction with other non-invasive physiologic testing to characterize the anatomy and physiology of the varicose vein network prior to injection or surgical intervention. However, duplex scans are also sometimes utilized during the sclerotherapy procedure itself. Their purported usefulness in this regard includes the localization of deep or inaccessible injection sites, such as when there are extensive networks of large deep varicosities, areas of significant reflux between superficial and deep systems, or risks to arterial structures. Ultrasound has also been used to monitor the effectiveness of compressive sclerotherapy in obliterating the lumen of the target vein and reducing reflux/retrograde flow. However, these indications have not been scientifically validated. There is little evidence, in the form of randomized prospective clinical trials, to support that ultrasound makes a significant difference in optimizing outcome or decreasing complications, from sclerotherapy for varicose veins, when compared to non-

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ultrasound-guided techniques. A structured evidence review conducted by the Alberta Heritage Foundation for Medical Research (AHFMR) (2003) concluded that "the reviewed evidence does not adequately address the questions; which sclerosant is superior and which technique with or without ultrasound guidance is most efficacious."

Venous reflux can be elicited manually by calf muscle compression and release, by the Valsava maneuver, or by pneumatic tourniquet release (Markovic & Shortell, 2014). If saphenofemoral reflux lasting longer than 500 ms is present, the diameter of the great saphenous veins (GSV) is recorded 2.5 cm distal to the saphenofemoral junction. The size of the vein has been correlated with the presence of significant saphenous reflux. The compliant GSV adjusts its luminal size to the level of transmural pressure, and measurement of its diameter has been shown to reflect the severity of hemodynamic compromise in limbs with GSV reflux. In a cohort study, Navarro, et al. (2002) evaluated the relationship of GSV diameter determined in the thigh and calf to clinical severity of reflux in 112 legs in 85 consecutive patients with saphenofemoral junction and truncal GSV incompetence. The authors stated that they found that the GSV diameter proved to be a relatively accurate measure of hemodynamic impairment and clinical severity in a model of saphenofemoral junction and GSV incompete ce, predicting not only the absence of abnormal reflux, but also the presence of critical venous incompetence. A GSV diameter of 5.5 mm or less predicted the absence of abnormal reflux, with a sensitivity of 78 %, a specificity of 87 %, positive and negative predictive values of 78 %, and an accuracy of 82 %.

Ligation and division of the saphenofemoral and/or saphenopopliteal junction is indicated in patients with symptomatic varicose veins who have failed conservative management, when reflux of greater than 0.5 seconds is demonstrated by Doppler examination or Duplex scanning. The literature states that operative excision of varicose veins in the leg(s) should be reserved for those that are very large (greater than 6 mm), extensive in distribution, or occur in large clusters. Ligation alone usually results in a high recurrence rate of the varicose vein, which may then require sclerotherapy treatment (MSAC, 2008). Stripping of the greater and/or lesser saphenous vein, performed in conjunction with ligation and division of their respective junctions, is indicated when the saphenous veins themselves show varicose changes (usually greater than 1 cm in diameter). Varicose vein surgery and/or sclerotherapy during pregnancy is not appropriate because dilatation of veins in the legs is physiologic and will revert to normal after delivery, at which time a more accurate appraisal can be made. Visible subcuticular veins (i.e., spider angiomas, and telangiectasias) less than 2.5 mm in size do not cause symptoms and their treatment is purely cosmetic.

Ambulatory phlebectomy (AP) (also known as microphlebectomy) is a minimally invasive procedure performed under local anesthesia, and is an accepted outpatient therapy for the removal of varicose veins. This treatment allows excision of almost all of the large varicose veins except the proximal long saphenous vein, which is better-managed by stripping. Non-refluxing varicose veins on the surface of the leg, not including the saphenous veins, may be treated as an outpatient procedure under local anaesthetic using ambulatory phlebectomy (MSAC, 2011). However, recurrence rates can be high if the source of the reflux is not treated (MSAC, 2011). The junctions themselves can not be treated with simple phlebectomy as junctional reflux must be addressed by endovenous ablation methods or rarely by surgical ligation and stripping (MSAC, 2011; Weiss, 2007). Patients can ambulate immediately after AP. Complications associated with AP include blister formation, localized thrombophlebitis, skin necrosis, hemorrhage, and persistent edema. The use of broad compression pads following AP reduces hemorrhage and enhances resorption.

The TriVex System (transilluminated powered phlebectomy) is an alternative method of providing ambulatory phlebectomy. This entails endoscopic resection and ablation of the superficial veins using an illuminator and a "powered vein rejector", a small powered surgical device. In this procedure, veins are marked with a magic marker. In order to enhance visualization of the veins, a bright light is introduced into the leg through a tiny incision. The powered vein rejector, which has a powered oscillating end, is then introduced to cut and dislodge the veins. The pieces of vein are then gently retrieved by suction down a tube. Transilluminated powered phlebectomy is usually performed in the hospital on an outpatient basis and under general anesthesia or using local anesthesia with sedation.

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The manufacturer of the TriVex System states that the unique illumination feature allows the surgeon to quickly and accurately target and remove the vein and then visually confirm its complete extraction. The manufacturer claims that this new process makes varicose vein removal more effective, complete and less traumatic for patients, by reducing the number of incisions required to perform the procedure and the duration of surgery. The manufacturer also claims that this method not only reduces the pain associated with varicose vein removal but also reduces the potential for post-operative infection. There is inadequate evidence, however, in the published peer-reviewed medical literature substantiating these claims. The potential advantages of the TriVex System over standard ambulatory phlebectomy have not been proven. Therefore, the TriVex procedure should be billed as any other varicose vein removal procedure.

The term endovenous catheter ablation (EVCA) has been used to refer to the several new catheter based minimally invasive alternatives to surgical stripping, including laser ablation and radiofrequency ablation. Endovenous catheter ablation and surgical ligation/stripping are indicated for treatment of the same general population: patients in whom the great and/or small saphenous veins have reflux or incompetence of 0.5 seconds or longer demonstrated on duplex scanning, and varicose vein symptoms significantly impinge on quality of life (MSAC, 2011). These patients have exhausted conservative treatment measures, and sclerotherapy is considered unlikely to provide successful results. Endovenous laser ablation and radiofrequency ablation are essentially identical except for the use of different specialized equipment and catheters, with thermal energy delivered through either a radiofrequency catheter or laser fiber (MSAC, 2011). The objectives of the two treatments are the same, being the destruction or ablation of a refluxing vein or segment of vein via application of thermal energy. The procedure to place the catheter within the vein is the same for radiofrequency ablation and endovenous laser ablation, also both procedures are conducted under duplex ultrasonography guidance (MSAC, 2011). The physiological mechanism of vein ablation is also the same, with thermal energy producing endothelial and vein wall damage, denaturing and occluding the vein to close the vein, abolishing venous reflux and visible varicosities (MSAC, 2011).

ECVA is performed with tumescent anesthesia (Markovic and Shortell, 2014). Tumescent anesthesia allows physicians to use large volumes (500 ml) of dilute (0.1%) lidocaine in a single session while achieving anesthesia levels equivalent to those achieved with 1% lidocaine. In this way, the entire thigh portion of the GSV can be safely anesthetized (and consequently obliterated) at one time. Epinephrine can be added to the solution to improve postoperative hemostasis, increase venous contraction around the heat-generating catheter, and lengthen the duration of postprocedural analgesia. A common formula for the tumescent anesthesia solution is 450 ml of normal saline mixed with 50 ml of 1% lidocaine with epinephrine (1:100,000 dilution) and 10 ml of sodium bicarbonate to buffer the acidity of the lidocaine.

Endovenous laser ablation of saphenous vein (ELAS) is a treatment alternative to surgical ligation and stripping of the greater saphenous vein. Endovenous laser therapy for varicose veins is indicated for patients with clinically documented primary venous reflux, confirmed by duplex ultrasound, of the great or small saphenous veins (MSAC, 2008). Endovenous laser ablation is only suitable for patients with large, saphenous varicose veins, as the catheter requires saphenous veins with a minimum 4.5mm in diameter. These patients have exhausted other conservative treatment measures and sclerotherapy is considered unlikely to be successful (MSAC, 2008). After ultrasound examination to confirm the site and extent of saphenous reflux, a catheter is introduced into the damaged vein along a guide wire via percutaneous puncture at the distal extent of the diseased saphenous vein (MSAC, 2008). Perivascular infiltration of dilute local anesthetic along the length of the vein is then performed under ultrasound guidance to collapse the lumen and compress the vein onto the catheter, to dissipate heat generated during the procedure so as to prevent tissue damage, and to anesthetize the vein (MSAC). The guide wire is replaced with a laser probe introduced through the catheter to just below the saphenofemoral or saphenopopliteal junction, with positioning confirmed by ultrasound. Laser energy is then applied as the fiber and catheter are slowly withdrawn so as to close the vein and abolish venous reflux. Pulses of laser light are emitted inside the vein, and the vein collapses, and seals shut. This procedure may be performed in the office under local anesthesia. A bandage or compression hose is placed on the treated leg following the treatment. The procedure is performed on an outpatient basis.

Endovenous laser treatment can only be used for large veins, as a catheter must be inserted into the lumen of the vein to be treated (MSAC, 2008). Endovenous laser treatment is not viable on saphenous veins smaller

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than 4.5 mm in diameter, and cannot be used for the treatment of small veins or telangiectases. Smaller veins may be treated with sclerotherapy or ambulatory phlebectomy.

A range of laser wavelengths can be used to achieve occlusion; there is no strong evidence to indicate that any particular wavelength is superior to any other (MSAC, 2008). One systematic evidence review reported that the short term (within 6 months) reported occlusion rates of the GSV and small saphenous vein (SSV) found in studies of endovenous laser therapy were all greater than 90%.

Absolute contraindications to ELAS treatment include occlusive deep venous thrombosis and pregnancy. Relative contraindications include occlusive arterial disease, hypercoabulability, tortuous veins, and inability to ambulate (MSAC, 2008).

Endoluminal radiofrequency thermal heating (VNUS Closure Procedure) has been used with or without ligation and division for treatment of incompetence of the saphenofemoral and saphenopopliteal junction. To perform the radiofrequency ablation (RFA) procedure, the affected leg is prepared and draped, and a superficial local anaesthetic agent is used to anesthetize the site of cannulation. A radiofrequency catheter is inserted into the lumen of the greater saphenous vein, starting at its junction with the femoral vein. Under some protocols, the placement of the catheter is guided by duplex ultrasonography. The radiofrequency catheter heats the inner lumen of the vein to 85°C, with subsequent scarring and closure of the treated vein. The procedure is performed in an office setting without general anesthesia; treatment time averages 20 mins. Adverse sequelae include purpura, erythema and pain, which generally resolve days or weeks after treatment, and indurated fibrous cords that may remain for several months.

Upon completion of the RFA procedure, the site of venous puncture is dressed, and compression stockings and/or bandages are applied as appropriate to reduce the risk of venous thromboembolism and to reduce postoperative bruising and tenderness (MSAC, 2011). Non-steroidal anti-inflammatory drugs are commonly used for post-procedural pain relief. For most patients additional procedures such as sclerotherapy or phlebectomy are required for the treatment of superficial veins below the knee, any tributary varicose veins, and telangiectases. These procedures may be performed during the RFA or endovenous laser treatment procedure, or over one or two follow-up visits.

Radiofrequency ablation is designed as a single-use therapeutic intervention, delivered as a single course of treatment per affected leg to obliterate the great or small saphenous veins through the application of thermal energy (MSAC, 2011). While generally indicated for primary varicose veins, re-treatment of varicose veins with RFA may be possible in some patients where neovascularisation or revascularisation has occurred. However, revascularization in the short term following treatment is uncommon. Studies reporting on radiofrequency ablation with the more efficient second generation catheters report ablation rates close to 100% at 6-month follow-up with no major adverse events (MAS, 2011).

Prospective case series extending to 24 months have shown success rates with RFA similar to those reported for vein ligation and stripping. Weiss and Weiss (2002) reported complete disappearance of the treated saphenous vein in 90 % of 21 patients followed for 24 months. Endothermal radiofrequency thermal heating may be performed with or without high ligation of the greater saphenous vein. Chandler et al (2000) found no statistically significant difference in 1-year success rates from endovenous radiofrequency catheter ablation in 120 limbs treated without saphenofemoral ligation and 60 limbs treated with saphenofemoral ligation. The authors concluded that "these early results suggest that extended sapheno-femoral junction (SFJ) ligation may add little to effective GSV [greater saphenous vein] obliteration, but our findings are not sufficiently robust to warrant abandonment of SFJ ligation as currently practiced in the management of primary varicose veins associated with GSV reflux."

Pivotal studies of endovenous catheter ablation (endovenous laser ablation and endovenous radiofrequency ablation) procedures have focused on junctional incompetence. There is a lack of evidence of the effectiveness of endovenous catheter ablation procedures for treatment of varicose tributaries and perforator veins. In addition, there are no studies comparing endovenous catheter ablation procedures to standard methods of treating varicose tributaries and perforator veins with sclerotherapy and ambulatory phlebectomy.

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The Society for Interventional Radiologists (2003) has a position statement on VNUS that states that "(d)uplex ultrasound is necessary to map the anatomy of the venous system prior to the procedure, and imperative during the procedure for correct catheter placement and for proper tumescent anesthetic administration to minimize potential complications. Duplex ultrasound also is necessary for follow-up after endovenous ablation."

Sadick (2000) has noted that the new less-invasive technologies for treatment of varicose veins must be evaluated with caution. "Long-term studies with other technologies must be compared with surgical ligation of the incompetent SFJ (sapheno-femoral junction). Six-month and 5-year follow-ups are two different end-points. The latter is a more accurate time interval of therapeutic efficacy."

Subfascial endoscopic perforator vein surgery (SEPS) is a minimally invasive endoscopic procedure that eliminates the need for a large incision in the leg. It has been explored as an alternative to the traditional open surgical treatment of chronic venous insufficiency. The aim of the procedure is to interrupt incompetent medial calf perforating veins to reduce venous reflux and decrease ambulatory venous hypertension in critical areas above the ankle where venous ulcers most frequently develop. Kalra and Gloviczki (2002) stated that available evidence confirmed the superiority of SEPS over open perforator ligation, but do not address its role in the surgical treatment of advanced chronic venous insufficiency (CVI) and venous ulceration. Ablation of superficial reflux by high ligation and stripping of the greater saphenous vein with avulsion of branch varicosities is concomitantly performed in the majority of patients undergoing SEPS. The clinical and hemodynamic improvements attributable to SEPS thus are difficult to ascertain. As with open perforator ligation, clinical and hemodynamic results are better in patients with primary valvular incompetence (PVI) than in those with the post-thrombotic (PT) syndrome. Until prospective, randomized, multicenter clinical studies are performed to address lingering questions regarding the effectiveness of SEPS, the procedure is recommended in patients with advanced CVI secondary to PVI of superficial and perforating veins, with or without deep venous incompetence. The performance of SEPS in patients with PT syndrome remains controversial.

Contraindications for SEPS include associated arterial occlusive disease, infected ulcer, a non-ambulatory patient, and a medically high-risk patient. Diabetes, renal failure, liver failure, morbid obesity, ulcers in patients with rheumatoid arthritis, or scleroderma, and presence of deep vein obstruction at the level of the popliteal vein or higher on pre-operative imaging are relative contraindications. Patients with extensive skin changes, circumferential large ulcers, recent deep vein thrombosis, severe lymphedema, or large legs may not be suitable candidates (Kalra and Gloviczki, 2002).

McDonagh et al (2002, 2003) has reported on the effectiveness of ultrasound-guided foam sclerotherapy (comprehensive objective mapping, precise image-guided injection, anti-reflux positioning and sequential sclerotherapy (COMPASS) technique) in the treatment persons with varicosities of the greater saphenous vein with saphenous vein reflux. Published studies of the COMPASS technique involve relatively short-term follow up. Study subjects were followed for 3 years, and for only 2 years after completion of a series of repeat sclerotherapy injections that were administered over 1 year. In addition, these studies do not include a comparable group of subjects treated with surgery, which has been the primary method of treating incompetent long saphenous veins. Thus, it is not possible to reach definitive conclusions about the durability of results of the COMPASS technique or its effectiveness compared with surgery for treatment of greater saphenous vein varicosities and saphenofemoral incompetence. In addition, published studies of the COMPASS technique come from a single group of investigators. In reviewing the study by McDonagh (2002), Allegra (2003) commented: "Surgical treatment has a long history with 5-20 year follow-ups being routine. The 3 year follow-up in the present study is certainly not comparable …. This study does not answer questions raised against ultrasound guided sclerotherapy. It would be important to have the relevant aspects of this study duplicated, reproduced, and verified."

Published long-term randomized controlled clinical studies have demonstrated that surgery plus sclerotherapy is more effective than surgery alone for treatment of varicosities associated with incompetence of the saphenofemoral junction. Belcaro et al (2003) reported on the results from the Venous Disease International Control (VEDICO) trial, the first long-term randomized controlled clinical trial of foam sclerotherapy. The VEDICO trial involved 749 patients with varicose veins and saphenous vein

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incompetence who were randomly treated by six different approaches: standard sclerotherapy, high-dose sclerotherapy, surgical ligation, stab avulsion, foam sclerotherapy, and combined surgery (ligation or stab avulsion) and high dose sclerotherapy. At 10 years, the occurrence of new veins was 56 % for standard sclerotherapy, 51 % for foam sclerotherapy, 49 % for high-dose sclerotherapy, 41 % for stab avulsion, 38 % for ligation, and 27 % for combined surgery and sclerotherapy.

Belcaro et al (2000) reported on the results of a randomized controlled clinical study comparing ultrasound-guided sclerotherapy with surgery alone or surgery combined with sclerotherapy in 96 patients with varicose veins and superficial venous incompetence. Although all approaches were reported to be effective in controlling the progression of venous incompetence, surgery appeared to be the most effective method on a long-term basis, and that surgery combined with sclerotherapy may be more effective than surgery alone. After 10 years follow-up, no incompetence of the saphenofemoral junction was observed in both groups assigned to surgery, compared to 18.8 % of limbs of subjects assigned to ultrasound-guided sclerotherapy. Of limbs treated with ultrasound-guided sclerotherapy, 43.8 % of the distal venous systems were incompetent, compared to 36 % of limbs of subjects treated with surgery alone, and 16.1 % of limbs of subjects treated with surgery plus sclerotherapy.

The L'Agence Nationale d' Accreditation et d'Evaluation en Sante (l'ANAES) (Grange et al, 1998) conducted a systematic review of the literature on the indications of surgery for varicose veins of the legs. Given the lack of good scientific evidence on the various treatments for primary varicose veins, the working group made recommendations based on professional agreement. They concluded that surgery is the treatment of choice for saphenous veins with reflux. An evidence review of surgical treatments for deep venous incompetence by the Alberta Heritage Foundation for Medical Research (Scott and Corabain, 2003) stated that "(s)clerotherapy is particularly effective in superficial venous incompetence when there is a large vein located in close proximity to the ulcer. However, surgery is indicated when there is substantial proximal incompetence in a saphenous vein."

A comprehensive evidence review of sclerotherapy for varicose veins conducted by the Alberta Heritage Foundation for Medical Research (2003) concluded that "the reviewed evidence does not adequately address the questions; which sclerosant is superior and which technique with or without ultrasound guidance is most efficacious … In recent years, new methods such as ES (endovascular sclerotherapy) and foam sclerotherapy (using ultrasound guidance) have been developed and proposed to improve the safety and efficacy of sclerotherapy for various types of varicose veins. Evidence about these new techniques for treating patients with incompetence of the long saphenous vein is limited." The assessment concluded that although " (s)clerotherapy appears to be the treatment of choice for reticular varicosities, telangiectasia and other small, unsightly blood vessels … (t)he place of sclerotherapy as the first treatment for larger varicose veins (saphenous or non-saphenous) remains controversial."

There is a lack of reliable evidence that one type of sclerosant is significantly better than any other (Tisi 2007; Jia et al, 2006). Jia and colleagues (2007) evaluated the safety and effectiveness of foam sclerotherapy for varicose veins. The authors concluded that serious adverse events associated with foam sclerotherapy are rare. However, there is insufficient evidence to allow a meaningful comparison of the effectiveness of this treatment with that of other minimally invasive therapies or surgery.

Kendler and associates (2007) noted that "(r)ecently the use of foam sclerotherapy had a renaissance. Several studies have documented the efficacy of foam sclerotherapy in selected patients. The possibility of treating patients in an outpatient setting, with low costs and rapidly, makes foam sclerotherapy very attractive compared to invasive and minimally invasive methods. However long-term follow-ups in properly controlled randomized trials are needed before foam sclerotherapy can be recommended as a routine procedure".

The FDA has approved Asclera (polidocanol) injection (BioForm Medical Inc., Franksville, WI) to close spider veins (tiny varicose veins less than 1 millimeter in diameter) and reticular veins (those that are 1 to 3 millimeters in diameter). As these small veins have not been demonstrated to cause symptoms, treatment of these small veins is considered cosmetic.

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There is emerging evidence for the Ambulatory Conservative Hemodynamic Management of Varicose Veins (CHIVA) method. In an open-label, randomized controlled trial, Pares and colleagues (2010) compared the effectiveness of the Ambulatory Conservative Hemodynamic Management of Varicose Veins (CHIVA) method for the treatment of varicose veins with respect to the standard treatment of stripping. According to the authors, CHIVA consists of minimally invasive surgical procedures under local anesthesia that are based on hemodynamic analysis of the legs with pulsed Doppler ultrasound. A total of 501 adult patients with primary varicose veins were treated in a single center. They were assigned to an experimental group, the CHIVA method (n = 167) and 2 control groups: stripping with clinic marking (n = 167) and stripping with Duplex marking (n = 167). The outcome measure was clinical recurrence within 5 years, assessed clinically by previously trained independent observers. Duplex ultrasonography was also used to assess recurrences and causes. In an intention-to-treat analysis, clinical outcomes in the CHIVA group were better (44.3 % cure, 24.6 % improvement, 31.1 % failure) than in both the stripping with clinic marking (21.0 % cure, 26.3 % improvement, 52.7 % failure) and stripping with Duplex marking (29.3 % cure, 22.8 % improvement, 47.9 % failure) groups. The ordinal odds ratio between the stripping with clinic marking and CHIVA groups, of recurrence at 5- year follow-up, was 2.64, (95 % confidence interval (CI): 1.76 to 3.97, $p < 0.001$). The ordinal odds ratio of recurrence at 5-year follow-up, between the stripping with Duplex marking and CHIVA group, was 2.01 (95 % CI: 1.34 to 3.00, $p < 0.001$). The authors concluded that these findings indicated that the CHIVA method is more effective than stripping with clinical marking or stripping with Duplex marking to treat varicose veins. Furthermore, when carrying out a stripping intervention, Duplex marking does not improve the clinical results of this ablative technique.

In a randomized study, Rasmussen et al (2011) compared 4 treatments for varicose GSVs. A total of 500 consecutive patients (580 legs) with GSV reflux were randomized to endovenous laser ablation (EVLT, 980 and 1,470 nm, bare fiber), radiofrequency ablation (RFA), ultrasound-guided foam sclerotherapy (USGFS) or surgical stripping using tumescent local anesthesia with light sedation. Mini-phlebectomies were also performed. Patients were examined with duplex imaging before surgery, and after 3 days, 1 month and 1 year. At 1 year, 7 (5.8 %), 6 (4.8 %), 20 (16.3 %) and 4 (4.8 %) of the GSVs were patent and refluxing in the laser, radiofrequency, foam and stripping groups respectively ($p < 0.001$). One patient developed a pulmonary embolus after foam sclerotherapy and 1 a deep vein thrombosis after surgical stripping. No other major complications were recorded. The mean (S.D.) post-intervention pain scores (scale 0 to 10) were 2.58 (2.41), 1.21 (1.72), 1.60 (2.04) and 2.25 (2.23), respectively ($p < 0.001$). The median (range) time to return to normal function was 2 (0 to 25), 1 (0 to 30), 1 (0 to 30) and 4 (0 to 30) days, respectively ($p < 0.001$). The time off work, corrected for weekends, was 3.6 (0 to 46), 2.9 (0 to 14), 2.9 (0 to 33) and 4.3 (0 to 42) days, respectively ($p < 0.001$). Disease-specific quality-of-life and Short Form 36 (SF-36) scores had improved in all groups by 1-year follow-up. In the SF-36 domains bodily pain and physical functioning, the radiofrequency and foam groups performed better in the short-term than the others. The authors concluded that all treatments were efficacious. The technical failure rate was highest after foam sclerotherapy, but both RFA and foam were associated with a faster recovery and less post-operative pain than EVLT and stripping.

In a Cochrane review, Nesbitt et al (2011) reviewed available randomized controlled trial (RCT) data comparing USGFS, RFA and EVLT to conventional surgery (high ligation and stripping (HL/S)) for the treatment of great saphenous varicose veins. The Cochrane Peripheral Vascular Diseases (PVD) Group searched their Specialised Register (July 2010) and CENTRAL (The Cochrane Library 2010, Issue 3). In addition the authors performed a search of EMBASE (July 2010). Manufacturers of EVLT, RFA and sclerosant equipment were contacted for trial data. All RCTs of EVLT, RFA, USGFS and HL/S were considered for inclusion. Primary outcomes were recurrent varicosities, re-canalization, neovascularization, technical procedure failure or need for re-intervention, patient quality of life (QOL) scores and associated complications. Secondary outcomes were type of anesthetic, procedure duration, hospital stay and cost. A total of 13 reports from 5 studies with a combined total of 450 patients were included. Rates of re-canalization were higher following EVLT compared with HL/S, both early (within four months) (5/149 versus 0/100; odds ratio (OR) 3.83, 95 % CI: 0.45 to 32.64) and late re-canalization (after 4 months) (9/118 versus 1/80; OR 2.97 95 % CI: 0.52 to 16.98), although these results were not statistically significant. Technical failure rates favored EVLT over HL/S (1/149 versus 6/100; OR 0.12, 95 % CI: 0.02 to 0.75). Recurrence following RFA showed no difference when compared with surgery. Re-canalization within 4 months was observed more frequently following RFA compared with HL/S although not statistically significant (4/105 versus 0/88; OR 7.86, 95 % CI: 0.41 to 151.28); after 4 months no difference was

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observed. Neovascularization was observed more frequently following HL/S compared with RFA, but again this was not statistically significant (3/42 versus 8/51; OR 0.39, 95 % CI: 0.09 to 1.63). Technical failure was observed less frequently following RFA compared with HL/S although this was not statistically significant (2/106 versus 7/96; OR 0.48, 95 % CI: 0.01 to 34.25). No RCTs comparing HL/S versus USGFS met the study inclusion criteria. QOL scores and operative complications were not amenable to meta-analysis. The authors concluded that currently available clinical trial evidence suggests RFA and EVLT are at least as effective as surgery in the treatment of great saphenous varicose veins. There are insufficient data to comment on USGFS. They stated that further randomized trials are needed; and they should aim to report and analyze results in a congruent manner to facilitate future meta-analysis.

Mueller and Raines (2013) stated that the ClariVein system is the first venous ablation technique to employ a hybrid (dual-injury) technique built into 1 catheter-based delivery system. Endo-mechanical abrasion is produced by the tip of the catheter's rotating wire (mechanical component); and EVCA is via simultaneous injection of sclerosant over the rotating wire (chemical component). The author was an early adopter of this technique and via experience has developed a detailed step-by-step protocol. To date, there have been 2 pivotal clinical studies published using the ClariVein system. These data were compared with the results using other methods of endovenous ablation. The authors concluded that the ClariVein system has the potential to become a first-line treatment.

Lawson et al (2013) noted that less invasive endovenous techniques have been shown to be as effective as open surgery in the treatment of varicose veins. Furthermore, they cause less post-operative bruising and pain and enable early return to normal activities and work. Tumescent anesthesia is safe and obviates complications of general or spinal anesthesia. Drawbacks are a steep learning curve and painful administration during treatment. Tumescentless techniques like ClariVein or VenaSeal Sapheon Closure System are recently under investigation. Short-term results of VenaSeal are comparable with thermal ablation. The procedure is safe without serious adverse events. Peri-operative pain and patient discomfort with this tumescentless approach is minimal but post-operative recovery is temporarily hindered by thrombophlebitis in 14 to 15 % of patients. One-year results in a small feasibility study has demonstrated durable closure at this end-point. No longer-term results are available. A randomized control trial between VenaSeal and Covidien ClosureFast is in a preparatory phase.

A randomized controlled trial comparing foam sclerotherapy to laser ablation and surgery found that laser ablation and surgery had better outcomes, and that laser had the fewest procedural complications. Brittenden et al (2014) stated that ultrasound-guided foam sclerotherapy and endovenous laser ablation are widely used alternatives to surgery for the treatment of varicose veins, but their comparative effectiveness and safety remain uncertain. In a randomized trial involving 798 participants with primary varicose veins at 11 centers in the United Kingdom, these researchers compared the outcomes of foam, laser (laser ablation of truncal saphenous veins, followed if needed by foam sclerotherapy) and surgical treatments (proximal ligation and stripping of the great saphenous vein with concurrent phlebectomy). Study participants had varicose veins larger than 3 mm in diameter and reflux of the saphenous veins of more than 1 second by duplex ultrasound. The participants mean age was 49 years, 57% were women, and approximately 30% had bilateral varicose veins. Those with recurrent varicose veins after previous treatment were excluded. Primary outcomes at 6 months were disease-specific quality of life and generic quality of life, as measured on several scales. Secondary outcomes included complications and measures of clinical success. After adjustment for baseline scores and other covariates, the mean disease-specific quality of life was worse after treatment with foam than after surgery ($p = 0.006$) but was similar in the laser and surgery groups. There were no significant differences between the surgery group and the foam or the laser group in measures of generic quality of life. At 6 months, approximately 80% of patients in the laser and surgery groups showed complete ablation of the great saphenous vein on duplex ultrasound, compared with only 43% in the foam group ($p < 0.001$). The frequency of procedural complications was similar in the foam group (6 %) and the surgery group (7 %); but was lower in the laser group (1 %) than in the surgery group ($p < 0.001$); the frequency of serious adverse events (approximately 3 %) was similar among the groups. At 6 months, lumpiness and staining of the skin were somewhat more common in the foam group.

On November 26, 2013, the FDA approved Varithena (polidocanol injectable foam) for the treatment of patients with incompetent veins and visible varicosities of the great saphenous vein (GSV) system. The

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prescribing information states: "Varithena (polidocanol injectable foam) is indicated for the treatment of incompetent great saphenous veins, accessory saphenous veins and visible varicosities of the great saphenous vein (GSV) system above and below the knee. Varithena improves the symptoms of superficial venous incompetence and the appearance of visible varicosities." Although the FDA approval does not exclude use of Varithena foam sclerotherapy for the treatment of SF or SP junctional (SPJ) reflux, there are a lack of studies comparing Varithena to endovenous ablation procedures for SF or SPJ reflux. In addition, there is a paucity of evidence examining the long-term durability of results of Varithena treatment of junctional reflux.

Todd et al (2014) reported on a RCT to determine efficacy and safety of polidocanol endovenous microfoam in treatment of symptoms and appearance in patients with saphenofemoral junction incompetence due to reflux of the great saphenous vein or major accessory veins. Patients were randomized equally to receive polidocanol endovenous microfoam 0.5 %, polidocanol endovenous microfoam 1.0 % or placebo. The primary efficacy end-point was patient-reported improvement in symptoms, as measured by the change from baseline to Week 8 in the 7-day average electronic daily diary VVSymQ™ score. The co-secondary end-points were the improvement in appearance of visible varicosities from baseline to Week 8, as measured by patients and by an independent physician review panel. In 232 treated patients, polidocanol endovenous microfoam 0.5 % and polidocanol endovenous microfoam 1.0 % were superior to placebo, with a larger improvement in symptoms (VVSymQ (-6.01 and-5.06, respectively, versus -2.00; $p < 0.0001$) and greater improvements in physician and patient assessments of appearance ($p < 0.0001$). These findings were supported by the results of duplex ultrasound and other clinical measures. Of the 230 polidocanol endovenous microfoam-treated patients (including open-label patients), 60 % had an adverse event compared with 39 % of placebo; 95 % were mild or moderate. No pulmonary emboli were detected and no clinically important neurologic or visual adverse events were reported. The most common adverse events in patients treated with polidocanol endovenous microfoam were retained coagulum, leg pain and superficial thrombophlebitis; most were related to treatment and resolved without sequelae.

Brittenden and colleagues (2015) stated that foam sclerotherapy (foam) and endo-venous laser ablation (EVLA) have emerged as alternative treatments to surgery for patients with varicose veins (VV), but uncertainty exists regarding their effectiveness in the medium to longer term. These investigators evaluated the clinical effectiveness and cost-effectiveness of foam, EVLA and surgery for the treatment of VV. A total of 798 patients with primary VV (foam, $n = 292$; surgery, $n = 294$; EVLA, $n = 212$) were included in this study. Patients were randomized between all 3 treatment options (8 centers) or between foam and surgery (3 centers). Primary outcome measures included disease-specific [Aberdeen Varicose Vein Questionnaire (AVVQ)] and generic [European Quality of Life-5 Dimensions (EQ-5D), Short Form questionnaire-36 items (SF-36) physical and mental component scores] quality of life (QoL) at 6 months. Cost-effectiveness as cost per quality-adjusted life-year (QALY) gained. Secondary outcome measures entailed QoL at 6 weeks; residual VV; Venous Clinical Severity Score (VCSS); complication rates; return to normal activity; truncal vein ablation rates; and costs. The results appeared generalizable in that participants' baseline characteristics (apart from a lower-than-expected proportion of females) and post-treatment improvement in outcomes were comparable with those in other RCTs. The health gain achieved in the AVVQ with foam was significantly lower than with surgery at 6 months [effect size -1.74, 95 % CI: -2.97 to -0.50; $p = 0.006$], but was similar to that achieved with EVLA. The health gain in SF-36 mental component score for foam was worse than that for EVLA (effect size 1.54, 95 % CI: 0.01 to 3.06; $p = 0.048$) but similar to that for surgery. There were no differences in EQ-5D or SF-36 component scores in the surgery versus foam or surgery versus EVLA comparisons at 6 months. The trial-based cost-effectiveness analysis showed that, at 6 months, foam had the highest probability of being considered cost-effective at a ceiling willingness-to-pay ratio of £20,000 per QALY. EVLA was found to cost £26,107 per QALY gained versus foam, and was less costly and generated slightly more QALYs than surgery. Markov modelling using trial costs and the limited recurrence data available suggested that, at 5 years, EVLA had the highest probability (approximately 79 %) of being cost-effective at conventional thresholds, followed by foam (approximately 17 %) and surgery (approximately 5 %). With regard to secondary outcomes, health gains at 6 weeks ($p < 0.005$) were greater for EVLA than for foam (EQ-5D, $p = 0.004$). There were fewer procedural complications in the EVLA group (1 %) than after foam (7 %) and surgery (8 %) ($p < 0.001$). Participants returned to a wide range of behaviors more quickly following foam or EVLA than following surgery ($p < 0.05$). There were no differences in VCSS between the 3 treatments. Truncal ablation rates were higher for surgery ($p < 0.001$) and EVLA ($p < 0.001$) than for foam, and were similar for surgery and EVLA. The authors concluded that considerations of both the 6-month

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clinical outcomes and the estimated 5-year cost-effectiveness suggested that EVLA should be considered as the treatment of choice for suitable patients.

Marsden et al (2015) investigated the cost-effectiveness of interventional treatment for VV in the United Kingdom National Health Service (UK NHS), and informed the national clinical guideline on VV, published by the National Institute of Health and Care Excellence (NICE). An economic analysis was constructed to compare the cost-effectiveness of surgery, endo-thermal ablation (ETA), ultrasound-guided foam sclerotherapy (UGFS), and compression stockings (CS). The analysis was based on a Markov decision model, which was developed in consultation with members of the NICE guideline development group (GDG). The model had a 5-year time horizon, and took the perspective of the UK NHS. Clinical inputs were based on a network meta-analysis (NMA), informed by a systematic review of the clinical literature. Outcomes were expressed as costs and quality-adjusted life years (QALYs). All interventional treatments were found to be cost-effective compared with CS at a cost-effectiveness threshold of £20,000 per QALY gained; ETA was found to be the most cost-effective strategy overall, with an incremental cost-effectiveness ratio of £3,161 per QALY gained compared with UGFS. Surgery and CS were dominated by ETA. The authors concluded that interventional treatment for VV is cost-effective in the UK NHS. Specifically, based on current data, ETA is the most cost-effective treatment in people for whom it is suitable. The results of this research were used to inform recommendations within the NICE guideline on VV.

Compression Following Treatment for Varicose Veins

El-Sheikha et al (2015) stated that consensus regarding compression following treatment of VV has yet to be reached. This systematic review aimed to establish the optimal compression regimen after venous treatment. A systematic review of MEDLINE, Embase and CENTRAL was performed to identify RCTs investigating different compression strategies following treatment for superficial venous insufficiency. A total of 7 RCTs comparing different durations and methods of compression fulfilled the inclusion criteria. The treatment modality was open surgery in 3 trials, foam sclerotherapy in 2 and EVLA in 2 trials. The quality of the studies was variable, and significant sources of potential bias were present. Both the studies and compression regimens used were heterogeneous; 10 products were used in 6 general regimens for a duration of 0 to 42 days. One study suggested that 7 days rather than 2 days of stockings following EVLA was associated with superior QoL and less pain at 1 week. Another study reported that, following surgery, application of a compression stocking after 3 days of bandaging was associated with a slightly longer recovery than no compression after 3 days. One study recorded compliance clearly, finding it to be only 40 %. The quality and heterogeneity of the studies precluded meta-analysis. The authors concluded that there is currently little quality evidence upon which to base any recommendations concerning compression following treatment for VV.

Micronized Purified Flavonoid Fraction Therapy

Pietrzycka and colleagues (2015) stated that the etiology of VV involves various factors and pathomechanisms including endothelial cell activation or dysfunction, venous hypertension, vein wall hypoxia, shear stress disturbances, inflammatory reaction activation or free radical production. To improve the understanding of the mechanisms of potential pharmacological interventions for chronic venous disease, these researchers evaluated the influence of micronized purified flavonoid fraction (MPFF) on the relationship between antioxidant enzyme balance, endothelin-1 (ET-1) and tumor necrosis factor-alpha (TNF-α) levels. Blood samples were obtained from 89 women with primary VV; 34 were treated with MPFF and 55 did not receive any phlebotropic drug treatment. For the evaluation of the blood antioxidant enzyme balance, catalase (CAT) and superoxide dismutase (SOD) activity was assessed and the CAT/SOD ratio was calculated. Patients taking MPFF had significantly lower ET-1 levels than those not taking MPFF [median (25 to 75th quartile): 24.2 (22.30 to 27.87) versus 37.62 (24.9 to 44.58) pg/ml; $p < 0.05$]. In those taking MPFF, a higher CAT/SOD ratio [39.8 (24.7 to 72.6) versus 28.8 (16.3 to 57.7); $p < 0.05$] and a lower TNF-α concentration [6.82 (4.42 to 13.39) versus 12.94 (6.01 to 27.33) pg/ml; $p < 0.05$] was also observed. In women not taking MPFF, ET-1 levels increased with the CAT/SOD ratio. In those taking MPFF, the ET-1 level was stable at about 25.0 pg/ml; up to a CAT/SOD ratio of 100. TNF-α level increased continuously with an increasing CAT/SOD ratio; however, the highest levels of TNF-α were observed in women not taking

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MPFF. The authors concluded that they demonstrated the ability of MPFF to effectively lower the levels of ET-1 and TNF-α in patients with chronic venous disease. They sated that further investigations are needed to define the therapeutic potential of MPFF including the potential effect on chronic subclinical inflammation, antioxidant imbalance and vascular dysfunction during the development of chronic venous disease.

Cyanoacrylate Tissue Adhesive (e.g., the VariClose Vein Sealing System and the VenaSeal Closure System)

The VenaSeal Closure System:

The VenaSeal Closure System (Sapheon Inc., Morrisville, NC) is a minimally invasive, non-tumescent, non-thermal and non-sclerosant procedure that uses a medical adhesive to close the diseased vein in patients with symptomatic venous reflux disease. Unlike other treatments, the VenaSeal Closure System does not require tumescent anesthesia, allowing patients to return to their normal activities following the procedure; it also eliminates the risk of nerve or other heat-related injuries associated with thermal-based procedures, and thus may reduce the need for compression stockings post-procedure.

Toonder et al (2014) noted that percutaneous thermo-ablation techniques are still being used today and seem more effective than non-thermal techniques. However, thermal techniques require anesthesia and potentially may cause inadvertent damage to surrounding tissues such as nerves. Cyanoacrylate adhesive has a proven record, but not for the treatment of chronic venous disease of the leg. Innovation has led to the development of the VenaSeal Sapheon Closure System, which has been designed to use a modified cyanoacrylate glue as a new therapy for truncal vein incompetence. These researchers examined the feasibility of ultrasound-guided cyanoacrylate adhesive perforator embolization (CAPE). The authors stated that results of this feasibility study showed a 76 % occlusion rate of incompetent perforating veins without serious complications; further investigation with a dedicated delivery device in a larger patient population is needed.

McHugh and Leahy (2014) stated that endothermal treatment of the great saphenous vein has become the first line of treatment for superficial venous reflux. Newer treatments, especially non-thermal ablation have potential benefits both for patient acceptability and decreased risk of nerve injury. These researchers described the current non-thermal options available including advantages and disadvantages. Ultrasound-guided foam sclerotherapy avoids the risk of nerve injury, however it is not as effective as endothermal ablation. Mechano-chemical endovenous ablation combines mechanical endothelial damage using a rotating wire, with the infusion of a liquid sclerosant (the ClariVein System). Reports suggested that this system is safe and effective, eliminating the need for tumescent anesthesia with no reported case of nerve injury. Finally the VenaSeal Closure System comprises the endovenous delivery of cyanoacrylate tissue adhesive to the vein causing fibrosis. Peri-operative discomfort seems to be minimal but the complication of thrombophlebitis has been reported in up to 15 % of patients. The authors concluded that non-thermal options promise comparable treatment efficacy without the added morbidity associated with high thermal energies. They stated that the potential of treating venous reflux without the risk of nerve damage may change how surgeons approach venous disease.

On February 20, 2015, the FDA granted pre-market approval of the VenaSeal Closure System to treat superficial varicosities of the legs through endovascular embolization and is intended for adults with clinically symptomatic venous reflux diagnosed by duplex ultrasound. The FDA approval was based on a multi-center, RCT by Morrison et al (2015)

Morrison and colleagues (2015) noted that preliminary evidence suggests that CAPE may be effective in the treatment of incompetent GSVs. These investigators reported early results of a RCT of CAPE versus RFA for the treatment of symptomatic incompetent GSVs. A total of 222 subjects with symptomatic GSV incompetence were randomly assigned to receive either CAPE (n = 108) with the VenaSeal Closure System or RFA (n = 114) with the ClosureFast System. After discharge, subjects returned to the clinic on day 3 and again at months 1 and 3. The study's primary end-point was closure of the target vein at month 3 as assessed by duplex ultrasound and adjudicated by an independent vascular ultrasound core laboratory. Statistical testing focused on showing non-inferiority with a 10 % delta conditionally followed by superiority testing.

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No adjunctive procedures were allowed until after the month 3 visit, and missing month 3 data were imputed by various methods. Secondary end-points included patient-reported pain during vein treatment and extent of ecchymosis at day 3. Additional assessments included general and disease-specific quality of life surveys and adverse event rates. All subjects received the assigned intervention. By use of the predictive method for imputing missing data, 3-month closure rates were 99 % for cyanoacrylate embolization (CAE) and 96 % for RFA. All primary end-point analyses, which used various methods to account for the missing data rate (14 %), showed evidence to support the study's non-inferiority hypothesis (all $p < 0.01$); some of these analyses supported a trend toward superiority ($p = 0.07$ in the predictive model). Pain experienced during the procedure was mild and similar between treatment groups (2.2 and 2.4 for CAPE and RFA, respectively, on a 10-point scale; $p = 0.11$). At day 3, less ecchymosis in the treated region was present after CAPE compared with RFA ($p < 0.01$). Other adverse events occurred at a similar rate between groups and were generally mild and well-tolerated. The authors concluded that CAPE was proven to be non-inferior to RFA for the treatment of incompetent GSVs at month 3 after the procedure. Both treatment methods showed good safety profiles; CAPE does not require tumescent anesthesia and is associated with less post-procedure ecchymosis. While these findings supported non-inferiority, the reliability of this approach is unclear. These early results need to be validated by well-designed studies with lower rates of data loss and longer follow-up.

Proebstle et al (2015) noted that cyanoacrylate (CA) embolization of refluxing GSVs has been previously described. The outcomes from a multi-center study are still lacking. A prospective, multi-center study was conducted in 7 centers in 4 European countries to abolish GSV reflux by endovenous CA embolization. Neither tumescent anesthesia (Ta) nor post-interventional compression stockings were used. Varicose tributaries remained untreated until at least 3 months after the index treatment. Clinical examination, quality of life assessment, and duplex US evaluation were performed at 2 days and after 1, 3, 6, and 12 months. In 70 patients, of whom 68 (97.1%) were available for 12-month follow-up, 70 GSVs were treated. Two-day follow-up showed 1 proximal and 1 distal partial re-canalization; 3 additional proximal re-canalizations were observed at 3-month ($n = 2$) and 6-month ($n = 1$) follow-up. Cumulative 12-month survival free from re-canalization was 92.9 % (95 % CI: 87.0 % to 99.1 %). Mean (standard deviation) Venous Clinical Severity Score improved from 4.3 ± 2.3 at baseline to 1.1 ± 1.3 at 12 months. Aberdeen Varicose Vein Questionnaire score showed an improvement from 16.3 at baseline to 6.7 at 12 months ($p < 0.0001$). Side effects were generally mild; a phlebitic reaction occurred in 8 cases (11.4 %) with a median duration of 6.5 days (range of 2 to 12 days). Pain without a phlebitic reaction was observed in 5 patients (8.6 %) for a median duration of 1 day (range of 0 to 12 days). No serious adverse events (AEs) occurred; and paresthesia was not observed. The authors concluded that endovenous CA embolization of refluxing GSVs was safe and effective without the use of TA or compression stockings. Moreover, they stated that further work is needed to compare CA against endothermal ablation in RCTs.

Kolluri et al (2016) noted that CAC was shown in the recently published VenaSeal Sapheon Closure System versus Radiofrequency Ablation for Incompetent Great Saphenous Veins (VeClose) randomized clinical study to be an effective and non-inferior option (in terms of both safety and effectiveness) to RFA in the treatment of incompetent GSVs. These investigators reported the safety and effectiveness outcomes of the VeClose roll-in (training) group treated with CAC by physicians who had received device use training, but had no prior treatment experience with the technique and compared the outcomes with those from the randomized RFA and CAC groups. The first 2 subjects at each participating site ($n = 20$) were roll-in cases (i.e., not randomized but instead treated with CAC) to ensure the physician's familiarity with the procedure. Subsequent eligible subjects were randomized to either CAC or RFA. After treatment, all subjects returned for assessment on day 3, and months 1, 3, 6, and 12. The study's primary endpoint was complete closure of the target vein at month 3. Secondary analyses included patient-reported intra-procedural pain and investigator-rated ecchymosis at day 3. Additional assessments included procedure time, QOL surveys, and AEs. Mean procedure time was longer in the roll-in group (31 mins) compared with the randomized groups (24 mins for CAC and 19 mins for RFA; $p < 0.0001$). Procedure time decreased in both the CAC and RFA groups over time, plateauing at about 22 mins for CAC and 15 mins for RFA. In the roll-in group, the 3-month complete occlusion rate was 100 %. Reported intra-procedural pain ratings were similar between the roll-in and randomized groups. Other clinical assessments, including QOL improvement and AEs, were similar between the roll-in and randomized groups. Increasing procedure number did not affect clinical outcomes. The authors concluded that these findings suggested that in the hands of experienced endovenous

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physicians, this new technique was straight-forward, rapidly learned, and reliably produced good safety and effectiveness results.

The authors stated that this study had 2 main drawbacks. First, the number of subjects (n = 20) in the roll-in group was relatively small, which limited power to detect important differences in safety and effectiveness compared with the larger number of randomized subjects. Second, the number of sites that performed more than 20 CAC cases was small (n = 2), limiting the ability to precisely define the learning curve. However, the data from the roll-in subjects provided reasonably strong evidence that the VenaSeal Closure System, in the hands of experienced endovenous physicians (after undergoing 2 proctored roll-in cases), produced good safety and effectiveness results despite initial unfamiliarity with this new technology. Study outcomes were unrelated to procedure number, and these investigators observed a plateauing of procedure time after the 10th procedure. At this time, the FDA has required the company to offer this product only to physicians who are already trained in endothermal techniques, and to proctor the first 2 cases. Based on this study's results, this training program appeared reasonable.

Lam and colleagues (2017) stated that the treatment of incompetent truncal veins has been innovated by the introduction of minimally invasive non-thermal non-tumescent (NTNT) techniques. One of these consists of the use of cyanoacrylate glue to occlude the vein lumen by means of the VenaSeal device. These investigators evaluated NTNT ablation of incompetent saphenous trunks using the VenaSeal device. They concluded that cyanoacrylate adhesive embolization of incompetent truncal veins using the VenaSeal device is a safe and effective innovative technique. Moreover, they stated that further studies are needed to evaluate anatomical and clinical outcomes at long-term.

Morrison et al (2017) noted that endovenous CA closure (CAC) is a new FDA-approved therapy for treatment of clinically symptomatic venous reflux in saphenous veins. The device is indicated for the permanent closure of lower extremity superficial truncal veins, such as the GSV. Early results from a randomized trial of CAC have been reported previously. These investigators reported 1-year outcomes. There were 222 subjects with symptomatic GSV incompetence randomly assigned to receive either CAC (n = 108) or RFA (n = 114). After the month 3 visit, subjects could receive adjunctive therapies aimed at treating visible varicosities and incompetent tributaries. Vein closure was assessed at day 3 and months 1, 3, 6, and 12 using duplex US. Additional study visit assessments included the Venous Clinical Severity Score; Clinical, Etiology, Anatomy, and Pathophysiology classification; EuroQol-5 Dimension; and Aberdeen Varicose Vein Questionnaire. Both time to closure and time to first re-opening of the target vein were evaluated using survival curve analysis; AEs were evaluated at each visit. Of 222 enrolled and randomized subjects, a 12-month follow-up was obtained for 192 (95 CAC and 97 RFA; total follow-up rate, 192/222 [86.5 %]). By month 1, 100 % of CAC subjects and 87 % of RFA subjects demonstrated complete occlusion of the target vein. By month 12, the complete occlusion rate was nearly identical in both groups (97.2 % in the CAC group and 97.0 % in the RFA group); 12-month freedom from re-canalization was similar in the CAC and RFA groups, although there was a trend toward greater freedom from re-canalization in the CAC group (p = 0.08). Symptoms and quality of life improved equally in both groups. Most AEs were mild-to-moderate and not related to the device or procedure. The authors concluded that in patients with incompetent GSVs, treatment with both CAC and RFA resulted in high occlusion rates. Time to complete occlusion was faster with CAC, and freedom from re-opening was higher after CAC; quality of life scores improved equally with both therapies.

This study had several drawbacks:

I. this trial included a modest drop-out rate, with month 12 data unavailable for 13 of 108 (12.0 %) subjects in the CAC group (9 withdrawn and 4 visits not done) and 19 of 114 (16.6 %) subjects in the RFA group (8 withdrawn and 11 visits not done),
II. blinding, although potentially advantageous, was not feasible because RFA requires TA administration, and CAC has characteristic findings on ultrasound. However, the primary study outcome (anatomic closure) was easily judged with ultrasound and is objective,
III. ultrasound interpretations performed by study investigators could have introduced bias; however, the core laboratory had no knowledge of the sites' findings at the time of the readings, and their findings

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agreed with those of the investigators (there was 100 % agreement between investigator reads and core laboratory reads; k statistic was 1.0), and

IV. to minimize confounding due to non-device-related post-intervention factors, subjects in both groups were asked to use compression stockings after the index procedure for 7 days.

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This was done solely for the trial, but it was not done for 2 prior studies of CAC. Whether compression stockings improve complete occlusion rates could be the subject of further study.

In a prospective, single-arm, single-center, feasibility study, Almeida et al (2017) evaluated the long-term safety and effectiveness of endovenous cyanoacrylate (CA)-based closure of incompetent GSV. This trial was conducted at the Canela Clinic (La Romana, Dominican Republic) to assess the effectiveness and safety of a CA-based adhesive for GSV closure at 36 months after treatment. A total of 38 subjects were treated by injection of small boluses of CA under US guidance and without the use of peri-venous TA or post-procedure graduated compression stockings. Periodic scheduled follow-up was performed during 36 months. At month 36, there were 29 subjects who were available for follow-up. Complete occlusion of the treated veins was confirmed by duplex US in all subjects with the exception of 2 subjects showing re-canalization at month 1 and month 3. Kaplan-Meier analysis revealed an occlusion rate at month 36 of 94.7 % (95 % CI: 87.9 % to 100 %). The mean Venous Clinical Severity Score (VCSS) improved from 6.1 ± 2.7 at baseline to 2.2 ± 0.4 at month 36 ($p < 0.0001$). Pain, edema, and varicosities (VCSS subdomains) improved in 75.9 %, 62.1 %, and 41.4 % of subjects, respectively, at month 36. Overall AEs were mild or moderate and self-limited. The authors concluded that CA adhesive appeared to be an effective and safe treatment for saphenous vein closure, with long-term occlusion rates comparable to those of other thermal and non-thermal methods and with no reported serious AEs. This was a small study (n = 38) with a rather high drop-out rate (23.7 %; 9 out of 38).

Gibson and Ferris (2017) noted that CA closure of the GSV with the VenaSeal Closure System is a relatively new modality. Studies have been limited to moderate-sized GSV and some have mandated post-operative compression stockings. These investigators reported the results of a prospective study of CA closure for the treatment of GSV, SSV, and/or accessory saphenous veins (ASV) up to 20 mm in diameter. A total of 50 subjects with symptomatic GSV, SSV, and/or ASV incompetence were each treated at a single session. Compression stockings were not used post-procedure. Subjects returned to clinic at week 1 and again at 1 month. Post-procedure evaluations were performed at 7 days and 1 month and included numerical pain rating score, revised venous clinical severity score, the Aberdeen Varicose Vein Questionnaire score, and time to return to work and normal activities. Duplex US was performed at each visit. Procedural pain was mild (numerical pain rating scale 2.2 ± 1.8). All treated veins (48 GSV vein, 14 ASV, and 8 SSV) had complete closure by duplex US at 7 days and 1 month. Mean time to return to work and normal activities was 0.2 ± 1.1 and 2.4 ± 4.1 days, respectively. The revised venous clinical severity score was improved to 1.8 ± 1.4 ($p < 0.001$) and Aberdeen Varicose Vein Questionnaire score to 8.9 ± 6.6 ($p < 0.001$) at 1 month. Phlebitis in the treatment area or side branches occurred in 10 subjects (20 %) and completely resolved in all but 1 subject (2 %) by 1 month; 98 % of subjects were "completely" or "somewhat" satisfied, and 2 % "unsatisfied" with the procedure at 1 month, despite the protocol disallowance of concomitant side branch treatment. The authors concluded that CA closure was safe and effective for the treatment of 1 or more incompetent saphenous or accessory saphenous veins. Closure rates were high even in the absence of the use of compression stockings

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or side branch treatment. Time back to work or normal activities was short and improvements in venous severity scores and QOL were significant, comparing favorably with alternative treatment methods.

The drawbacks of this study included:

I. its single-arm design,
II. relatively small sample size (n = 50) at a single center,
III. some end-points may be biased positively or negatively by the absence of a concurrent comparator group, and both the patients and physicians were aware that CA closure is a relatively novel procedure, and
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III. some end-points may be biased positively or negatively by the absence of a concurrent comparator group, and both the patients and physicians were aware that CA closure is a relatively novel procedure, and
IV. the short-term follow-up (1 month).

Zierau (2018) reported about 6-years results of a retrospective comparative study of the VenaSeal-Closure System in the treatment of 1,950 saphenous veins (1,349 GSV, 517 SSV, VSAL in 53 cases, VSAM in 30 and Giacomini vein in 2 cases); treatment also included 11 cases of ulcera crures. The present research paper shed light on the advantages and disadvantages and presented the 72-month results of a single-center ambulatory clinical study with retrospective design. The author stated that in light of 17 years of experience, they recommended that every vein center that applies endo-venous treatment should have at least 2 alternative treatment procedures at its disposal. For them, this meant that in practical work with VenaSeal, all insufficient saphenous veins should as far as possible always be treated in 1 session. Independently of this, these investigators at Saphenion meanwhile regarded the VenaSeal-Closure System as the first choice in the range of catheter-supported therapeutic procedures for GSV and SSV or VSAA-varicosis.

In a retrospective, single-center, comparative study, Zierau (2019) reported the 5 year-results with the VenaSea- Closure System in 2,085 veins (1,515 GSV, 570 SSV) versus RF-induced thermo-therapy (RFITT) ablation in 282 veins (181 GSV, 101 SSV). This study compared the advantages and disadvantages of both methods and presented the 5-year findings of the author's single-center study. The author recommended sealing veins in typical truncal varicose veins of GSV, SSV or VSAA (vena safena accessoria anteriore). This was a retrospective, single-center study; and it should be noted that the author is the founder and CEO of Saphenion.

In a prospective, single-center study, Tang et al (2019) examined the effectiveness and patient experience of the VenaSeal Closure System, which uses cyanoacrylate glue to occlude the refluxing truncal superficial veins to treat varicose veins and chronic venous insufficiency. A total of 77 patients (93 legs; 103 procedures) underwent VenaSeal Closure System ablation – 49 (63.6 %) for GSV incompetence, 16 (20.8 %) bilateral GSV, 2 (2.6 %) SSV, and 10 (13.0 %) combined unilateral GSV and SSV/anterior thigh vein reflux. Furthermore, 65/93 legs (69.9 %) had C4 to C6 disease. Patients were reviewed at 2 weeks, 3, 6 and 12 months post-procedure. There was 100 % technical success; 28/77 (36.4 %) underwent concomitant phlebectomies. All procedures were well-tolerated with a mean post-operative pain score of 3.0 (range of 0 to 5). After 3 months, median patient satisfaction was 9.0 (IQR: 7.0 to 10.0). At 2-week follow-up, the GSV was completely occluded in 88/88 (100 %) veins and SSV completely closed in 11/11 (100 %) veins. At 3-month follow-up, the GSV was occluded in 51/53 (96.2 %) veins and SSV completely closed in 5/5 (100 %) veins. At 6-month follow-up, the GSV was completely occluded in 42/45 (93.3 %) veins and SSV completely closed in 5/7 (71.4 %) veins. At 1 year, GSV and SSV occlusion rates were 54/59 (91.5 %) and 5/8 (62.5 %), respectively; and there was 1 DVT. Transient superficial phlebitis was reported in 10/93 (10.8 %) legs, which were all self-limiting. There were 9/103 (8.7 %) anatomical recurrences, but no patients required re-intervention as they were asymptomatic. The authors concluded that cyanoacrylate glue was a safe and effective modality to ablate refluxing saphenous veins in Asian patients in the short-term. There

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was a high satisfaction rate and peri-procedural pain was low. These researchers stated that early results were promising; however, further evaluation and longer term follow-up are needed.

Almeida et al (2020) noted that VenaSeal is a cyanoacrylate polymer adhesive for the treatment of patients with CVI. As an implanted device, questions remain regarding how long cyanoacrylate persists after closure. In this report, a 65-year old man was examined 5.5 years after CAC, and a segment of GSV was excised for histopathologic analysis. Findings were typical of a foreign body reaction. The vessel was occluded with collagenized mature fibrous tissue and polymer remnants, which were encapsulated by multi-nucleated giant cells. Focal areas of granulomatous inflammation were present in the vein wall extending to the adventitia.

Kolluri and colleagues (2020) noted that several RCTs have compared different interventions for CVI management, but mixed comparison of these interventions is lacking. In a meta-analysis, these researchers compared the VenaSeal Closure System with EVLA, RFA, MOCA, sclerotherapy, and surgery for management of CVI to achieve anatomic success (complete closure of treated vein within 6 months after intervention) as the primary outcome and health-related QOL (HRQoL; EuroQol-5 Dimension, Aberdeen Varicose Vein Questionnaire), VCSS, pain scores, and AEs as secondary outcomes. They carried out a systematic review of journal data-bases, and RCTs between January 1996 and September 2018 comparing different therapeutic options. Risk of bias and quality of publications were evaluated using the Cochrane bias tool; Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines were used for study selection and reporting. A total of 20 RCTs comprising 4,570 patients were analyzed. Data for anatomic success, VCSS, HRQoL, pain score, and AEs were extracted and analyzed using mixed treatment comparison in a network meta-analysis. A Bayesian fixed or random effects model was selected for analysis. Rank probability graphs were generated for various treatments and corresponding ranks obtained to estimate their probability of being best. Relative treatment effects were calculated in terms of log ORs for anatomic success and AEs; mean difference (MD) was calculated for VCSS, HRQoL, and pain score. For the primary outcome measure (anatomic success), the VenaSeal Closure System had the highest probability of being ranked 1st ($p = 0.980$); RFA was ranked 2nd ($p = 0.365$), EVLA 3rd ($p = 0.397$), surgery 4th ($p = 0.290$), MOCA 5th ($p = 0.695$), and sclerotherapy 6th ($p = 0.982$). For secondary outcome measures, the VenaSeal Closure System ranked 3rd for VCSS ($p = 0.332$), 5th for EuroQol-5 Dimension ($p = 0.420$), and 3rd for Aberdeen Varicose Vein Questionnaire ($p = 0.300$). Although, the VenaSeal Closure System was slightly inferior to some of the other interventions for HRQoL, the 95 % CI of log OR indicated insufficient evidence for any concrete conclusion to be drawn. The VenaSeal Closure System ranked 1st in reduction of post-operative pain score from baseline ($p = 0.690$) and was lowest in occurrence of AEs ($p = 0.650$). Odds of occurrence of AEs was 3.3 times in the sclerotherapy arm, 2.7 times in the EVLA arm, 1.6 times with surgery, and 1.1 times with RFA versus the VenaSeal Closure System arm. The authors concluded that the VenaSeal Closure System is a promising therapeutic option for anatomic success at 6 months, with fewer occurrences of AEs (wound and groin infection, pulmonary embolism) in CVI patients compared with other interventions in this study. Moreover, these researchers stated that additional economic analysis including cost-effectiveness analysis would provide interesting perspectives on real-world insights to patients, payers, and providers.

The authors stated that drawbacks of this analysis included restricted data availability in terms of time-points and pooling of data. Therefore, the analysis could be performed only at the available time-points. Another drawback of this analysis was that it compared the methods for the first 6 months after treatment, which was a relatively short period. Results were reported in different manners in the included studies; hence, fewer data-points could not be pooled and used in this analysis. Neither of the 2 included MOCA RCTs (as allowed by this study's methodology) reported all outcomes that were analyzed in this network meta-analysis.

Morrison et al (2020) noted that the proprietary CAC System versus RFA Trial (VenaSeal Sapheon Closure System Pivotal Study [VeClose]) showed that CAC was effective and non-inferior to RFA in vein closure with good 36-month outcomes. Conducted under a separate protocol, the aim of this extended follow-up was to evaluate the long-term safety and effectiveness of CAC and RFA for the treatment of incompetent GSV at 5 years (60 months) of follow-up. This 60-month extension study was conducted for all patients who completed the randomized VeClose study and who were willing to participate. The pivotal VeClose study included patients with symptomatic moderate-to-severe varicosities (Clinical, Etiology, Anatomy, and

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Pathophysiology [CEAP] class C2 to C4b) and symptomatic GSV incompetence who were randomly assigned (1:1) to either CAC or RFA. The primary outcome measure of this 60-month extension study was complete closure of the target vein, with planned exploratory analysis of non-inferiority. Secondary outcomes included CEAP class; completion of the Venous Clinical Severity Score, EuroQoL-Five Dimension survey, and Aberdeen Varicose Vein Questionnaire; patient satisfaction with treatment; AEs related to target GSV; and details of adjunctive procedures. A total of 89 patients completed the 60-month visit, which included 47 from the CAC group, 33 from the RFA group, and 9 CAC roll-in patients. No new re-canalization events have been observed in the groups between 36 and 60 months of follow-up. At 60 months, Kaplan-Meier estimates for freedom from re-canalization in the randomized CAC and RFA groups were 91.4 % and 85.2 %, respectively, demonstrating non-inferiority of CAC compared with RFA. Both groups demonstrated sustained improvements in EuroQoL-Five Dimension and QOL measures through 60 months. Whereas patients assigned to C0 or C1 clinical class were excluded from the original study, more than 50 % of all returning patients (64 % [57/89]) were now assigned to C0 or C1, suggesting an improved clinical class from baseline. Furthermore, 41.1 % of returning CAC patients and 39.4% of returning RFA patients were presently at least 2 CEAP clinical classes lower than at baseline. No long-term device- or procedure-related serious AEs occurred in either group between 36- and 60-month follow-up. The authors concluded that CAC and RFA were effective in achieving complete target vein closure of the GSV at long-term follow-up, with CAC demonstrating continued non-inferiority to RFA. These investigators stated that the CAC System was also associated with sustained improvements in symptoms and QOL, lower CEAP class, and high level of patient satisfaction without serious AEs between 36 and 60 months.

The authors stated that this study had several drawbacks. One drawback of the original investigational device exemption (IDE) study was that blinding of physicians, patients, and outcome assessors was not possible owing to the nature of the interventions because RFA requires tumescent anesthesia (TA) and CAC has characteristic findings on US. The extension study was performed under a separate protocol and included only patients from the IDE study who were able to be contacted and agreed to participate. The original consent for the IDE study did not include this 60-month visit, so the patients had no expectation of longer follow-up and may have had less motivation to participate (especially if they were doing well). Furthermore, there was a 2-year gap between the 3-year and 5-year follow-up, making it that much more difficult to locate patients. Additionally, it is widely recognized that recruiting patients for a follow-up visit, especially at 5 years, is challenging. There may be various reasons for patients not attending follow-up data collection -- patients' lack of interest, patients' expectations for the outcome of interest were met, or patients' having satisfactory outcomes and hence losing interest in the trial. Follow-up may also be lost if participants changed their names, addresses, and phone numbers or because of some unforeseen personal circumstances that may prevent them from completing the trial. Previous research has demonstrated that losses to follow-up were higher when no treatment is needed after surgery, especially with a longer follow-up period when no specific treatment is needed. In patients lost to follow-up, identifying the next of kin or other important contacts was a challenge. Despite the challenges that contributed to a smaller sample size, non-inferiority of CAC compared with RFA was still demonstrated (non-inferiority established by 1-sided 97.5 % CI with lower limit of -3.5 %). The patient satisfaction reported in this study was subjective. Even though there are many standardized and validated patient satisfaction instruments to assess specific aspects of care, these do not apply for the entire spectrum of care as they have limited potential of validity and reliability. As a result, the method of measure of patient satisfaction in this study has not been validated.

In an invited commentary on the afore-mentioned study by Morrison et al (2020), Gohel (2020) stated that "The ultimate role of cyanoacrylate closure in the management of superficial reflux will depend on many factors. Clearly, cost and reimbursement considerations will be powerful drivers for change either in favor of or against novel interventions. Venous specialists now have many, highly effective superficial venous interventions at their disposal. Just as different endovenous modalities will have different advantages and disadvantages, patients with venous disease are unique, and the use of multiple venous modalities on the same patient is increasingly common. We are entering an era where clinicians need to be skilled in a range of superficial venous interventions and have the ability to select and adapt the intervention to the specific needs of each individual patient. Do nonthermal, nontumescent modalities such as cyanoacrylate glue closure have a place in the multimodal venous toolkit of the future Almost certainly".

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Chan et al (2020) noted that patients with venous leg ulcers (VLUs) represent the worse spectrum of CVI. The Early Venous Reflux Ablation (EVRA) landmark trial published in 2018 demonstrated that early endo-venous intervention resulted in faster healing of VLUs. These researchers described their post-EVRA experience using endo-venous cyanoacrylate glue ablation (ECGA) to treat superficial venous reflux on an early basis and examined its safety and efficacy in the setting of VLUs. There were 37 patients (39 legs, 43 truncal veins) with 43 discrete venous ulcers who underwent ECGA for CVI symptoms and VLUs. They received compression therapy and regular dressings for the VLUs post-operatively and were reviewed at 1 week, 3 months, 6 months, and 12 months after the procedure. Post-operative healing time for VLUs and complications were recorded along with the patient's satisfaction and post-procedure pain scores. The venous ulcers were all less than 30 cm2 before ECGA. The mean time for VLU healing from operation was 73.6 ± 21.9 days, and the primary occlusion rate of the CVI at both 1 week and 3 months was 100 %. No major AEs were observed except for 1 case of DVT. There was significant improvement in the revised VCSS post-operatively from 11 ± 1.63 (baseline) to 5.6 ± 1.37 ($p < 0.001$) at 3-month follow-up (on a scale of 0 to 27, with the severity of symptoms at a maximal 27). The VAS scores for pain were low post-operatively, decreasing from a pre-operative score of 6.84 ± 1.42 to 2.72 ± 1.59 ($p < 0.001$) at the 3-month follow-up (on a scale of 1 to 10, with 10 being the most severe pain). The median time to return to normal activities was 7 days (IQR, 5 to 7 days). The authors concluded that ECGA together with compression therapy for VLUs was both safe and effective in this population of Asian patients. ECGA for patients with VLUs had excellent patient acceptability, minimal morbidity, and low re-canalization rates at 12 months. Moreover, these researchers stated that larger studies with longer follow-up periods are needed to validate the preliminary outcomes of this study, and if it is proven to significantly improve ulcer healing rates, this will change the way clinicians approach chronic venous ulceration.

Masuda et al (2020) stated that stimulated by published reports of potentially inappropriate application of venous procedures, the AVF and its Ethics Task Force in collaboration with multiple other professional societies including the SVS, American Vein and Lymphatic Society (AVLS), and the Society of Interventional Radiology (SIR) developed the appropriate use criteria (AUC) for chronic lower extremity (LE) venous disease to provide clarity to the use of venous procedures, DUS imaging, timing, and reimbursements. The AUC were developed using the RAND/UCLA Appropriateness Method, a validated method of developing appropriateness criteria in healthcare. By carrying out a modified Delphi exercise and incorporating best available evidence and expert opinion, AUC were developed and scored. There were 119 scenarios rated on a scale of 1 to 9 by an expert panel, with 1 being never appropriate and 9 being appropriate. The majority of scenarios consisted of symptomatic indications were deemed appropriate for venous intervention. For scenarios with anatomically short segments of reflux and/or no symptoms, the indications were rated less appropriate. For the indication of edema, a wide dispersion of ratings was observed especially for short segments of saphenous reflux or stenting for iliac/inferior vena cava disease, noting that there are multi-factorial causes of edema, some of which could co-exist with venous disease and possibly impact effectiveness of treatment. Several scenarios were considered never appropriate, including treatment of saphenous veins with no reflux, iliac vein or inferior vena cava stenting for iliac vein compression as an incidental finding by imaging with minimal or no symptoms or signs, and incentivizing sonographers to find reflux. The authors concluded that the AUC statements were intended to serve as a guide to patient care, especially in areas where high-quality evidence is lacking to aid clinicians in making day-to-day decisions for common venous interventions. This may also prove useful when applied on a population level, such as practice patterns, and not necessarily to dictate decision-making for individual cases. As a product of a collaborative effort, it is hoped that this could be used by physicians and multiple stake-holders committed toward improving patient care and to identify and stimulate future research priorities.

Cho et al (2020) noted that several modalities are used for the treatment of varicose veins. Open surgical treatment with ligation and stripping of the saphenous vein has been the standard of care for many years. Endo-venous thermal ablation has been shown to be a safe and effective alternative with high, long-term, target-vein closure rates. Despite this, there is the possibility of thermal injury to surrounding structures. The recently introduced CAC is also considered to be a good alternative and the risk of injury to surrounding structures is minimal. In a prospective, multi-center, open-label RCT, these researchers attempt to demonstrate the non-inferiority of CAC with the VenaSeal Closure System compared to surgical stripping in terms of clinical outcomes for the treatment of incompetent GSV. After baseline measurements, subjects will

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be randomly allocated into either the CAC group or the surgical-stripping group. The primary end-point of the study is the complete closure rate of the target vein in the CAC group, and the absence of venous reflux or residual venous tissue after surgical stripping in the surgical-stripping group. These end-points will be measured by Doppler US carried out by qualified vascular technologists or investigators at 3 months after treatment. Secondary outcomes will include peri-operative pain, post-operative ecchymosis, clinical assessment (including general and disease-specific QOL evaluations), complete closure rate, and absence of venous reflux or residual venous tissue at the 12- and 24-month follow-ups, as well as all AE rates during the 24-month follow-up period. The authors concluded that this multi-center RCT is designed to show non-inferiority in terms of complete closure rate of cyanoacrylate compared to surgical stripping for the treatment of incompetent saphenous veins. The approximate date of completion of recruitment would be February 29, 2020.

Dimech and Cassar (2020) noted that 1/3 of adults in the U.S. and U.K. suffer from varicose veins; NBCA glue is a novel endovascular, non-tumescent, non-thermal ablation technique for the treatment of this condition. In a systematic review, these investigators examined the efficacy of NBCA in ablating primary truncal varicose veins and eliminating reflux compared with existing endovascular techniques. Secondary outcomes include complications and QOL. PRISMA was used as a guide and studies were screened for risk of bias and methodological quality. Subjects had to be greater than or equal to 18 years of age and followed-up post-treatment with color Duplex US (DUS). Eligibility criteria included SFJ or SPJ incompetence with reflux down truncal veins lasting greater than 0.5 seconds on DUS interrogation and a Clinical, Etiological, Anatomical, and Pathophysiological classification of venous disorders ranging between C1 and C6. Out of 2,910 patients (3,220 veins) in 17 studies, 1,981 were administered NBCA, 445 RFA, and 484 EVLA with mean procedure times of 25.7, 23.2, and 28.7 mins, respectively. Mean recruitment period was 9 months (1 to 36 months) and followed-up for an average of 12.3 months (1 to 36 months). The majority were C2 to C3; 2-year occlusion rates were 93.7, 90.9, and 91.5 % for NBCA, RFA, and EVLA, respectively. NBCA-treated patients experienced the least complications, with bruising, phlebitis, and pain being the most prevalent; QOL improved equally in all 3 modalities. The authors concluded that NBCA is simple to administer, safe, and effective even without compression stockings. Moreover, these researchers stated that further studies are needed to examine longer-term benefit and the effect of anti-coagulation on vein obliteration.

The authors stated that this systematic review had several drawbacks. The comprehensive literature search and data extraction were performed by 1 author. It excluded MOCA and the period of follow-up was short. A meta-analysis would have been ideal but as highlighted in a recent article, the scarcity and heterogeneity of RCTs made this difficult. As most patients were not sedated, double-blinding was impossible. Outcome assessors were often the same ones recruiting, performing the treatments and/or following-up patients. This was taken into consideration in part by modifying the Cochrane risk of bias tool. Some methodologies opted for an induction period to cater for the “learning curve” but others did not. One major inconsistency was in the duration of procedures. There were no set standards as to when time-keeping should start and stop. The lack of reproducibility made these measurements unreliable. In terms of patient characteristics, 1 study included more smokers in the NBCA group and another deviated its protocol to include a patient with higher body mass index (BMI). No differentiation between unilateral or bilateral treatment of varicose veins was made. “Return to normal activities” needs better definition, as these activities were different in an elderly or morbidly obese patient compared with a healthy fit subject. Reflux was best detected in the standing posture on DUS as recommended by the European Society for Vascular Surgery, but some measured this supine. Lastly, it would be interesting to perform a clinical trial addressing the use of NBCA for varicose veins in anti-coagulated patients.

Furthermore, National Institute of Health and Care Excellence’s interventional procedures guidance on “Cyanoacrylate glue occlusion for varicose veins” (NICE, 2020) provides the following recommendations:

- Evidence on the safety and efficacy of cyanoacrylate glue occlusion for varicose veins is adequate to support the use of this procedure provided that standard arrangements are in place for clinical governance, consent and audit.
- The procedure should only be done by clinicians with appropriate training in this procedure and experience in the use of venous ultrasound.

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In addition, the consensus document on "Cyanoacrylate closure for peripheral veins" by the Australasian College of Phlebology (Parsi et al, 2020) concluded that "Cyanoacrylate adhesive closure appears to be an effective endovenous procedure, with short-term closure rates comparable to ETA and therefore greater efficacy than traditional surgery for treating superficial veins of the lower limbs. Ongoing data collection is required to establish the long-term safety".

Wilczko et al (2021) stated that the current treatment of venous disease is focused on the minimally invasive exclusion of the affected vein. Besides widely used thermal ablation, chemical ablation with cyanoacrylate, reported as safe and highly effective, has been gaining increasing interest. These investigators presented data from a 2-year observation in 89 patients (61 women/28 men, mean age of 44.3 ± 13.5 years) suffering from venous insufficiency (C2 to C4), treated either using short-chain cyanoacrylate, the VenaBlock system (n = 43) or laser thermos-ablation with ELVeS 1470 (n = 46). The assessment comprised the occurrence of venous disease-related symptoms and the US examination of the leg venous system. The frequency of re-canalization after 2 years from the VenaBlock procedure was significantly higher than after laser treatment (37.2 % versus 8.7 %). Apart from re-canalization, in some patients from both groups, the symptoms of recurrence and/or disease progression, including the development of insufficiency in other veins of treated or contralateral legs (9.3 % versus 15.2 % and 9.3 % versus 17.4 %, respectively), were observed. Unexpectedly, the general prevalence of the disease progression did not differ significantly between the VenaBlock and ELVeS groups (44.2 % versus 34.8 %, respectively). The authors concluded that despite the higher re-canalization rate of VenaBlock compared to ELVeS, the overall effectiveness of cyanoacrylate and laser thermo-ablation after 2 years was similar; thus, both methods similarly failed to prevent recurrence and disease progression, which appeared to be method-independent.

These researchers stated that the findings of this study were consistent with observations from several previous studies with mid-term follow-up. In addition, a significantly higher rate of re-canalization and recurrence has been demonstrated in long-term studies regarding varicose vein treatment. The possible explanation for that phenomenon could be the under-estimated role of alternative reflux sources in superficial veins, mainly of pelvic origin; thus, major attention should be reserved for the proper identification of incompetent veins, both in the abdomino-pelvic and in the lower limb venous system, before any treatment.

Proebstle et al (2021) noted that cyanoacrylate closure of refluxing saphenous veins has demonstrated safety and effectiveness results in feasibility and pivotal studies. These investigators provided the 36-month follow-up results of a prospective, non-randomized, multi-center, cohort study. A total of 70 patients were enrolled in a prospective, multi-center study conducted at 7 centers in 4 European countries and underwent treatment of a solitary refluxing great saphenous vein with endo-venous cyanoacrylate embolization without the use of tumescent anesthesia or post-procedure compression stockings. The primary effectiveness end point was freedom from re-canalization (closure rate) of the great saphenous vein at 6 months. Safety was assessed by occurrence of adverse events (AEs) after the procedure and during the 6-month follow-up period. Quality of life (QOL) and clinical improvement parameters were measured before and after the procedure and through a 12-month follow-up period. Anatomic success and clinical improvement were assessed through 36 months after the procedure. Of 70 treated patients, 64 (91 %) were available for the 3-year follow-up. The closure rates by Kaplan-Meier life table methods at 6-, 12-, 24-, and 36-month time-points were 91.4 %, 90.0 %, 88.5 %, and 88.5 %, respectively. Through 36 months, the improvement in change of the mean venous clinical severity score over time was statistically significant by dropping from 4.3 at baseline to 0.9 at the 36-month follow-up ($p < 0.001$). The authors concluded that the 3-year follow-up results of the prospective, multi-center eSCOPE study demonstrated the continued anatomic and clinical effectiveness of cyanoacrylate embolization over an extended follow-up period.

The authors stated that the comparatively low closure rate observed in the eSCOPE study may be attributed to the inherent limitations of this study such as small sample size that resulted in a high statistical impact of treatment failure in a single patient on GSV closure rates. Another noteworthy factor would be the fact that this device was being used by clinicians with no prior experience in cyanoacrylate occlusion, who were selected to participate in this study based on their experience in thermal ablation. Although the roll-in phase analysis of the VeClose Trial suggested that the learning curve of the usage of the device was relatively short and its impact on clinical outcomes was negligible, it was essential to remember that the eSCOPE study was

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initiated much earlier than the VeClose trial and almost all clinicians involved in eSCOPE study had no prior contact to cyanoacrylate closure technology.

Kiguchi et al (2021) stated that they have previously demonstrated that VenaSeal adhesive, compared with radiofrequency ablation (RFA), in the treatment of refluxing saphenous veins in Clinical, Etiology, Anatomy, Pathophysiology class 6 (CEAP 6) limbs, led to shorter healing times of venous ulcers. These investigators hypothesized that the longer treated length possible with VenaSeal's non-thermal modality may affect the number of critical refluxing perforators contributing to the non-healing wound. This follow-up study compared the need for follow-up treatment of perforator veins following saphenous vein treatment with either RFA (ClosureFast) or adhesive closure (VenaSeal). This study was a retrospective, multi-center review of CEAP 6 patients who had closure of their saphenous veins from 2015 to 2020. Patients who underwent follow-up treatment of perforator veins were grouped according to their method of initial management of their saphenous veins . The primary endpoint was incidence of a perforator procedure after ClosureFast or VenaSeal ablation. Secondary endpoints included sclerotherapy to facilitate wound healing. Bi-variate analysis used the $\chi 2$ test, Fisher exact test, t-test, and Wilcoxon rank sum test. A p value of < 0.05 defined statistical significance. There were 119 CEAP 6 patients with saphenous closure: 51 limbs treated with VenaSeal and 68 with RFA. Median follow-up was 105 days (inter-quartile range [IQR]: 44 to 208). All limbs achieved wound healing during the study period. Mean time to wound healing post index procedure was shorter for VenaSeal than RFA (72 versus 293.8 days, $p > 0.0009$), as was median time (43 versus 104 days, $p = 0.001$). More limbs treated with RFA had previous known deep vein thrombosis (DVT; 29 % versus 10 %, $p = 0.009$), deep venous insufficiency (82 % versus 51 %, $p = 0.0003$), and perforator reflux (57 % versus 29 %, $p = 0.002$). Limbs with identified follow-up perforator reflux treated with RFA had a higher prevalence of initially treated saphenous veins with RFA compared with those treated with VenaSeal (49 % versus 27 %, $p = 0.003$). There was no difference between the methods of vein closure and use of concurrent sclerotherapy. The authors concluded that ClosureFast and VenaSeal were both safe and effective modalities of saphenous ablation; however, VenaSeal treatment was associated with less perforator reflux than RFA intervention.

In a prospective, multi-racial Singaporean Asian cohort study, Tang et al (2021) examined the performance of CAC for varicose veins. These investigators reported early clinical and patient satisfaction outcomes at 3 months after intervention. There were 100 patients (151 legs; 156 truncal veins) who underwent CAC between April and December 2018. Of the 151 legs, 49 (32.5 %) were treated for GSV incompetence, 96 (63.6 %) for bilateral GSV incompetence, 1 (0.7 %) for SSV incompetence, and 5 (3.3 %) for combined unilateral GSV and SSV reflux. At baseline, 68 legs (45.0 %) had C4 to C6 disease, and 67 legs (44.4 %) had concomitant multiple stab avulsions. Patients were reviewed at 2 and 12 weeks by Duplex US to check for venous re-canalization (defined as greater than or equal to 5 cm in length), pain score, revised VCSS, EuroQoL-5 Dimension questionnaire score, Aberdeen Varicose Vein Questionnaire score, 14-item Chronic Venous Insufficiency Questionnaire (CIVIQ-14) for QOL measures, and patient satisfaction. Time to return to work and normal activities were also recorded. There were 59 women, and mean age was 60.1 ± 12.7 years. There was 100 % technical success. Patients tolerated the procedure well and reported low peri-procedural pain (mean pain score, 2.9 [range of 0 to 9]). Patient surveys at the 3-month interval showed high satisfaction rates, with 72 of 91 (79.1 %) being extremely or very satisfied. By day 10, of 100 patients, 93 (93 %) resumed daily activities, whereas 36 (36 %) had returned to work. At 2 weeks, GSV and SSV were completely occluded in 150 of 150 (100 %) and 6 of 6 (100 %) veins, respectively. At 3 months, GSV and SSV occlusion rates were 140 of 141 (99.3 %) and 6 of 6 (100 %), respectively. Transient superficial phlebitis was reported in 27 of 151 (18 %) legs, which was self-limited. At 3 months, revised VCSS improved from 5.00 (range of 1.00 to 18.00) to 1.00 (0.00 to 10.00; $p < 0.001$); EuroQoL-5 Dimension score, from 0.686 (-0.382 to 1.00) to 1.00 (0.12 to 1.00; $p < 0.001$); Aberdeen Varicose Vein Questionnaire score, from 17.14 (1.29 to 61.15) to 4.83 (0.00 to 57.12; $p < 0.001$); and 14-item Chronic Venous Insufficiency Questionnaire, from 19.64 (1.79 to 73.21) to 7.14 (0.00 to 51.79; $p < 0.001$). The authors concluded that CAC had a high safety profile and was an effective way to block refluxing truncal veins in Asian patients at early follow-up. Patients were highly satisfied and reported low post-operative pain. There was significant continued QOL improvement at 3 months. Moreover, these researchers stated that early results are encouraging, but they await further prospective long-term follow-up from the study.

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Belramman et al (2022) noted that endovenous thermal ablations (ETAs) are recommended as 1st-line treatment for truncal vein reflux, have a short recovery time, and are cost-effective; however, ETAs are associated with discomfort during tumescent anesthesia infiltration. To minimize discomfort, non-thermal, non-tumescent ablation techniques had emerged in the form of MOCA and cyanoacrylate adhesive CAE injection. In a randomized clinical trial, these investigators examined pain scores immediately after truncal ablation using a 100-mm VAS and 10-point number scale to compare pain-related outcomes following MOCA versus CAE treatment. The Multicenter Randomized II Clinical Trial Comparing Mechanochemical Ablation versus Cyanoacrylate Adhesive for the Treatment of Primary Truncal Saphenous Veins Incompetence study was a prospective, multi-center randomized clinical trial carried out at 3 sites between November 2017 and January 2020. Inclusion criteria were primary great or small saphenous varicose veins; exclusion criteria included recurrent varicose veins, current DVT, or serious arterial disease. There were 392 participants screened, 225 were excluded, and 167 participants underwent randomization; 4 participants did not receive allocated intervention and were included in the intention-to-treat (ITT) analysis. Follow-up took place at 2 weeks, and 3, 6, and 12 months. The primary outcome measure was pain score immediately after completing truncal ablation using a 100-mm visual analog scale (VAS) and a 10-point number scale. Secondary outcome measures included degree of ecchymosis, occlusion rates, clinical severity, and generic and disease-specific QOL scores. Of 167 study participants, 99 (59.3 %) were women, and the mean (SD) age was 56 (15.8) years. Overall, 155 truncal veins treated (92.8 %) were GSV. Demographic data and baseline status were comparable between treatment groups. A total of 73 patients (47 %) underwent adjunctive treatment of varicosities. Overall median (IQR) maximum pain score after truncal treatment was 23 mm (10 to 44) on the VAS and 3 (2 to 5) on the number scale, showing no significant difference in median (IQR) pain measured by VAS (MOCA, 24 [11.5 to 44.7] mm versus CAE, 20 [9.0 to 42.0] mm; p = 0.23) or by number scale (MOCA, 4 [2 to 5] versus CAE, 3 [2 to 4]; p = 0.18). Both groups showed significant and comparable improvement in clinical severity, generic and disease-specific QoL scores, and complete occlusion rates. Four patients treated with CAE developed minor complications (superficial thrombophlebitis and thrombus extensions). The authors concluded that to their knowledge, this was the 1st randomized clinical trial directly comparing non-tumescent ablation techniques. The findings of this study showed that the MOCA and CAE techniques exhibited a similar peri-procedural pain score. Moreover, these researchers stated that additional well-designed randomized clinical trials with longer-term follow-up are needed to determine the longer-term effectiveness and cost-effectiveness of using NTNT techniques in treatment of saphenous veins reflux.

The authors stated that this study had several drawbacks. First, the lack of interventional blinding owing to the different device techniques, although this was mitigated by blinded follow-up assessments. Second, because of the COVID-19 pandemic, 70 patients (41 %) were lost to follow-up or dropped-out by 12 months. However, the proportion did not differ between those who dropped-out and those who did not, and this limitation was controlled by performing ITT analysis with the primary outcome recorded on day zero. This would suggest that the confidence in the primary outcome could be strong. Third, although both groups were examined for a pain score 10 days post-operatively, the use of analgesia was not reported.

Farah et al (2022) stated that several diagnostic tests and therapeutic options for patients with LE varicose veins have existed for decades. In a systematic review and meta-analysis, these investigators examined the latest evidence to support the forthcoming updates of the clinical practice guidelines on the management of varicose veins for the SVS, the AVF, and the AVLS. They searched multiple data-bases for studies that addressed 4 clinical questions identified by the AVF and the SVS guideline committee regarding evaluation and treatment of patients with varicose veins. Studies were selected and appraised by pairs of independent reviewers. A meta-analysis was conducted when feasible. These researchers included 73 original studies (45 were RCTs) and 1 systematic review from 12,915 candidate references. Moderate certainty of evidence supported the usefulness of DUS examination as the gold standard test for diagnosing saphenous vein incompetence in patients with varicose veins and CVI (clinical, etiological, anatomic, pathophysiological classification [CEAP] class C2 to C6). High ligation and stripping (HL/S) was associated with higher anatomic closure rates at 30 days and 5 years when compared with RFA and UGFS (moderate certainty), while no significant difference was observed when compared with EVLA at 5 years. UGFS was associated with an increased risk of recurrence compared with HL/S. EVLA was associated with lower anatomic closure rates at 30 days than CAC and higher rates at 1 and 5 years when compared with UGFS. Thermal interventions were associated with lower generic QoL scores and an increased risk of AEs when compared

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with CAC or n-butyl cyanoacrylate (low certainty). Thermal interventions were associated with a lower risk of recurrent incompetence when compared with UGFS and an increased risk of recurrent incompetence than CAC. The evidence for GSV ablation alone to manage perforator disease was inconclusive. The authors concluded that this systematic review examined the available evidence to develop and support forthcoming updated SVS/AVF/AVLS clinical practice guideline recommendations. The evidence supports duplex US scanning for evaluating patients with varicose veins and confirms that HL/S resulted in similar long-term saphenous vein closure rates as EVLA and in better rates than RFA and UGFS. Thermal interventions were associated with inferior generic QoL scores than non-thermal interventions, but had a lower risk of recurrent incompetence than UGFS. The recommendations in the guidelines should consider this information as well as other factors such as patients' values and preferences, anatomic considerations of individual patients, and surgical expertise.

Bahi et al (2023) noted that CVI is a common vascular condition with significant resultant patient morbidity. There has been a shift towards minimally invasive treatment modalities with VenaSeal endovenous ablation among the more recent treatment modalities introduced. Long-term outcome data for this treatment modality is not widely available yet. In a retrospective, single-center study, these researchers reported 6-week patient outcomes over a 5-year period from a high-volume tertiary vascular center. They reported short-term outcomes following VenaSeal endovenous ablation for symptomatic saphenous incompetence. Patients were followed-up at 6-weeks post-procedurally by tele-medicine or in-person clinic appointment without routine venous US assessment. These investigators reported outcomes for 235 patients during this study period. All patients tolerated the procedure under local anesthesia. Average age was 60.5 years (29 to 82 years) with slight male predominance (55.7 %). The majority were New Zealand European (63.8 %); BMI was 28.5 (22.2 to 41.4). These researchers reported a 21 % rate of self-limiting phlebitis, and 33 minor complication events, which included 15 cases of residual varicose veins, 9 saphenous nerve neuropraxia, 6 cases of puncture-site cellulitis, and 3 DVTs. Patient demographics and primary surgeon did not have a statistically significant outcome on development of complications. The authors concluded that that VenaSeal endovenous ablation was a safe and effective method of treatment for symptomatic truncal saphenous vein incompetence. They reported safely managing post-operative phlebitis conservatively and found a mixture of clinical and phone clinic follow-up sufficient without requirement for objective Duplex US following the procedure to ensure objective saphenous vein closure. These investigators stated that the next step will be gathering prospective data to attain more information regarding patient perception of this method of treatment.

The authors stated that this study had several drawbacks. Being a retrospective study, these researchers did not have the advantage of prospective data gathering. They did not routinely conduct patient questionnaires regarding post-operative symptoms in their center, and as such these data would be helpful; however, were unavailable. In addition, being a center where cyanoacrylate endovenous ablation is the current sole method of treatment of symptomatic varicose veins, these researchers did not have readily available data for other methods including EVLT or RFA for comparison. Without objective post-procedural Duplex US, these investigators relied on clinical assessment and self-reporting by patients as a proxy for procedural technical success.

Cutler et al (2023) noted that chronic venous insufficiency is a costly pathology affecting 40 % or less of the U.S. population. Previous studies have highlighted the substantial burden of resource usage and work loss costs associated with this disease, with an estimate of upward of 2 million workdays lost annually in the U.S. alone. Endo-venous closure of refluxing superficial veins has been proven to improve patients' health-related QOL (HR-QOL). Despite the transition from inpatient to outpatient ablation procedures, these minimally invasive techniques could still result in the loss of working days and economic disability. These investigators compared the time to return-to-work after treatment between 2 techniques of superficial vein closure: RFA and cyanoacrylate adhesive closure (CAC). They carried out a retrospective, multi-center review of actively employed patients with CEAP class 2 to 5 disease who had undergone closure of their truncal veins from 2019 to 2020. The patients who had undergone RFA were compared with those who had undergone CAC. Patients with active venous leg ulcers were excluded. The primary endpoint was the time to return-to-work. The secondary endpoints included infection, thrombophlebitis, and post-procedure DVT. Multi-variate linear regression analysis was used to examine the factors affecting the time to return-to-work in the most predictive model. The analysis was carried out to exclude holidays and weekends. Statistical

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significance was defined as $p < 0.05$. A total of 97 patients were included in the study cohort, of whom 58 had undergone RFA and 39, CAC. Most of the patients in the cohort had had CEAP class 2 disease (60 %). The patients in the RFA group were more frequently men (57 % versus 33 %; $p = 0.023$). Significantly more patients who had undergone CAC had had post-procedure thrombophlebitis (5 % versus 33 %; $P < .001$). The mean time to return-to-work was significantly shorter for the CAC group than for the RFA group (0.8 ± 1.4 days versus 1.3 ± 1.4 days; $p = 0.003$). On multi-variate analysis, the treatment modality was the only modifiable factor in the time to return-to-work ($p = 0.028$). The development of post-procedure thrombophlebitis significantly increased the time to return-to-work for both groups ($p = 0.038$). No DVT or infection had developed following the procedure in either cohort. The authors concluded that RFA and CAC have both been shown to be safe and effective treatments to eliminate truncal venous insufficiency. Despite the significantly higher rates of post-procedure thrombophlebitis, the CAC patients were able to return-to-work significantly more quickly, reducing the opportunity cost of missed workdays, compared with RFA.

In a retrospective study, Arslanturk and Gumus (2023) compared the early and mid-term results of RFA and cyanoacrylate ablation (CA) used in the treatment of small saphenous insufficiency. A total of 84 patients with isolated small saphenous vein insufficiency who underwent either CA (Group 1, n = 40) or RFA (Group 2, n = 44) were analyzed. The occlusion rate of target vessel was 95 % in Group 1 and 93.1 % in Group 2 patients, respectively, at 1-year follow-up without any significant difference. Sural nerve injury was observed in 3 (6.8 %) patients in Group 2 due to the thermal damage of the RFA device. The authors concluded that while both techniques could be used with satisfactory and safe results in 1-year follow-up period, CA may have a better safety profile compared to RFA due to lower complication rates in terms of paresthesia and sural nerve damage with similar occlusion rates. The authors concluded that this study showed successful results with both modalities of SSV in the treatment of patients with symptomatic disability; CA may have a better safety profile compared to RFA due to lower complication rates in terms of paresthesia and sural nerve damage with similar occlusion rates. These researchers believe that this study will make a great contribution to the literature on this subject and it will lead future investigations.

The authors stated that this study had 2 main drawbacks. First, due to the nature of the interventions, blinding was not possible in this study; however, in an effort to minimize possible bias, evaluator-reported results were produced by protocol-based assessments or objective and validated scoring systems. Patient-reported results were also included and completed independently before assessments to further reduce bias. Another drawback was the sample size. Although this trial was powered to detect differences in technical effectiveness, further prospective, multi-center studies are needed to simultaneously compare each treatment modality to generalize the outcome.

Falvo et al (2023) stated that endovenous thermal methods are superseding surgical stripping for treating chronic superficial venous disease but require tumescent anesthesia and can cause heat-related nerve injuries. Endovenous CA is a more recent technique that does not share these limitations. These researchers carried out a retrospective, observational, single-center study of consecutive adults managed with endovenous CA; the follow-up was 18 months. They identified factors associated with target vein closure at 18 months and measured changes in QOL using the generic 36-item Short-Form (SF-36) tool and the venous disease-specific VEINES-QOL/Sym questionnaire; and AEs were collected. In the 55 participants with 67 treated veins, the closure rate at 18 months was 94 % (95 % CI: 85 % to 98 %). Target vein diameter of 9.5 mm or greater had 81 % sensitivity and 75 % specificity for predicting re-canalization; and QOL scores improved significantly ($p < 0.001$ for both surveys). The only AE was a type-IV allergic reaction to cyanoacrylate that was resolved with corticosteroid and histamine-antagonist therapy. The authors concluded that endovenous CA was highly effective and was safe in experienced hands. Moreover, these researchers stated that further studies are needed to examine if changes in the protocol would increase the closure rate in patients with target veins diameter of 9.5 mm or greater. These investigators noted that they are currently conducting a randomized, multi-center study of VenaSeal therapy versus endovenous thermal ablation or surgical stripping in this indication. This will be a large patient enrollment study (about 800 subjects) entailing at least 40 centers worldwide, which will also evaluate time to ulcer healing in the CEAP's six subjects.

The authors stated that the drawbacks of this study included the retrospective design, which carried a risk of information bias. It included a relatively small sample size (n = 55 subjects) at a single center. This trial was observational, and endovenous CA was not compared to other treatment methods. The recruitment at a

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single university hospital may have induced selection bias. A single experienced operator carried out all the procedures. No standardized assessment tools were employed to measure pain intensity or bruising. Finally, 8 participants were unable to complete the 18-month questionnaire at the 18-month visit, and their data were collected using less reliable methods. It was of utmost importance to highlight the fact that a worldwide randomized trial, called the Spectrum Study, is ongoing, comparing surgery, thermal ablation, and glue therapy. This is the largest study on this topic, which should render accurate short- and long-term outcomes in the near future. Moreover, these researchers noted that a formal cost–benefit analysis has not been performed, especially since this product remains expensive and is not reimbursed by the French social insurance system.

An UpToDate review on "Comparison of methods for endovenous ablation for chronic venous disease" (Dillavou and Kiguchi, 2023) states that "Cyanoacrylate glue sealing (i.e., VenaSeal) was approved for use in the United States in 2015. This technique seals the vein from the saphenofemoral/popliteal junction distally by injecting microbeads of cyanoacrylate glue during segmental pullback of a catheter combined with ultrasound probe compression … A global multi-armed, post-market trial is underway (VenaSeal Spectrum) to further characterize cyanoacrylate glue closure compared with vein stripping and endothermal closure across the range of subjects with chronic venous disease".

Tang et al (2024) stated that medium-term clinical outcome data are lacking for cyanoacrylate glue (CAG) ablation for symptomatic varicose veins, especially from the Asian population. By means of the prospective, Singaporean ASVS registry, these researchers determined the 3-year symptomatic relief gained from using the VenaSeal device to close refluxing truncal veins from. The revised VCSS (rVCSS) and 3 QOL questionnaires were completed to evaluate clinical improvement in venous disease symptoms along with a dedicated patient satisfaction survey. A total of 70 patients (107 limbs; 40 women; mean age of 60.9 ± 13.6 years) were included at 3 years. At 3 years, rVCSS showed sustained improvement from baseline (5.00 to 0.00; $p < 0.001$) and 51/70 (72.9 %) had improvement by at least 2 or more CEAP categories. Freedom from re-intervention was 90 %; and 85.7 % patients were extremely satisfied with the treatment outcome. No further reports of further hyper-sensitivity reactions after 1 year. The authors concluded that the 3-year follow-up results of the ASVS registry showed continued and sustained clinical effectiveness with few re-interventions following CAG embolization in Asian patients with chronic venous insufficiency.

The authors stated that the drawbacks of the study included 30 % of patients were unable to be contacted during this time as well as the lack of truncal occlusion data by Duplex U because of the COVID-19 crisis making it difficult for research patients to come to the hospital for imaging. These researchers plan to address this when they conduct 5-year ASVS data analysis.

Lim et al (2024) noted that CVI is a common medical condition that afflicts over 30 million adults in the U.S. Treatment and management have changed dramatically over the past 2 decades with the introduction of various endovenous therapies, including RFA and CAC using the Venaseal system. These investigators examined a direct comparison of outcomes between RFA and VenaSeal conducted by a single surgeon within a managed care organization. Between May 2020 and December 2021, there were 87 patients undergoing 97 total procedures who were treated with either VenaSeal (n = 55) or RFA (n = 42). Primary outcomes included successful GSV closure, post-operative DVT, embolization, SFJ thrombosis, skin reaction (allergic or thrombophlebitis), or procedure-related emergency room (ER) and urgent care (UC) visits. Retrospective chart review was carried out to describe patient demographics, indications for treatment, treatment details (number of access sites and indication for treatment), as well as follow-up adjunctive vein procedures (sclerotherapy and stab phlebectomy). All patients had post-procedural US performed within 7 days by the operating surgeon and mean overall patient follow-up from index procedure was 12 months; 2 patients were lost to follow-up due to death. All operations used intra-operative US, and SFJ thrombosis was not observed in any patient post-procedure. The average age of VenaSeal and RFA groups was 59.05 and 59.51 years, respectively. The average number of access sites during the procedure for VenaSeal and RFA was 1.56 and 1.20, respectively. Of the 55 VenaSeal treatments, 9.1 % of patients reported post-operative skin reactions, and 9.1 % of patients underwent subsequent stab phlebectomy (22 stabs on average). Of the 42 RFA treatments, 7.1 % of patients reported post-operative skin reactions, and 9.5 % underwent subsequent stab phlebectomy (36 stabs on average). The percentage of patients who had post-operative UC or ER visits related to the procedure in the VenaSeal and RFA groups was 3.6 % and 0 %, respectively. The authors

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concluded that both VenaSeal and RFA showed effective vein closure of the GSV at 12-month follow-up, with VenaSeal showing continued non-inferiority to RFA. Based on these findings, post-operative skin reactions did not appear to be significantly higher with VenaSeal treatment, as previously reported. Moreover, these researchers stated that although under-powered to assess this, these results suggested a possible benefit with the VenaSeal procedure as patients requiring post-procedural phlebectomy required fewer overall stab incisions as compared to RFA patients; however, further study is needed to compare the 2 endovenous treatment techniques, especially in patients with venous ulceration and long-term re-canalization rates

The authors stated that this study had several drawbacks. First, this trial was limited to a single center and a single surgeon's experience. Second, the COVID-19 pandemic may have created biases within this study. While there was a trend toward fewer post-GSV closure stab phlebectomies in the VenaSeal group, it was noted that the VenaSeal patients had a duration of 5 months between GSV closure and subsequent stab phlebectomy. By comparison, the RFA patients had a 3-month gap between GSV ablation and subsequent phlebectomy. Outpatient surgical delays due to COVID-19 may have caused greater delays in the VenaSeal group versus the RFA group, allowing greater time for venous decompression and need for few stab phlebectomies. Third, this study was not sufficiently powered to evaluate outcomes in patients with venous ulceration. These investigators stated that future research should expand this work to include multiple experienced surgeons across several medical centers, as well as lengthen follow-up time to examine for long-term outcomes and potential complications.

In a "Letter to the Editor", Dinc (2024) stated that cyanoacrylates are safe and effective in the treatment of varicose vein; however, it did not provide any clinical data on the effectiveness of the VenaBlock and VenaSeal.

The VariClose Vein Sealing System

Bozkurt and Yılmaz (2016) stated that cyanoacrylate ablation is the newest non-thermal vein ablation technique. In a prospective comparative study, these investigators presented the 1-year results of a new cyanoacrylate glue versus endovenous laser ablation for the treatment of venous insufficiency. A total of 310 adult subjects were treated with cyanoacrylate ablation or endovenous laser ablation. The primary end-point of this study was complete occlusion of the great saphenous vein; secondary end-points were procedure time, procedural pain, ecchymosis at day 3, AEs, changes from baseline in VCSS, and AVVQ. Operative time was shorter (15 ± 2.5 versus 33.2 ± 5.7, $p < 0.001$), and peri-procedural pain was less (3.1 ± 1.6 versus 6.5 ± 2.3, $p < 0.001$) in cyanoacrylate ablation group compared to the endovenous laser ablation group. Ecchymosis on the 3rd day was also significantly less in cyanoacrylate ablation group ($p < 0.001$). Temporary or permanent paresthesia developed in 7 patients in endovenous laser ablation group and none in cyanoacrylate ablation group ($p = 0.015$). Closure rates at 1, 3, and 12 months were 87.1, 91.7, and 92.2 % for endovenous laser ablation and 96.7, 96.6, and 95.8 % for cyanoacrylate ablation groups, respectively. Closure rate at 1st month was significantly better in cyanoacrylate ablation group ($p < 0.001$). Although there was a trend of better closure rates in cyanoacrylate ablation patients, this difference did not reach to the statistical difference at 6th and 12th month ($p = 0.127$ and 0.138, respectively). Both groups had significant improvement in VCSS and AVVQ post-operatively ($p < 0.001$), but there was no significant difference in VCSS and AVVQ scores between the groups at 1st, 6th, and 12 months. Only a slightly better well-being trend was noted in cyanoacrylate ablation group in terms of AVVQ scores ($p = 0.062$). The authors concluded that the safety and effectiveness analysis showed that cyanoacrylate ablation is a safe, simple method that can be recommended as an effective endovenous ablation technique. Moreover, they stated that the follow-up data more than 1 year will clarify the future role of cyanoacrylate ablation for the treatment incompetent great saphenous veins.

Tekin and colleagues (2016) noted that endothermal treatment of the great saphenous vein has become the 1st line of treatment for superficial venous reflux. A new technique for venous insufficiency is non-thermal ablation with vein sealing system that comprises the endovenous delivery of cyanoacrylate tissue adhesive to the vein causing fibrosis. In a single-center, prospective study, these researchers examined the effectiveness of treatment of great saphenous vein incompetence in 62 patients with vein sealing system (VariClose). All cases were implemented under local anesthesia. Tumescent anesthesia was not required. Patients were not

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given any NSAID post-operatively; advised to wear elastic bandages for 1 day; and compression stockings were not offered. Treatment success was defined as complete occlusion of treated vein or re-canalized segment shorter than 5 cm. Subtotal re-canalization was defined as great saphenous vein flow containing 5 to 10 cm segment of treated vein. A re-canalized great saphenous vein or treatment failure was defined as an open part of the treated vein segment more than 10 cm in length. At 1 week and 1 month control, duplex scans showed total occlusion for all patients (100 %), total occlusion for 58 patients (93.5 %), and subtotal occlusion for 4 patients (6.5 %) at 3rd month. At the end of 6 months, total occlusion 56 patients (90.3 %) and subtotal occlusion for 2 patients (3.2 %). For 4 (6.5 %) patients, no occlusion was observed, and the diameter was greater than 11 mm. Embolization of great saphenous vein with cyanoacrylate has been performed since the beginning of this decade. Combined chemical and physical mechanism of action resulted in permanent vein closure. In a recently published study, a 24-month occlusion rate of 92 % was demonstrated. The most commonly reported complications of cyanoacrylate use for the treatment of varicose vein disease, so far, include ecchymosis and phlebitis. Almeida et al. reported that phlebitis is the most frequent side effect at a rate of 16 %. In this study, phlebitis rate was not as high as reported. It may be caused due to shorter time of follow-up in the hospital. The authors concluded that endovenous ablation of incompetent great saphenous vein with cyanoacrylate-based glue is feasible. Operation time is short, and tumescent anesthesia is unnecessary as post-procedure compression stockings; lack of significant side effects and an yearly success rate of 100 % are benefits of the system. These findings need to be validated by well-designed studies with larger sample size and longer follow-up.

In a retrospective study, Yasim and associates (2017) presented the early results of the use of N-butyl cyanoacrylate (VariClose)-based non-tumescent endovenous ablation for the treatment of patients with varicose veins. A total of 180 patients with varicose veins due to incompetent saphenous veins were treated with the VariClose endovenous ablation method between May 2014 and November 2014. Participants consisted of 86 men and 94 women, with a mean age of 47.7 ± 11.7 years; they had a great saphenous vein diameter greater than 5.5 mm and a small saphenous vein diameter greater than 4 mm in conjunction with reflux for more than 0.5 s. Patients with varicose veins were evaluated with venous duplex examination, CEAP, and their VCSS were recorded. The median CEAP score of patients was 3, and the saphenous vein diameters were between 5.5 and 14 mm (mean of 7.7 ± 2.1 mm). A percutaneous entry was made under local anesthesia to the great saphenous vein in 169 patients and to the small saphenous vein in 11 patients. Duplex examination immediately after the procedure showed closure of the treated vein in 100 % of the treated segment. No complications were observed. The mean follow-up time was 5.5 months (ranging from 3 to 7). Re-canalization was not observed in any of the patients during follow-up. The average VCSS was 10.2 before the procedure and decreased to 3.9 after 3 months ($p < 0.001$). The authors concluded that the application of N-butyl cyanoacrylate (VariClose) is an effective method for treating varicose veins; it yielded a high endovenous closure rate, with no need for tumescent anesthesia. However, long-term results are currently unknown.

Furthermore, Bootun and colleagues (2016b) stated that the early results of 2 recently launched non-thermal, non-tumescent methods, mechanochemical endovenous ablation (MOCA) and cyanoacrylate glue, are promising.

Koramaz and associates (2017) retrospectively compared an n-butyl cyanoacrylate (NBCA)-based ablation method with EVLA for the management of incompetent GSV. Between May 2013 and August 2014, there were 339 patients with incompetent varicose veins who were treated with either the endovenous application of NBCA (VariClose Vein Sealing System [VVSS]; Biolas, Ankara, Turkey) or EVLA. The pre-procedural, intra-procedural, post-procedural, and follow-up data of the patients were collected and retrospectively compared. The mean age was 45.09 ± 12 years in the VVSS group and 47.08 ± 11 years in the EVLA group ($p = 0.113$). The average ablated vein length was 31.97 ± 6.83 cm in the VVSS group and 31.65 ± 6.25 cm in the EVLA group ($p = 0.97$). The average tumescent anesthesia use was 300 ml (range of 60 to 600 ml) in the EVLA group. The average procedure time was 7 minutes (range of 4 to 11 minutes) in the VVSS group and 18 minutes (range of 14 to 25 minutes) in the EVLA group ($p < 0.01$). On the basis of US examinations performed at the end of the procedure, all procedures in both groups were successful, and the target vein segments were fully occluded. The 12-month total occlusion rates in the VVSS and EVLA groups were 98.6 % and 97.3 %, respectively ($p = 0.65$). In both the VVSS and EVLA groups, the VCSS declined significantly with no difference between groups. There were fewer AEs after VVSS treatment compared

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with EVLA treatment (pigmentation, $p \leq 0.002$; phlebitis, $p \leq 0.015$). There was no need for tumescent anesthesia in the VVSS group. The authors concluded that the NBCA-based vein sealing system was a fast and effective therapeutic option for the management of incompetent saphenous veins that did not involve tumescent anesthesia, compression stockings, paresthesia, burn marks, or pigmentation. Moreover, they stated that further large-scale studies with long-term outcomes are needed to identify the optimal treatment modalities for patients with SVI.

Vos and co-workers (2017) performed a systematic review and meta-analysis to evaluate the effectiveness of MOCA and cyanoacrylate vein ablation (CAVA) for GSV incompetence. Medline, Embase, Cumulative Index to Nursing and Allied Health Literature, and Cochrane databases were searched for papers published between January 1966 and December 2016. Eligible articles were prospective studies that included patients treated for GSV incompetence and described the primary outcome. Exclusion criteria were full text not available, case reports, retrospective studies, small series (n less than 10), reviews, abstracts, animal studies, studies of SSV incompetence, and recurrent GSV incompetence. Primary outcome was anatomic success; secondary outcomes were initial technical success, VCSS, AVVQ score, and complications. A total of 15 articles met the inclusion criteria. Pooled anatomic success for MOCA and CAVA was 94.7 % and 94.8 % at 6 months and 94.1 % and 89.0 % at 1 year, respectively; VCSS and AVVQ score significantly improved after treatment with MOCA and CAVA. The authors concluded that these findings were promising for these novel techniques that could serve as alternatives for thermal ablation techniques. However, they stated that to determine their exact role in clinical practice, high-quality RCTs comparing these novel modalities with well-established techniques are needed.

Eroglu and colleagues (2017) presented mid-term results of patients with varicose veins treated with N-butyl cyanoacrylate (VariClose), a non-tumescent endovenous ablation technique. Endovenous ablation was performed on 180 patients with saphenous vein incompetence between May and October 2014. A total of 168 subjects capable of being followed-up for 30 months were included. Patients' pre- and post-operative data were recorded. Procedures were performed on the GSV in 159 patients and on the SSV in 9 patients. Saphenous vein diameters ranged between 5.5 mm and 14 mm. Full ablation was achieved in all patients following the procedure. No complications were encountered. Patients were monitored for 30 months. Ablation rates were 100 % at the 3rd month, 98.3 % at the 6th month, 96.6 % at 1 year, and 94.1 % at 30 months. Mean VCSS was 10.2 before procedures, decreasing to 3.9 at 3 months, 4.2 at 6 months, 2.9 at 12 months, and 2.7 at 30 months ($p = 0.000$). The authors concluded that d to its high success rate, absence of complications, no tumescent anesthesia requirement and high patient satisfaction, endovenous ablation with N-butyl cyanoacrylate is a good method. However, they stated that long-term follow-up results are needed.

Prasad and associates (2018) noted that recurrent lower limb venous insufficiency is often a challenge in clinical practice and is most commonly due to incompetent perforators. Many of these patients do not have adequate symptom relief with compression and require some form of treatment for incompetent perforator interruption. Various methods have been tried with different efficiencies. These investigators evaluated the feasibility, safety and effectiveness of an out-patient combined cyanoacrylate adhesion-sodium tetradecyl sulphate sclerotherapy for the treatment of patients with symptoms of persistent or recurrent lower limb venous insufficiency secondary to incompetent perforators. A total of 83 limbs of 69 patients with symptoms of persistent or recurrent lower limb venous insufficiency secondary to incompetent perforators were treated with cyanoacrylate embolization of incompetent perforators and sclerotherapy of dilated collateral veins (surface branch varicose veins). Technical success, procedural pain, perforator occlusion, venous occlusion, clinical improvement and ulcer healing were assessed. Follow-up was done 3- and 6-month post-procedure. Procedure could be successfully performed in all patients; a total of 191 perforators were treated. Perforator and varicose veins occlusion rate was 100 %. Deep venous extension of cyanoacrylate occurred in 4 (4.8 %) patients, with no adverse clinical outcome. Venous clinical severity score improved from a baseline of 8.18 ± 3.60 to 4.30 ± 2.48 on 3-month follow-up and 2.42 ± 1.52 on 6-month follow-up ($p < 0.0001$). All ulcers showed complete healing within 3 months. Significant prolonged thrombophlebitis occurred in 38.5 % of limbs. The authors concluded that combined cyanoacrylate adhesion and sodium tetradecyl sulphate sclerotherapy was technically easy, had a lot of advantages including being an out-patient procedure and highly effective but with a guarded safety profile. The main drawbacks of this study were its relatively small sample size ($n = 69$) and short-term follow-up (6 months); and the findings were confounded by the combined use of cyanoacrylate adhesion and sodium tetradecyl sulphate sclerotherapy.

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Gibson and colleagues (2018) previously reported 3-month and 12-month occlusion rates after treatment of clinically symptomatic saphenous vein reflux with either cyanoacrylate closure (CAC) using the VenaSeal Closure System (Medtronic, Dublin, Ireland) or radiofrequency ablation (RFA) in a randomized, multi-center, clinical trial, VenaSeal Sapheon Closure System vs Radiofrequency Ablation for Incompetent Great Saphenous Veins (VeClose). These investigators reported the 24-month follow-up results of the VeClose trial. There were 222 patients with symptomatic great saphenous vein (GSV) incompetence who were randomly assigned to receive either CAC (n = 108) or RFA (n = 114). Patients were not allowed to receive adjunctive treatment of tributary varicosities until after the 3-month visit. Duplex ultrasound (US) of the target vein was performed at day 3 and months 1, 3, 6, 12, and 24 after treatment, and closure was assessed by US by the treating physician. Overall 24-month success rates were compared; in addition, time to first reopening of the target vein was evaluated using survival analysis. End-points such as Venous Clinical Severity Score, EuroQoL-5 Dimension, and Aberdeen Varicose Vein Questionnaire were evaluated. Of 222 randomized patients, 171 completed the 24-month follow-up, which included 87 from the CAC group and 84 from the RFA group. The 24-month complete closure rate was 95.3 % in the CAC group and 94.0 % in the RFA group, demonstrating continued non-inferiority of CAC compared with RFA (p = 0.0034). Symptoms and quality of life (QOL) improved similarly in both groups. No clinically significant device- or procedure-related late adverse events (AEs) occurred. The authors concluded that both CAC and RFA were effective in closure of the target GSV, resulting in similar and significant improvements in the patient's QOL through 24 months. They stated that these results suggested that CAC of the GSV is safe and durable out to 2 years. The investigators will continue to observe the patient cohorts, and longer term data will be collected at 3 and 5 years.

The authors stated that this trial was limited by the moderate withdrawal rate at 24 months 21 of the 108 subjects (19.4 %) in the CAC withdrew). In addition, blinding of patients was impossible with the tumescent use requirements (because RFA requires tumescent anesthesia but CAC does not), as was blinding of physicians (CA is clearly evident on long-term US examination). As the patients' expectations regarding the control treatment (RFA) and new treatment (CAC) were unknown at the time of treatment, it is unclear whether "excitement" regarding the latter would bias patient-reported outcomes. In addition, the impact of adjunctive procedures on patients' perception of success or failure of their treatment and on QOL is not known.

Radak and associates (2019) noted that in the last 20 years, varicose veins surgery went through a great turbulence and various innovations. A number of new techniques have been introduced with a goal to increase the success rate, reduce the peri-procedural complications and, overall, to improve patients' QOL. The latest of them, namely the CAE technique, threatens to shake the glory of the currently well-established endovenous methods. These investigators analyzed all previous studies by searching Medline base using PubMed. Although the idea of using NBCA glue in medical purposes was not new, the very first in-vivo and animal experiments using NBCA for vein closure were conducted just at the beginning of this millennium. The results of these studies gave warranty to begin with the first interventions in humans. Early studies reported very high success rates of more than 90 %, with the longest follow-up period of 36 months. There were no major AEs reported, while the minor ones -- mostly phlebitic reactions -- were defined as mild-to-moderate. The newest head-to-head studies showed that CAE is a non-inferior technique to other endovenous methods, with higher occlusion rates and fewer AEs. Short procedure time and no need for tumescent anesthesia or compressive stockings reduce patients discomfort to the minimum and definitely appeared to be a step forward answering the modern "walk in - walk out surgery" demands. The authors concluded that as an easily handling technique, CAE undoubtedly pierced its path to the top of varicose veins surgery; however, more head-to-head clinical trials with longer follow-up periods are needed to obtain a clearer picture.

Belramman and co-workers (2018) stated that thermal ablation techniques have become the 1st-line treatment of truncal veins in the management of chronic venous disease (CVD). Despite excellent outcomes, these methods are often associated with pain; generally due to their use of heat and the necessity of fluid infiltration around the vein. More recently, novel non-thermal techniques, such as MOCA and CAE have been developed to overcome these unwelcome effects. So far, the novel techniques have been found to have similar efficacy to thermal methods, yet no direct comparisons between the non-thermal treatment techniques have been conducted to-date, giving rise to this study. This is a prospective, multi-center, randomised

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clinical trial, recruiting patients with truncal saphenous incompetence. Patients will be randomised to undergo either MOCA or CAE truncal ablation, followed by treatment of any varicosities. All patients will be required to wear compression stockings for 4 days post-intervention. The primary outcome measure is the pain score immediately following completion of truncal ablation, measured by a 100-mm VAS. The secondary outcomes are entire treatment pain scores, clinical scores, QOL scores, occlusion rates, time to return to usual activities/work at 2 weeks, 3, 6 and 12 months. Re-intervention rate will be considered from the 3rd month. Cost-effectiveness will be assessed for each intervention at 12 months. The study is powered to detect a mean 10-mm difference in maximum pain score. Allowing for loss to follow-up, the total target recruitment is 180 patients. The study will be the first study to compare MOCA against CAE and is designed to determine which method causes less pain. Completion of this study is expected to be the end of 2019.

In a multi-center, prospective, RCT, Morrison et al (2019) evaluated the 36-month safety and efficacy of CAC for the treatment of incompetent GSVs in comparison with RFA. A total of 222 symptomatic subjects with incompetent GSVs were assigned to either CAC or RFA. The primary end-point, complete closure of the target GSV, was determined using duplex US examination starting from 3-month visit. At month 36, the GSV closure rates were 94.4 % for the CAC group and 91.9 % for the RFA group. Stable improvement in symptoms and QOL was observed in both groups; AEs rates between the 24- and 36-month visits were similar between the groups as were serious AEs, which were infrequent and judged unrelated to either the device or the procedure in both groups. The authors concluded that this trial continued to demonstrate the safety and efficacy of CAC for the treatment of GSV incompetence with GSV closure rate at 36 months similar to that of RFA, indicating non-inferiority of CAC to RFA. The improvement in QOL outcomes were also sustained and similar between the 2 treatment groups. These investigators also noted that follow-up of the patient cohorts post-procedure would continue up to 60 months.

The authors noted that this study had several drawbacks. Although blinding was potentially advantageous, it was not entirely feasible, as RFA requires tumescent anesthesia administration. The trial was also limited by relatively high number of patient with missing data from each treatment group (the lack of data for approximately 1/3 of patients in each group was due to patient drop-out or the data could not be collected in the time period dictated by the study period). And finally, potential advantages of CAC over RFA were purposefully not addressed in this study, as the present report was only intended to demonstrate 36-month parity of CAC with RFA. Cost analysis comparisons of CAC and RFA were not included in the design of the study, but could be forthcoming in future studies in addition to analyses of the potential advantages of CAC described above.

Furthermore, an UpToDate review on “Liquid, foam, and glue sclerotherapy techniques for the treatment of lower extremity veins” (Scovell, 2019) states that “Cyanoacrylate glue -- A system that ablates the treated vein using an adhesive agent (VenaSeal) has been approved for use in the United States. The use of glue was initially described for treatment of saphenous incompetence in 2013. The procedure is performed like radiofrequency and laser ablation, but without the need for tumescent anesthesia. In a randomized trial comparing this system with radiofrequency ablation, short-term outcomes at 3 months were similar. Longer-term follow-up is needed to determine the durability of the results”.

ClariVein

Witte et al (2015) noted that endovenous mechano-chemical occlusion using the ClariVein catheter is a new technique combining mechanical injury to the venous endothelium coupled with simultaneous catheter-guided infusion of a liquid sclerosant. This produces irreversible damage to the endothelium resulting in fibrosis of the vein. The technique is related to a low complication rate and a success rate of 96 % at 2 years and sustained quality of life improvement. This closure rate is comparable to endothermal techniques, but significantly less post-operative pain and earlier return to normal activities and work has been reported with endovenous mechano-chemical occlusion. The authors concluded that mechano-chemical occlusion using ClariVein has proven to be safe and effective and has several advantages compared to endothermal techniques. The possibility of retrograde ablation of distal SSV insufficiency in C6 ulceration is considered a significant advantage. Moreover, they stated that randomized comparative studies with long-term follow-up will continue to define the definite place of mechano-chemical occlusion.

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Deijen et al (2016) stated that mechano-chemical endovenous ablation is a novel technique for the treatment of GSV and SSV incompetence which combines mechanical injury of the endothelium with simultaneous infusion of liquid sclerosant. The main objective of this study was to evaluate early occlusion. All consecutive patients who were eligible for the treatment with mechano-chemical endovenous ablation were included. Inclusion period was from the introduction of the device in the hospitals (September 2011 and December 2011) until December 2012. A total of 449 patients were included representing 570 incompetent veins. In 506 treated veins, duplex ultrasonography was performed at follow-up: 457 veins (90 %) were occluded at a follow-up of 6 to 12 weeks. In univariate and multivariate analysis, failure of treated GSV was associated with sapheno-femoral junction incompetence (OR 4; 95 % CI: 1.0 to 17.1, $p = 0.049$). The authors concluded that the ClariVein device appeared to be safe and had a high short-term technical effectiveness. Long-term clinical outcomes are needed to ascertain the clinical value of the ClariVein.

In a RCT, Lam et al (2016) identified the ideal polidocanol dosage and form for mechano-chemical ablation in order to occlude the GSV. When adhering to safe dosage levels, sclerosants with higher concentrations potentially limit the extent of treatment. It has been demonstrated that this problem may be overcome by using polidocanol as a microfoam. This paper was established on findings of a preliminary analysis. The initial study was a single-blinded multi-center RCT where patients are allocated to 3 treatment arms. Group 1 consisted of mechano-chemical ablation + 2 % polidocanol liquid, group 2: mechano-chemical ablation + 3 % polidocanol liquid, and group 3: mechano-chemical ablation + 1 % polidocanol foam. A total of 87 (34 males and 53 females (60.9 %)), mean age of 55 years; S.D. 16.0 (range of 24 to 84), were enrolled in the study. Treatment length was 30 cm (range of 10 to 30) for 95.2 % of the patients. Mean operating time was 16 minutes (range of 5 to 70). The mean sapheno-femoral junction diameter (7.7 mm) was similar in all 3 groups. At 6 weeks post-treatment duplex ultrasound showed that 25 out of 25 (100 %), 27 out of 28 (96.4 %) and 13 out of 23 (56.5 %) were occluded in the mechano-chemical ablation + 2 % polidocanol liquid, mechano-chemical ablation + 3 % polidocanol liquid and mechano-chemical ablation + 1 % polidocanol microfoam respectively ($p < 0.001$). However, stricter scrutiny showed that the anatomical success rate defined as occlusion of at least 85 % of the treated length to be 88.0 %, 85.7 % and 30.4 % respectively ($p < 0.001$). The authors concluded that mechano-chemical ablation using ClariVein combined with 1 % polidocanol microfoam is significantly less effective and should not be considered as a treatment option of incompetent truncal veins. They stated that further investigation to determine the ideal polidocanol liquid dosage with mechano-chemical ablation is advocated and is being conducted accordingly.

Vun and colleagues (2015) evaluated the effectiveness of the ClariVein system of MOCA of superficial vein incompetence. ClariVein treatment uses a micro-puncture technique and a 4-Fr sheath to allow a catheter to be placed 1.5 cm from the SFJ. Unlike EVLT or RFA, no tumescence is required. The technique depends on a wire rotating at 3,500 r/min causing endothelial damage while liquid sclerosant (1.5 % sodium tetradecyl sulphate) is infused. The wire is pulled back while continuously infusing sclerosant along the target vessel's length. Initially, 8 ml of dilute sclerosant was used, but this was subsequently increased to 12 ml. No routine post-op analgesia was prescribed and specifically no NSAIDs. Procedure times and pain scores (visual analog scale [VAS]) were recorded and compared to EVLT and RFA. All patients were invited for duplex post-procedure. A total of 51 GSV and 6 SSV were treated and followed-up with duplex in the 10 months from July 2011. No major complications or deep vein thrombosis were reported. Duplex showed patency of 3 treated veins with 2 more veins having only a short length of occlusion, giving a technical success rate of 91 %. Comparison with 50 RFA and 40 EVLT showed procedure times were significantly less for ClariVein (23.0 ± 8.3 mins) than for either RFA (37.9 ± 8.3 mins) or EVLT (44.1 ± 11.4 mins). Median pain scores were significantly lower for ClariVein than RFA and EVLT (1 versus 5 versus 6, $p < 0.01$). The authors concluded that MOCA with the ClariVein system is safe and effective. After some initial failures, the use of 12 ml of dilute sclerosant resulted in a very high technical success rate greater than 90 % which accorded with the limited published literature; and procedure times and pain scores were significantly better than for RFA and EVLT. These researchers stated that they await the long-term clinical outcomes.

Bootun and associates (2016a) noted that endovenous techniques are, at present, the recommended choice for truncal vein treatment. However, the thermal techniques require tumescent anesthesia, which can be uncomfortable during administration. Non-tumescent, non-thermal techniques would, therefore, have potential benefits. In a RCT, these investigators compared the degree of pain that patients experience while receiving MOCA or RFA. The early results of this RCT were reported here. Patients attending for the

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treatment of primary varicose veins were randomized to receive MOCA (ClariVein) or RFA (Covidien Venefit). The most symptomatic limb was randomized. The primary outcome measure was intra-procedural pain using a validated VAS. The secondary outcome measures were change in quality of life and clinical scores, time to return to normal activities and work as well as the occlusion rate. A total of 119 patients were randomized (60 in the MOCA group). Baseline characteristics were similar. Maximum pain score was significantly lower in the MOCA group (19.3 mm, SD ± 19 mm) compared to the RFA group (34.5 mm ± 23 mm; $p < 0.001$). Average VAS was also significantly lower in the MOCA group (13.4 mm ± 16 mm) compared to the RFA (24.4 mm ± 18 mm; $p = 0.001$); 66 % attended follow-up at 1 month, and the complete or proximal occlusion rates were 92 % for both groups. At 1 month, the clinical and quality of life scores for both groups had similar improvements. The authors concluded that early results showed that MOCA is less painful than the RFA procedure, and clinical as well as quality of life scores were similarly improved at 1 month. The long-term data including occlusion rates at 6 months and quality of life scores are being collected. Furthermore, Bootun and colleagues (2016b) stated that the early results of 2 recently launched non-thermal, non-tumescent methods, MOCA and cyanoacrylate glue, are promising.

Lam and colleagues (2016) noted that the ClariVein system is an endovenous technique that uses MOCA to treat incompetent truncal veins. This study was conducted to identify the ideal Polidocanol dosage and form for MOCA in order to occlude the GSV. When adhering to safe dosage levels, sclerosants with higher concentrations potentially limit the extent of treatment. It has been demonstrated that this problem may be overcome by using Polidocanol as a microfoam. This paper was established on findings of a preliminary analysis. The initial study was a single-blinded, multi-center RCT where patients were allocated to 3 treatment arms:

I. group 1 consisted of MOCA + 2 % Polidocanol liquid,
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Leung and colleagues (2016) stated that endovenous thermal techniques, such as EVLA, are the recommended treatment for truncal varicose veins. However, a disadvantage of thermal techniques is that it requires the administration of tumescent anesthesia, which can be uncomfortable. Non-thermal, non-tumescent techniques, such as MOCA have potential benefits; MOCA combines physical damage to endothelium using a rotating wire, with the infusion of a liquid sclerosant. Preliminary experiences with MOCA showed good results and less post-procedural pain. The Laser Ablation versus Mechanochemical Ablation (LAMA) trial is a single-center RCT in which 140 patients will be randomly allocated to EVLA or MOCA. All patients with primary truncal superficial venous insufficiency (SVI) who meet the eligibility criteria will be invited to participate in this trial. The primary outcomes are intra-procedural pain and technical efficacy at 1 year, defined as complete occlusion of target vein segment and assessed using duplex ultrasound. Secondary outcomes are post-procedural pain, analgesia use, procedure time, clinical severity, generic and disease-specific quality of life, bruising, complications, satisfaction, cosmesis, time taken to return to daily activities and/or work, and cost-effectiveness analysis following EVLA or MOCA. Both

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groups will be evaluated on an intention-to-treat basis. The aim of the LAMA trial is to establish whether MOCA is superior to the current first-line treatment, EVLA. The 2 main hypotheses are:

I. MOCA may cause less initial pain and disability allowing a more acceptable treatment with an enhanced recovery, and
II. this may come at a cost of decreased efficacy, which may lead to increased recurrence and affect longer term quality of life, increasing the requirement for secondary procedures.

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II. this may come at a cost of decreased efficacy, which may lead to increased recurrence and affect longer term quality of life, increasing the requirement for secondary procedures.

In a large, single-center study, Tang and co-workers (2017) examined the effectiveness and patient experience of the ClariVein® endovenous occlusion catheter for varicose veins. A total of 300 patients (371 legs) underwent ClariVein treatment for their varicose veins; 184 for GSV incompetence, 62 bilateral GSV, 23 SSV, 6 bilateral SSV and 25 combined unilateral GSV and SSV. Patients were reviewed at an interval of 2 months post-procedure and underwent Duplex ultrasound assessment. Post-operative complications were recorded along with patient satisfaction. All 393 procedures were completed successfully under local anesthetic. Complete occlusion of the treated vein was initially achieved in all the patients, but at 8 weeks' follow-up, there was only partial obliteration in 13/393 (3.3 %) veins. These were all successfully treated with ultrasound-guided foam sclerotherapy. Procedures were well-tolerated with a mean pain score of 0.8 (0 to 10), and no significant complications were reported. The authors concluded that ClariVein can be used to ablate long and short saphenous varicose veins on a walk-in-walk-out basis. Bilateral procedures can be successfully performed, and these were well-tolerated as can multiple veins in the same leg. They stated that early results are promising but further evaluation and longer term follow-up are needed.

Witte and colleagues (2016) reported the midterm results of mechanochemical ablation (MOCA) for treating GSV insufficiency. In a 1-year period, 85 consecutive patients (median age of 51.4 years; 71 women) undergoing MOCA with polidocanol in 104 limbs were enrolled in a prospective registry. Patients were evaluated at baseline and during follow-up (4 weeks and 1, 2, and 3 years) using duplex ultrasound, the CEAP (clinical, etiologic, anatomic and pathophysiologic) classification, the Venous Clinical Severity Score (VCSS), the RAND Short Form 36-Item Health Survey (RAND-SF36), and the Aberdeen Varicose Vein Questionnaire (AVVQ). Primary outcome measures were clinical and anatomic success; secondary outcome measures included general and disease-specific quality of life and re-interventions. Technical success (99 %) was achieved in all but 1 patient in whom technical problems with the device led to conversion to another method for treatment of 2 limbs. After a median follow-up of 36 months (interquartile range [IQR] 12.5, 46.3), re-canalization occurred in 15 (15 %) of 102 successfully treated vein segments. Anatomic success was 92 %, 90 %, and 87 % after 1, 2, and 3 years, respectively. The VCSS improved at all time-intervals compared to the pre-procedure median. The clinical success at 3 years was 83 %. The AVVQ and RAND-SF36 scores showed an improvement at all time-intervals compared to baseline values. Between 12 and 36 months, however, a significant deterioration was observed in VCSS, which was accompanied by worsening of disease-specific and general quality of life. The authors concluded that in the longest follow-up of MOCA to-date, this study showed MOCA to be an effective treatment modality for GSV insufficiency at midterm follow-up, but clinical results appeared to drop over time. The major drawbacks of this study were:

I. the results were affected by the chosen definitions of success. Although the definition used is in accordance with previous landmark trials, heterogeneity in the definition among studies was a major problem in comparing results and emphasized the need for standardization of outcome measures, and
II. follow-up was not completed for every patient, and questionnaires were not always complete.

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II. follow-up was not completed for every patient, and questionnaires were not always complete.

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Lane and associates (2017) noted that endovenous thermal ablation has revolutionized varicose vein treatment. New non-thermal techniques such as MOCA allow treatment of entire trunks with single anesthetic injections. Previous non-randomized work has shown reduced pain post-operatively with MOCA. This study presented a multi-center, RCT assessing the difference in pain during truncal ablation using MOCA and radiofrequency endovenous ablation (RFA) with 6 months' follow-up. Patients undergoing local anesthetic endovenous ablation for primary varicose veins were randomized to either MOCA or RFA. Pain scores using VAS and number scale (0 to 10) during truncal ablation were recorded. Adjunctive procedures were completed subsequently. Pain after phlebectomy was not assessed. Patients were reviewed at 1 and 6 months with clinical scores, quality of life scores and duplex ultrasound assessment of the treated leg. A total of 170 patients were recruited over a 21-month period from 240 screened. Patients in the MOCA group experienced significantly less maximum pain during the procedure by VAS (MOCA median of 15 mm (IQR 7 to 36 mm) versus RFA 34 mm (IQR 16 to 53 mm), p = 0.003) and number scale (MOCA median of 3 (IQR 1 to 5) versus RFA 4 mm (IQR 3 to 6.5), p = 0.002). "Average" pain scores were also significantly less in the MOCA group; 74 % underwent simultaneous phlebectomy. Occlusion rates, clinical severity scores, disease specific and generic quality of life scores were similar between groups at 1 and 6 months. There were 2 deep vein thromboses, 1 in each group. The authors concluded that pain secondary to truncal ablation was less painful with MOCA than RFA with similar short-term technical, quality of life and safety outcomes. They stated that further work with larger studies and extended follow-up is needed to evaluate the long-term outcomes and recurrence rates.

The authors noted that "This study was limited by lack of treatment blinding for the patients and interventional clinicians. This was due to the technical differences between devices, i.e., tumescent injections in the RFA group and device vibration in the MOCA group. Follow-up appointments and ultrasound scanning were treatment blind. A further limitation of this study is the lack of long-term follow-up -- only short-term occlusion rates are assessed in this study, with the primary outcome obtained at the time of procedure. Operating time was not recorded in this study; however, all cases were performed in standardized theatre sessions in single slots with one surgeon performing all tasks, and 74 % of patients also underwent simultaneous phlebectomy. A major limitation of all tumescentless techniques is how to treat varicosities left after truncal ablation, with level one evidence now supporting combined treatment with phlebectomies. This study was designed and commenced prior to the completion of latest trial, but took into consideration the fact that phlebectomies cause pain, and so pain scores taken after truncal ablation but before any phlebectomies were completed. This, therefore, represents a significant limitation to the outcomes of this trial, as the pain scores reported above do not assess the complete treatment, except for those patients who did not undergo phlebectomy". Also, this study did not assess pain scores after phlebectomy or after the peri-procedural period.

National Institute for Health and Care Excellence's guideline on "Endovenous mechanochemical ablation for varicose veins" (2016) stated that " Current evidence on the safety and efficacy of endovenous mechanochemical ablation for varicose veins appears adequate to support the use of this procedure provided that standard arrangements are in place for consent, audit and clinical governance. Clinicians are encouraged to collect longer-term follow-up data".

Elias and Raines (2012) evaluated the safety and efficacy of the ClariVein system that employs mechanochemical ablation (MOCA) of the great saphenous vein (GSV). Patients eligible for ablation of the GSV underwent micro-puncture access with only local anesthesia to insert a 4 or 5 Fr sheath. The ClariVein catheter was placed through the sheath, the wire was extruded, and the distal tip of the wire positioned 2 cm from the sapheno-femoral junction under ultrasound (US) guidance. Catheter wire rotation was then activated for 2 to 3 seconds at approximately 3,500 rpm. With the wire rotating, infusion of the sclerosant was started simultaneously with catheter pullback. The sclerosant used was 1.5 % liquid sodium tetradecyl sulphate. A total of 30 GSVs in 29 patients were treated. All patients have reached 6-month follow-up; the average number of post-operative days was 260. No adverse events (AERs) have been reported; the primary closure rate was 96.7 %. The authors concluded that MOCA appeared to be safe and efficacious. The ClariVein technique eliminated the need for tumescent anesthesia. The great majority of incompetent GSVs can be treated with this technique. (This was a small study (n= 29) with short-term follow-up (6 months))

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In a prospective cohort study, Boersma and associates (2013) evaluated the feasibility, safety and 1-year results of MOCA of small saphenous vein (SSV) insufficiency. A total of 50 consecutive patients were treated for primary SSV insufficiency with MOCA using the ClariVein device and polidocanol. Initial technical success, complications, patient satisfaction and visual analogue scale (VAS) pain score were assessed. Anatomic and clinical success was assessed at 6 weeks and at 1 year. Initial technical success of MOCA was 100 %. At the 6-week assessment, all treated veins were occluded. The 1-year follow-up duplex showed anatomic success in 94 % (95 % confidence interval [CI]:, 0.87 to 1). Venous clinical severity score (VCSS) decreased significantly from 3.0 (interquartile range (IQR) 2 to 5) before treatment to 1.0 (IQR 1 to 3, $p < 0.001$) at 6 weeks and to 1.0 (IQR 1 to 2, $p < 0.001$) at 1 year. Median procedural VAS score for pain was 2 (IQR 2 to 4). No major complications were observed, especially no nerve injury. The authors concluded that MOCA was a safe, feasible and efficacious technique for treatment of SSV insufficiency. One-year follow-up showed a 94 % anatomic success rate and no major complications.

One of the drawbacks of this study was that the maximum diameter of treated SSVs was 11 mm. The technical and clinical success of MOCA in larger-diameter varicose veins was still unknown. Pain scores during MOCA were very low. Post-procedural pain scores were not measured. The authors stated that further controlled studies are needed to compare pain with other techniques in SSV ablation. Patients on oral anti-coagulants were excluded; thus, these researchers could not provide data on the effect of anti-coagulant therapy on MOCA. In contrast to endothermal therapy, anti-coagulants might influence clot formation and lead to increased re-canalization.

In a prospective, observational, multi-center report, Bishawi and colleagues (2014) evaluated the efficacy of a tumescent-free technique using MOCA in selected patients with lower extremity chronic venous disease. Demographic information, clinical and procedural data were collected on a customized database. The distribution and extent of venous reflux and the closure rate of the treated veins were assessed with duplex US. Pain was evaluated during the procedure and post-operatively using an analog scale. The presence and severity of complications were recorded. Patient improvement was assessed by clinical-etiology-anatomy-pathophysiology (CEAP) class and venous clinical severity score (VCSS). There were 126 patients that were included at baseline, 81 % females, with a mean age of 65.5 ± 14 years. The average BMI was 30.5 ± 6. The mean diameter of the great saphenous vein in the upper thigh was 7.3 mm and the mean treatment length was 38 cm. Adjunctive treatment of the varicosities was performed in 11 % of patients during the procedure. Closure rates were 100 % at 1 week, 98 % at 3 months, and 94 % at 6 months. Post-procedure complications included hematoma 1 %, ecchymosis 9 %, and thrombophlebitis 10 %. There were no cases of venous thromboembolism. There was significant improvement in VCSS ($p < 0.001$) for all time intervals. The authors concluded that MOCA of the saphenous veins had the advantage of endovenous ablation without tumescent anesthesia, making it an almost pain-free procedure. High occlusion rates with significant clinical improvement can be achieved with this method at short-term.

Ozen et al (2014) evaluated the reliability and 2-year results of ClariVein device used in MOCA of GSV. In the authors' clinic, a total of 63 patients with GSV insufficiency had been treated using ClariVein device and polidocanol for 2 years. Both legs were treated in 10 of these patients. The anatomical and clinic success were assessed by Doppler US 6 months, 1 year, and 2 years later. The implementation success rate of the technique was 98 %. The anatomical success was found as 94 % at the end of 6 months, 95 % at the end of 1 year, and 95 % at the end of 2 years. The venous clinic severity score was found as 3.2 (IQR: IQR: 2 to 6) after 6 months, 1.2 (IQR: 1 to 3, $p < 0.001$) after 1 year, and 1.1 (IQR: 1 to 2, $p < 0.001$) after 2 years. No complications developed in any of the patients. The authors concluded that ClariVein was a simple, reliable, and efficient treatment method for GSV insufficiency. In 2-year follow-up, the anatomical success rate was found as 95 %, and no major complications were observed.

Stanisic et al (2016) stated that MOCA of the GSV and the SSV is an alternative to thermal ablation for the treatment of superficial venous reflux. These researchers evaluated the efficacy of MOCA for the treatment of incompetent GSV and SSV. They included 50 patients (60 legs) with incompetent GSV or SSV. Patients were aged 22 to 71 years, with median age of 41 years. Diameters of the saphenous veins treated were 4 to 16 mm, with median diameter 9 mm. Lengths of incompetent segments of the GSVs were 20 to 45 cm, with median length 36 cm. Lengths of incompetent segments of the SSVs were 12 to 25 cm, with median length of 17 cm. These investigators performed venous ablation using the ClariVein device with simultaneous

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injection of 2 % polidocanol in the dose of 0.2 ml/cm of the treated vein. All patients completed 12 months follow-up. In all patients the procedure resulted in complete occlusion of the incompetent segment of the saphenous vein. Additional foam sclerotherapy was needed in 41 legs (68.3 %). After 12 months partial or complete re-canalization was revealed in 1 GSV and 3 SSVs. The remaining veins (93.3 %) were completely occluded. During the procedure these researchers observed transient signs of polidocanol toxicity in 2 patients. The authors concluded that MOCA using the ClariVein device was a safe method for ablation of incompetent truncal veins in patients who prefer to be managed quickly, without pain and with satisfactory results after 1 year.

In a 2-year follow-up on the efficacy of MOCA in patients with symptomatic C2 or more advanced chronic venous disease, Kim and co-workers (2017) reported if early efficacy was maintained at 24 months. Patients with reflux in the great saphenous vein involving the sapheno-femoral junction and no previous venous interventions were included. Demographic information, clinical, and procedural data were collected. The occlusion rate of treated veins was assessed with duplex US. Patient clinical improvement was assessed by CEAP class and venous clinical severity score. Of the initial 126 patients, there were 65 patients with 24 month follow-up. Of these 65 patients, 70 % were women, with a mean age of 70 ± 14 years and an average BMI of 30.5 ± 6. The mean great saphenous vein diameter in the upper thigh was 7.6 mm and the mean treatment length was 39 cm. Adjunctive treatment of the varicosities was performed in 14 % of patients during the procedure. Closure rates were 100 % at 1 week, 98 % at 3 months, 95 % at 12 months, and 92 % at 24 months. There was 1 patient with complete and 4 with partial re-canalization ranging from 7 to 12 cm (mean length 9 cm). There was significant improvement in CEAP and venous clinical severity score ($p < 0.001$) for all time intervals. The authors concluded that early high occlusion rate with MOCA was associated with significant clinical improvement which was maintained at 24 months, making it a very good option for the treatment of great saphenous vein incompetence.

Whiteley et al (2017) examined the effects of MOCA using ClariVein on ex-vivo GSV with histology and immunofluorescent staining. Extra-fascial GSVs were harvested during surgery for varicose veins and were treated ex-vivo for 10 to 11 minutes with either liquid sclerotherapy or the use of ClariVein, with and without 3 % sodium tetradecyl sulfate. Veins were sectioned and subjected to hematoxylin and eosin staining and immunofluorescent staining for endothelial and smooth muscle cell markers (CD31 and α-actin) to assess overall damage and cell death in the vein wall compared with control sections. Histologic observations confirmed intimal damage from ClariVein, as has been previously shown; however, medial damage was also evident, which was not observed in control or liquid sclerotherapy sections. Immunofluorescent staining in the 3 sections studied showed a 42 % decrease in CD31 staining and 27 % mean reduction in α-actin staining up to a depth of 300 μm with liquid sclerotherapy. This cytotoxic effect was significantly enhanced by MOCA with a reduction in CD31 staining just above 60 % and a 46 % mean decrease in α-actin staining noted up to a depth of 300 μm. Far greater reductions in staining compared with sclerotherapy were observed up to a depth of 600 μm. The authors concluded that MOCA using 3 % sodium tetradecyl sulfate increased the penetration of the sclerosant and its effect into the vein wall and showed superior rates of tissue destruction compared with liquid sclerotherapy alone. In this model, it appeared not solely to damage the endothelium but also to shear the medial layer, creating small lesions into which sclerosant can flow and exert its cytotoxic effect. These investigators stated that short-term follow-up studies of MOCA showed results that were comparable to those of RFA or EVLA. Initial investigations into the short- to medium-term success rates of ClariVein for treating reflux in the GSV reported excellent closure rates that stand above 95 % up to 1 year after the procedure, with the longest follow-up of 2 years showing 92 % closure. Separate analyses also showed significantly less post-operative pain and faster recovery of the patient with MOCA compared with RFA. This showed a considerable advantage over US-guided foam sclerotherapy, which was associated with a high risk of re-canalization and recurrent reflux even as early as 1 year after the procedure. These researchers noted that as time progresses, the medium- and long-term success rates of MOCA will need to be evaluated and compared with existing treatment modalities.

This study had several drawbacks. Primarily, these investigations were conducted ex-vivo and involved only 3 extra-fascial veins as opposed to true GSV. In addition, as an ex-vivo model, there was no blood flow during our 10-min testing sessions, and although the freshly harvested veins did exhibit some spasm during the treatment, these researchers could not be sure that this was as intense as would be achieved in-vivo. However, they did not think that these factors invalidate the basic findings in this model. The presence of

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blood would inactivate only some of the sclerosant, and the fact that MOCA works in-vivo showed that this was not significant clinically. Also, any additional spasm would only be more likely to enhance the shearing effect observed in this study. The study did not look at other concentrations of sodium tetradecyl sulfate (STS) or other agents such as polidocanol as the study focused on the mechanism of MOCA rather than on the effectiveness of MOCA. These investigators used 3 % STS as a marker of cell death that they had already demonstrated and reported on to show the increase in depth of penetration of the cytotoxic effect into the media by adding the mechanical effect of the MOCA system. Similarly, foam sclerotherapy was not studied as foam was not used with MOCA as a sclerosing product, and this study was looking at the mechanism of action of MOCA rather than being a comparison between different methods of ablating veins with different sclerotherapy techniques.

Witte et al (2017) reported the mid-term results of MOCA for treating GSV insufficiency. In a 1-year period, 85 consecutive patients (median age of 51.4 years; 71 women) undergoing MOCA with polidocanol in 104 limbs were enrolled in a prospective registry. The patients were evaluated at baseline and during follow-up (4 weeks and 1, 2, and 3 years) using duplex US, the CEAP (clinical, etiologic, anatomic and pathophysiologic) classification, the Venous Clinical Severity Score (VCSS), the RAND Short Form 36-Item Health Survey (RAND-SF36), and the Aberdeen Varicose Vein Questionnaire (AVVQ). Primary outcome measures were clinical and anatomic success. Secondary outcome measures included general and disease-specific QOL and re-interventions. Technical success (99 %) was achieved in all but 1 patient in whom technical problems with the device led to conversion to another method for treatment of 2 limbs. After a median follow-up of 36 months (IQR 12.5 to 46.3), re-canalization occurred in 15 (15 %) of 102 successfully treated vein segments. Anatomic success was 92 %, 90 %, and 87 % after 1, 2, and 3 years, respectively. The VCSS improved at all time intervals compared to the pre-procedure median. The clinical success at 3 years was 83 %. The AVVQ and RAND-SF36 scores showed an improvement at all time intervals compared to baseline values. Between 12 and 36 months, however, a significant deterioration was observed in VCSS, which was accompanied by worsening of disease-specific and general QOL. The authors concluded that in the longest follow-up of MOCA to-date, this study showed MOCA to be an effective treatment modality for GSV insufficiency at mid-term follow-up, but clinical results appeared to drop over time. The authors stated that the results of the present study were affected by the chosen definitions of success. Although the definition used was in accord with previous landmark trials, heterogeneity in the definition among studies was a major problem in comparing results and emphasized the need for standardization of outcome measures. Furthermore, follow-up was not completed for every patient, and questionnaires were not always complete.

While a Cochrane review on “Endovenous ablation therapy (laser or radiofrequency) or foam sclerotherapy versus conventional surgical repair for short saphenous varicose veins” (Paravastu et al, 2016) did not address the use of MOCA, it is interesting to note that the authors stated that “Further RCTs for all comparisons are required with longer follow-up (at least 5 years). In addition, measurement of outcomes such as recurrence of reflux, time taken to return to work, duration of procedure, pain, etc., and choice of time points during follow-up should be standardised such that future trials evaluating newer technologies can be compared efficiently”.

Kugler and Brown (2017) conducted a systematic review of the literature on non-thermal ablative techniques using a Medline (Ovid) search from January 2000 to August 2016. Only prospective studies and literature review articles in the English language were included for final analysis. A total of 358 unique articles were identified, with a total of 60 articles meeting the stated inclusion and exclusion criteria. Historically, non-thermal ablative techniques have not demonstrated clinical results on par with thermal ablative interventions. However, 3 newer non-thermal ablative techniques have become available for use in the United States. Review of the literature demonstrated significant improvements in non-thermal ablative results, with intermediate-term data suggesting improved durability. The authors concluded that advances in non-thermal ablative techniques have led to a developing role and acceptance in the primary management of varicose veins and venous insufficiency, even in the setting of challenging cases. Moreover, they noted given advances in technology, non-thermal ablative techniques for the primary management of superficial venous insufficiency have acceptable success rates compared with thermal techniques and may be preferred in certain cases in which thermal techniques may have drawbacks, such as the below-knee saphenous vein or tortuous, superficial saphenous veins. However, long-term outcome data and head-to-head comparisons of

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the techniques are sparse in the literature. Whereas Clinical, Etiology, Anatomy, and Pathophysiology (CEAP) classifications help provide a basis for comparison regarding severity of disease, each trial has additional exclusion and inclusion criteria, which limits comparison. Finally, there are no good data, including no randomized trial, to allow comparison between non-thermal techniques.

Witte and colleagues (2017) systematically reviewed all available English literature on mechano-chemical endovenous ablation and reported on the anatomical, technical, and clinical success. A systematic literature search was performed in PubMed, Embase, and the Cochrane Library on mechano-chemical endovenous ablation for the treatment of insufficient great and/or small saphenous vein. Methodological quality of the included studies was evaluated using the MINORS score. The primary outcome measure was anatomical success, defined as closure of the treated vein on follow-up duplex ultrasound imaging. Secondary outcomes were technical and clinical success, and major complications defined as deep venous thrombosis, pulmonary embolisms or paresthesia. The literature search identified 759 records, of which 13 were included, describing 10 unique cohorts. A total of 1,521 veins (1,267 great saphenous vein and 254 small saphenous vein) were included, with cohort sizes ranging from 30 to 570 veins. The pooled anatomical success rate after short-term follow-up was 92 % (95 % CI: 90 to 94%) (n = 1,314 veins). After 6 and 12 months these numbers were 92 % (95 % CI: 88 to 95 %) (n = 284) and 91 % (95 % CI: 86 to 94 %) (n = 228), respectively. The long-term anatomical success rates at 2 and 3 years were 91 % (95 % CI: 85 to 95 %) (n = 136) and 87 % (95 % CI: 75 to 94 %) (n = 48), respectively. Major complications and especially nerve injury were very rare (less than or equal to 0.2 %). All studies were of moderate or good quality using the MINORS scoring scale. The authors concluded that mechano-chemical endovenous ablation using the ClariVein in combination with liquid sclerosant is associated with an anatomical success rate ranging from 87 % to 92 % and good clinical success. To-date, no randomized controlled trials (RCTs) are available studying the anatomical success after mechano-chemical ablation, compared to the endothermal ablation. The risk of major complications is very low after the procedure.

The authors stated that this study had several drawbacks. To-date, the long-term data remain limited and around 50 % of initially included patients are lost to follow-up at 2 to 3 years after treatment. Although it is widely accepted that the anatomical success should be assessed with follow-up duplex ultrasounds, there is still heterogeneity regarding the exact definition of occlusion; in 5 cohorts a description of the exact definition used was lacking. In the remainder, the definition of the anatomical success ranged from occlusion over 85 % of total length treated to less than 10 cm recanalization. Furthermore, it should be appreciated that all published data are derived from cohort studies or RCTs not adequately powered for anatomical success. Recently, a protocol for the LAMA trial, a RCT comparing anatomical success and pain after MOCA versus EVLA, has been published. This trial is currently recruiting participants. Unfortunately, the randomized controlled trials comparing MOCA and RFA in GSV and SSV stopped inclusion prematurely due to reimbursement issues in Netherlands.

Khor and colleagues (2018) examined the effectiveness and patient experience of ClariVein for varicose veins and CVI in a multi-ethnic Asian population from Singapore. A total of 121 patients underwent mechano-chemical ablation. Patients were reviewed at an interval of 1 week, and at 3, 6 and 12 months post-procedure and underwent Duplex US with patient satisfaction assessment. At 3 months of follow-up, the GSV and SSV occlusion rates were 90.8 % and 96.0 %, respectively. At 6 months of follow-up, the GSV and SSV occlusion rates were 86.9 % and 90.9 %, respectively. At 1 year, GSV and SSV occlusion rates were 84.8 % and 94.3 %, respectively. The authors concluded that early results were similar to what was described so far in the mechano-chemical ablation literature; but recurrences were more than expected at 1 year. These researchers stated that these findings were disappointing but were tempered by the fact that the majority of patients were asymptomatic and required no re-intervention.

In a single-center cohort study, Mohamed and associates (2019) examined the clinical effectiveness of mechano-chemical ablation for the treatment of SVI at 1 year. Patients with primary, unilateral, symptomatic axial incompetence were offered treatment with mechano-chemical ablation using ClariVein with 1.5 % sodium tetradecyl sulphate. Assessments including clinical examination, duplex ultrasound and patient-reported health-related quality of life were performed at baseline and weeks 1, 6, 26 and 52. A total of 32 patients were recruited to the study. Complete target vein occlusion at 1 year was found in 21 (75 %) patients; 6 patients (21.4 %) required secondary procedures, of which 3 had axial endovenous thermal

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ablation and 3 needed ambulatory phlebectomy with perforator ligation. There was a significant improvement in the median (interquartile range) Venous Clinical Severity Score from baseline 6 (5 to 8) to a score of 1 (0 to 2) at 1year ($p < 0.001$). There was also a significant improvement in health-related quality of life, both generic ($p = 0.001$) and disease specific ($p < 0.001$); 1 patient (3.1 %) had a post-procedural non-fatal pulmonary embolus. The authors concluded that mechanochemical ablation is a feasible and effective treatment for superficial venous incompetence. When using consensus definitions for anatomical closure, results for mechanochemical ablation may be less favorable than previously reported. These researchers stated that further studies are needed to compare clinical and technical outcomes following mechanochemical ablation to other methods of endovenous ablation.

Oud et al (2024) stated that the short-term anatomical success rates of MOCA using the Clarivein device in the treatment of GSV incompetence were high; however, the anatomical success rates appeared to decrease over time. In a prospective, single-center study, these researchers assessed the long-term outcomes of GSV treatment using the Clarivein and examined if specific anatomical features would better correlate with clinical or QOL related outcomes. The primary outcome was anatomical success (AS), defined as compete occlusion or a re-canalized segment, irrespective of reflux, of less than 10 cm in length. Furthermore, reflux-free AS (RF-AS) was determined and defined as a complete occlusion or a re-canalized segment with less than 10 cm of reflux. Clinical success was determined using the VCSS; and QOL was assessed using the Dutch version of the Aberdeen Varicose Vein Questionnaire (DAVVQ) and the SF-36. Subgroup analyses were carried out based on whether AS or RF-AS was achieved or not. A total of 109 patients (115 limbs) were included. The mean follow-up time was 8.4 ± 0.9 years (range of 5.5 to 10.3 years). AS was observed in 60.5 % of limbs and RF-AS was observed in 72.8 % of limbs. Compared to baseline, the overall mean VCSS improved from 5.3 ± 2.4 to 4.1 ± 2.4, and the overall median DAVVQ score from 13.1 (7.3 to 19.4) to 10.5 (4.8 to 15.8) ($p < 0.001$). Improvement in VCSS was only significant in patients with successful treatment: 5.5 ± 2.4 to 3.7 ± 2.5 ($p < 0.001$) if AS was achieved and 5.0 ± 1.7 to 4.5 ± 1.9 ($p = 0.20$) if AS was not achieved. The same results were found for DAVVQ scores: 13.5 (8.7 to 20.6) to 10.3 (3.0 to 14.5) ($p < 0.001$) if AS was achieved and 12.9 (8.3 to 19.3) to 10.8 (6.7 to 18.2) ($p = 0.35$) if AS was not achieved. Regarding the overall SF-36 scores, the domains vitality, mental health and general health worsened significantly. The authors concluded that in over 8 years of follow-up, anatomical success after the treatment of GSV incompetence using the Clarivein device appeared to steadily decrease; however, clinical scores and disease-specific QOL scores still improved significantly compared to baseline. Anatomical success rates were higher when using the absence of reflux as a criterion of treatment success; however, these researchers found no convincing evidence that the absence of reflux better correlated with clinical and QOL-related outcomes compared to re-canalization irrespective of reflux; thus, the gap in evidence regarding the correlation between anatomical outcomes and clinical outcomes remains. These researchers stated that since anatomical success appeared to be inferior to endovenous thermal ablation techniques, Clarivein may be considered in a selected group of patients with a specific preference for this type of MOCA treatment.

The authors stated that this study had several drawbacks. First, the relatively small sample size. These investigators only included patients treated at the Red Cross Hospital in the original trial and 64.1 % of all eligible patients returned. This might have resulted in an inclusion bias. Second, neovascularization and neo-reflux have not been assessed. These are well described causes for recurrent varicose veins and might interfere with clinical outcomes. Third, deterioration in most SF-36 domains may have other causes in this aging population unrelated to superficial venous incompetence, as previously mentioned. Fourth, changes in the clinical and QOL related outcomes did not merely reflect the effect of the primary treatment, but also the effect of additional ipsilateral venous interventions carried out during the follow-up study, and the use of compression stockings, as 28.7 % of patients did in this cohort.

Polymorphism Genotyping of Matrix Metalloproteinases Genes (e.g., MMP1, MMP2, MMP3, and MMP7) as Markers of Predisposition to Varicose Veins

Kurzawski and associates (2009) noted that several risk factors for varicose veins have been identified: female gender, combined with obesity and pregnancy, occupations requiring standing for long periods, sedentary lifestyle, history of deep-vein thrombosis (DVT) and family history. However, no specific gene variants related to a wide prevalence of varicosities in general population have been identified. Extracellular

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matrix composition, predominantly maintained by matrix metalloproteinases (MMPs), may affect the vein-wall structure, which may lead to dilation of vessels and cause varicosities. MMP-1 (tissue collagenase I) and MMP-3 (stromelysin I) expression was found to be raised in varicose veins compared with normal vessels. Thus, these investigators carried out a study to evaluate a potential association between MMP1 and MMP3 promoter polymorphisms and a risk of varicose veins. Genotyping for the presence of the polymorphisms -1607dupG (rs1799750) in MMP1 and -1171dupA (rs3025058) in the MMP3 promoter region was performed using polymerase chain reaction (PCR) and restriction-fragment length polymorphism assays in a group of 109 patients diagnosed with varicose veins and 112 healthy controls. The frequencies of the MMP1 and MMP3 alleles (minor allele frequency 0.440 in patients versus 0.451 in the controls for MMP1-1607*G and 0.514 versus 0.469 for MMP3-1171*dupA, respectively) and of genotypes did not differ significantly between patients and controls. The authors concluded that MMP1-1607dupG and MMP3-1171dupA promoter polymorphisms were not valuable markers of susceptibility for varicose veins.

Shadrina and colleagues (2017) examined the effects of single nucleotide polymorphisms (SNPs) in the promoter regions of MMP genes rs1799750 (-1607dupG) MMP1, rs243865 (C-1306T) MMP2, rs3025058 (-1171dupA) MMP3, and rs11568818 (A-181G) MMP7 on the risk of varicose vein of the lower extremities in ethnical Russians, residents of the Russian Federation. These researchers genotyped 536 patients with this pathology and 273 healthy participants without history of chronic venous disease. Association was examined using logistic regression analysis. None of the studied polymorphisms showed statistically significant association with the risk of varicose veins of the lower extremities. The authors concluded that these findings provided evidence that these polymorphisms are not involved in the pathogenesis of varicose veins and cannot serve as markers of predisposition to this pathology.

Matrix Metalloproteinases Inhibitors for the Treatment of Varicose Veins

Chen and colleagues (2017) noted that the veins of the lower extremity are equipped with efficient wall, contractile vascular smooth muscle (VSM), and competent valves in order to withstand the high venous hydrostatic pressure in the lower limb and allow unidirectional movement of deoxygenated blood toward the heart. The vein wall structure and function are in part regulated by MMPs, which are zinc-dependent endopeptidases that are secreted as inactive pro-MMPs by different cells in the venous wall including fibroblasts, VSM, and leukocytes. Pro-MMPs are activated by other MMPs, proteinases, and other endogenous and exogenous activators. Matrix metalloproteinases degrade various extracellular matrix (ECM) proteins including collagen and elastin, and could affect other cellular processes including endothelium-mediated dilation, VSM cell migration, and proliferation as well as modulation of calcium ion (Ca2+) signaling and contraction in VSM. It is believed that increased lower limb venous hydrostatic pressure increases hypoxia-inducible factors and other MMP inducers such as extracellular MMP inducer, leading to increased MMP expression/activity, ECM protein degradation, vein wall relaxation, and venous dilation. Vein wall inflammation and leukocyte infiltration cause additional increases in MMPs, and further vein wall dilation and valve degradation, that could lead to chronic venous disease and VVs, which are often presented as vein wall dilation and tortuosity, incompetent venous valves, and venous reflux. Different regions of VVs show different MMP levels and ECM proteins with atrophic regions showing high MMP levels/activity and little ECM compared to hypertrophic regions with little or inactive MMPs and abundant ECM. Treatment of VVs includes compression stockings, venotonics, sclerotherapy, or surgical removal. However, these approaches do not treat the cause of VVs, and other lines of treatment may be needed. The authors stated that modulation of endogenous tissue inhibitors of metalloproteinases (TIMPs), and exogenous synthetic MMP inhibitors may provide new approaches in the management of VVs.

Measurements of Plasma Growth Factors in Patients with Varicose Veins Treated with Endovenous Laser Ablation for Predicting Adequacy of Treatment and the Possibility of Recurrence

Al-Zoubi and colleagues (2018) measured pre- and post-operative plasma concentrations of growth factors (angiopoietin-1 [ANG1], angiopoietin-2 [ANG2], epidermal growth factor [EGF], platelet-derived growth factor [PDGF], and vascular endothelial growth factor [VEGF]) in patients with primary lower limb VVs treated with EVLA. Blood samples were obtained from 30 patients with primary VVs undergoing treatment

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with EVLA before and 1 week after treatment. Similar samples were obtained from 20 healthy matched adults as a control. Plasma concentrations of growth factor derivatives (ANG1, ANG2, EGF, PDGF, and VEGF) were measured through commercially available enzyme-linked immunosorbent assay (ELISA). There was statistically significant reduction in median plasma growth factor (ANG1, EGF, PDGF, and VEGF) levels in the pre-operative group ($p = 0.001$) when compared with the control group except for ANG2, which showed increased plasma level ($p = 0.001$). However, values of plasma concentration of those growth factors after treatment with EVLA were nearly equal to the control group, especially in EGF and VEGF ($p = 0.564$, 0.515, respectively). The authors concluded that altered plasma concentrations of growth factors ANG1, ANG2, EGF, PDGF, and VEGF among patients with VVs normalized 1 week after treatment with EVLA compared with the control group. This may support the role of these factors in the pathogenesis of the disease. Moreover, they stated that future studies may evaluate if these changes can play a prognostic and/or predictive value regarding the adequacy of treatment and the possibility of recurrence.

VeinOPlus Vascular Device for the Treatment of Muscle Atrophy Due to Varicose Veins

Le Tohic et al (2009) examined if electrostimulation (VeinOPlus) of lower limbs relieves lower limbs venous insufficiency-related symptoms during pregnancy. A 2-step study was conducted. First, a mono-centric, prospective preliminary study including 30 pregnant women was conducted to assess the effects of electrostimulation on fetal monitoring and uterine contractions. Then, a prospective, multi-centric, non-randomized study including 58 pregnant women with a gestational age between 23 and 33 weeks of amenorrhea was conducted to evaluate the electrostimulation treatment. This evaluation was based on a clinical examination performed pre- and post-treatment, a CIVIQ questionnaire filled out pre- and post-treatment and a daily diary filled out by the patient during treatment duration. Treatment duration was 21 days including 2 daily treatment sequences of 20 mins; 3 groups of patients were identified based on initial intensity of venous insufficiency-related symptoms (group 1 minor symptoms, group 2 moderate symptoms, group 3 severe symptoms). Preliminary study showed no interferences between electrostimulation and fetal cardiac rhythm, uterine contractions and maternal uterine and fetal umbilical arteries Doppler. Concerning the evaluation of the electrostimulation: in group 1, electrostimulation significantly reduced heavy legs sensation ($p < 0.001$) and calves pain ($p = 0.02$) between the beginning and the end of the treatment. The 4 scores calculated with the CIVIQ questionnaire decreased after treatment and a significant reduction was noted for generalized pain feeling ($p =0.04$) and psychological impact ($p = 0.03$). In group 2, a significant decrease was noted for tiredness ($p < 0.001$), heavy legs sensation ($p < 0.001$), calves pain ($p < 0.001$) and edema ($p = 0.02$) between the beginning and the end of the treatment. The 4 scores calculated with the CIVIQ questionnaire significantly decrease after 21 days of treatment. In group 3, a significant decrease of heavy legs sensation ($p = 0.03$) and calves and malleoli perimeters ($p < 0.05$) was noted. After 21 days of treatment, the four scores calculated with the CIVIQ questionnaire significantly decrease ($p < 0.05$). When comparing the 3 groups, beneficial effects of the treatment were most marked in group 2 regarding subjective symptoms, CIVIQ questionnaire scores and leg pain. According to the patients, effectiveness and tolerance of the treatment ranged from good-to-excellent in the 3 groups. The authors concluded that electrostimulation was an effective and well-tolerated treatment of lower limbs venous insufficiency-related symptoms in pregnant women. Its use during pregnancy did not show any effects on fetus and pregnancy.

Bogachev et al (2011) stated that electro-muscular stimulation (EMS) with VEINOPLUS has recently emerged as a new technique to activate the calf muscle pump, improving the symptoms of venous disease. These researchers examined the efficacy of EMS and its impact on evening edema, venous pain, venous outflow and patients quality of life (QOL). A total of 30 patients (32 legs) aged 19 to 50 years (mean of 45.2 ± 1.3) classified CEAP C3 with chronic evening venous edema took part in the study (22 limbs: C3, S, Ep, Asp, Pr and 10 limbs: C3, S, Es, Aspd, Pr). All patients were treated with CE-registered VEINOPLUS neuromuscular stimulator for 30 days; 3 sessions per day (each session lasted for 20 mins) for the first 10 days, the next 10 days 2 sessions per day and 1 session per day for the last 10 days. The main evaluated criteria for venous edema was the circumference of the supramaleoal shin segment, measured twice with a tape measure; in the evening the day before and then 5 days after the treatment. All measurements were taken during the interval between 6 and 8 P.M. All the patients were asked to evaluate venous pain using the visual analog scale (VAS), and to fill in a CIVIQ questionnaire (validated for Russian patients) for the QOL measurement. Also venous refilling time (RT) was measured by digital PPG. That was also made twice --

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before the treatment and in 35 days after the treatment. No other treatments were used. The EMS treatment was well-tolerated by patients. There were no drop-outs and patients had no need to change their lifestyles. After the treatment, total or partial reduction of evening edema was shown in 93.8 % of limbs, the circumference of the lower leg diminished by 20.3 mm ($p < 0.001$), the number of painful legs reduced from 28 to 12 and the severity score of venous pain was cut from 8.3 ± 1.1 to 3.8 points ± 0.9 ($p < 0.001$), QOL improved significantly as the score dropped from 34.5 ± 7.8 to 17.2 points ± 4.6 ($p < 0.001$) and RT increased from 17.3 ± 0.9 to 21.5 seconds ± 1.1 ($p < 0.001$). Three months after VEINOPLUS treatment, total remission of symptoms was observed in 50 % of legs, despite there being no other treatments. The authors concluded that VEINOPLUS stimulation was an effective and well-tolerated therapeutic method for the treatment of chronic venous disease. The introduced scheme of EMS application was shown to be useful for treatment of chronic edema, for reducing pain and improving QOL. It can be used as an additional mean in treatment and to prevent symptoms of CVI. This study also revealed that stimulation of calf muscles with VEINOPLUS could improve venous outflow and symptoms of CVI; this finding should be investigated and confirmed in further studies.

Abraham et al (2013) examined the effect of electrical stimulation of calf muscles on arterial inflow and tissue oxygen content in peripheral artery disease (PAD) in the area of stimulation. A total of 15 adult patients [mean (standard deviation) age of 62 (12) years; height of 165 (8) cm; weight of 76 (13) kg; lowest ankle-brachial index 0.66 (0.19)] with stable arterial claudication were recruited. All patients performed a treadmill test (3.2 km/hour, 10 % slope) associated with a transcutaneous oximetry test expressed as decrease from rest of oxygen pressure (DROP) index values (calf changes minus chest changes from rest) with a maximum walking distance (median [25th/75th percentiles]) of 295 [133 to 881] m. The DROP index on the symptomatic side was -25 [-18/-34] mm Hg. On another day the patients underwent electrical stimulation in the seated position on the leg that was the most symptomatic on the treadmill. After resting values were recorded, the gastrocnemius was stimulated for 20 mins at increasing contraction rates at 5-min steps of 60, 75, 86, and 100 bpm on the most symptomatic side. Arterial blood inflow with duplex Doppler ultrasound (US) scanning of the femoral artery, DROP transcutaneous oxygen pressure value, and oxygen concentration (O2Hb) from the near-infrared spectroscopic signal of the calf were recorded on both sides. Patients were instructed to report eventual contraction-induced pain in the stimulated calf. Results were given as mean (standard deviation) or median [25th/75th percentiles] according to distribution, and the level of statistical significance was set at $p < 0.05$ on 2-tailed tests. Lower limb inflow (ml/min) was 64 [48/86] versus 63 [57/81] ($p > 0.05$) before stimulation, 123 [75/156] versus 57 [44/92] ($p < 0.01$) at 60 bpm, 127 [91/207] versus 49 [43/68] ($p < 0.01$) at 75 bpm, 140 [84/200] versus 57 [45/71] ($p <0 .01$) at 86 bpm, and 154 [86/185] versus 55 [46/94] ($p < 0.01$) at 100 bpm on the stimulated versus non-stimulated limb, respectively. No apparent decrease or significant leg difference was observed in DROP index or O2Hb values. None of the patients reported contraction-induced pain in the leg. The authors concluded that electrical stimulation of calf muscle with the Veinoplus device resulted in a significant increase of arterial inflow without measurable muscle ischemia or pain. These researchers stated that potential use of this device as an adjuvant treatment to improve walking capacity in PAD patients remains to be evaluated.

In a prospective, non-randomized controlled study, Lobastov et al (2014) examined the potential effect of electrical calf muscle stimulation (EMS) in the prevention of post-operative deep vein thrombosis (DVT) in high risk patients and evaluated the safety and efficacy of EMS in patients with calf DVT. This trial involved 80 patients over the age of 40 having major surgery (44 abdominal and 36 cranial or spinal surgery; duration more than 60 mins under general anesthesia). Patients were divided into 2 comparable groups: main ($n = 40$) and control ($n = 40$). In both groups graduated middle stretch compression bandage with compression level 20 to 40 mmHg was applied and low-dose unfractioned heparin (LDUH) injections (5,000 U s.c. 3 t.i.d) were started on 1st, or 2nd to 5th day after surgery and continued until discharge. The time of starting LDUH was comparable in both groups. In addition, EMS with the Veinoplus device was performed for not less than 5 periods of 20 mins/day (total greater than 100 mins) in the main group. Control of venous patency was performed with duplex US obligatory at baseline (first 24 hours after surgery) and then every 3 days until discharge. The incidence of DVT was 2.5 % in the main group and 25 % in the control group ($p = 0.007$). In patients without DVT at baseline it was 3 % versus 21 % ($p = 0.025$). Patients with baseline thrombosis who underwent EMS did not have any new cases of DVT and PE, while in patients without EMS thrombosis progression was observed in 43 % cases also without pulmonary embolism (not significant). The authors concluded that EMS with the Veinoplus device at greater than 100 mins/day (greater than 5 sessions) can

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decrease the rate of post-operative DVT in high risk patients. Using of EMS in patients with calf DVT did not increase the rate of PE. These researchers stated that these findings need to be confirmed in a randomized controlled trial (RCT).

Bogachev et al (2015) analyzed the results of the EMS usage in patients with venous ulcers developed on top of a post-thrombotic syndrome (PTS). A total of 60 patients (60 legs) with active venous ulcer (C6EsAsdpPr according to CEAP classification) were divided into 2 groups. In addition to the background therapy consisting of a standardized compression with ULCER X and intake of micronized purified flavonoid fraction (MPFF 1,000 mg daily), all the patients in the main group underwent EMS with Veinoplus V.I. for at least 3 times a day. Follow-up examinations were performed on days 30, 60 and 90. These included pain severity assessment with 100-mm VAS, disease severity measurement with VCSS (Venous Clinical Severity Score) and ankle circumference above malleolus, as well as recording number of healed venous ulcers. At day-90 pain severity was reduced in both main and control groups. However, according to VAS pain reduction rates were significantly higher in patients of the main group (from 8.7 ± 0.6 to 1.9 ± 0.3 in the main group and 8.4 ± 0.6 to 3.9 ± 0.5 in the control group). At the end of the study, ankle circumference decreased from 270.9 ± 4.6 mm to 257.1 ± 4.2 mm in the main and from 269.7 ± 5.3 mm to 263.4 ± 5.2 in the control group. VCSS before treatment was 7.3 ± 0.6 in the main group and 6.8 ± 0.5 in the control group. By day-90 VCSS significantly decreased to 2.3 ± 0.4 and 4.6 ± 0.5 in the main and control groups, respectively. Healing rates were significantly higher in the main group. On day-90, the number of open venous ulcers in the main group was 3 times lower than in the control group (4 versus 12). The authors concluded that EMS demonstrated high efficacy and good tolerability and provided significant reduction in pain severity, VCSS score and ankle edema, as well as a 3-fold increase in the number of healed venous ulcers.

Mechanochemical Ablation with Flebogrif for the Treatment of Great Saphenous Vein Insufficiency

Ammollo and colleagues (2020) noted that Flebogrif (Balton, Poland) is a novel MOCA device for saphenous vein insufficiency. It combines endothelial damage performed by radial retractable cutting hooks together with chemical ablation through sclerosant injection of 3 % polidocanol foam according to its Instructions for Use (IFU). These researchers examined Flebogrif's efficacy in terms of re-canalization rate and recurrence by varying polidocanol foam concentrations. They carried out 24 MOCAs on 23 patients with Flebogrif between January and May 2019 . In 12 cases, the polidocanol foam was prepared at a 3 % concentration, and in another 12 at 1.5 %; GSV re-canalization and truncular recurrence were evaluated at 1 and 3 months with a DUS examination. At 1- and 3-month follow-ups, none of the 14 patients treated with the polidocanol 3 % foam were observed to have had GSV re-canalization and truncular recurrence. Only 2 of the 14 (14.3 %) cases treated with polidocanol 1.5 % foam showed evidence of re-canalization within the first centimeters from the sapheno-femoral junction ($p > 0.05$). All patients experienced clinical benefits without recurrence of symptoms. The authors concluded that MOCA with Flebogrif was a safe, relatively inexpensive and effective alternative to standard methods in the treatment of saphenous insufficiency with encouraging short-term results. Despite the relatively small patient sample, no statistical significance in evidence of recurrence in the group of patients treated with 3 % foam and those treated with 1.5 % foam was noted. Moreover, these researchers stated that longer term analysis of GSV patency and recurrence is needed to further evaluate Flebogrif's impact and actual indications in the treatment of chronic venous disease. Further investigation is also needed to broaden the number of treated cases, and include a comparison with surgical or thermal ablation techniques.

The authors stated that the drawbacks of this study included the small number of cases (n = 23 patients) and the short follow-up period (3 months). Furthermore, the Flebogrif procedures were performed in patients with relatively small incompetent GSV, and no comparison with other techniques has been made.

Mechanochemical Ablation (MOCA) / Cyanoacrylate Adhesive Closure (CAC).

Ontario Health (2021) performed a health technology assessment of non-thermal endovenous procedures (CAC and MOCA) for individuals with symptomatic varicose veins, which included an evaluation of effectiveness, safety, cost-effectiveness, the budget impact of publicly funding MOCA and CAC, and patient

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preferences and values. The authors concluded that CAC and MOCA produced similar patient-important outcomes, and slightly shorter recovery compared with thermal ablation. Furthermore, CAC yielded similar anatomical outcomes as thermal endovenous ablation, but the technical outcomes of MOCA were slightly poorer. Compared with surgical vein stripping, all endovenous treatments were more effective and less expensive.

The technology assessment provided the following conclusions:

The evidence suggested that MOCA for GSV insufficiency:

- Results in slightly poorer vein closure than RFA (Grade: Moderate) or EVLA (Grade: Moderate)
- Slightly increases re-canalization compared with RFA (Grade: Moderate) or EVLA (Grade: Moderate)
- Results in little to no difference in magnitude of improvement in clinical symptoms (Grade: Moderate), QOL (Grade: Low), or patient satisfaction (Grade: Moderate), compared with RFA
- Results in little to no difference in magnitude of improvement in clinical symptoms (Grade: High), QOL (Grade: High), or patient satisfaction (Grade: High), compared with EVLA
- May reduce recovery time compared with RFA (Grade: Low) or EVLA (Grade: Moderate)
- May facilitate greater healing of venous ulcers and similar time to healing, and similar ulcer recurrence compared with thermal ablation (Grade: Low).

The evidence suggested that CAC for GSV insufficiency:

- Results in little to no difference in vein closure (Grade: Moderate), re-canalization (Grade: Moderate), magnitude of improvement in clinical symptoms (Grade: Moderate), QOL (Grade: Moderate), or patient satisfaction (Grade: Low), compared with RFA
- Results in little to no difference in vein closure (Grade: Moderate), re-canalization (Grade: Moderate), magnitude of improvement in clinical symptoms (Grade: Moderate), QOL (Grade: Low), or patient satisfaction (Grade: Low), compared with EVLA
- Likely reduces recovery time compared with RFA (Grade: Low) or EVLA (Grade: Low)
- May result in slightly poorer improvements in clinical symptoms or QOL compared with surgical vein stripping; however, the evidence is very uncertain (Grade: Very low)
- In people with GSV diameter of 10 mm or greater, may result in slightly poorer vein closure, slightly more re-canalization, and comparable magnitude of improvement in clinical symptoms, compared with RFA, EVLA, and surgical vein stripping, but the evidence is very uncertain (Grade: Very low).

The evidence suggested that CAC for SSV insufficiency:

- May result in slightly lower re-canalization compared with 980 nm EVLA and surgical vein stripping, and similar re-canalization as RFA and 1,470 nm EVLA (Grade: Low)
- Results in a greater magnitude of improvement in clinical symptoms compared with surgical vein stripping, and similar improvement as EVLA and RFA (Grade: Low).

These researchers stated that this review had several drawbacks. While nearly all studies were of individuals with GSV insufficiency and treatment, heterogeneity across studies in terms of design, time-point of outcome measurement, and inclusion criteria precluded quantitative synthesis of most outcomes. Had these investigators chose to perform a network meta-analysis, evidence would have been extremely limited and restricted to RCTs, and clinical diversity and heterogeneity among data would most likely have challenged the core assumptions of network meta-analysis. The main findings from RCTs and non-randomized studies were generally similar in both direction and magnitude despite differences in study methods, subject characteristics (e.g., spectrum of disease severity), and treatment protocols. For instance, procedural factors that may affect success for vein closure in MOCA include pullback rate (the speed with which the catheter is gradually withdrawn) and the concentrations, volumes, and type of sclerosant used. These researchers did not pre-define inter-procedural or post-procedural pain as a discrete outcome; however, they did assess clinical symptoms, QOL, and AEs (including pain, if reported) that may capture some of the patient experience of pain and discomfort during and after treatment. No differences were found. This review summarized comparative evidence for the therapeutic options comprising most clinical practice in Ontario.

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These investigators did not identify any published studies comparing MOCA with CAC or comparing any non-thermal endovenous procedure with less common vein-sparing procedures (e.g., ambulatory conservative hemodynamic correction of venous insufficiency -- a procedure known by its French acronym, CHIVA [cure conservatrice et hémodynamique de l'insuffisance veineuse en ambulatoire]). There are several subpopulations and potential effect modifiers for which it would have been valuable to better understand the safety and effectiveness of non-thermal endovenous interventions. For instance, while these researchers did not specify them explicitly, it would have been of interest to compare patients with and without prior DVT, with and without significant deep vein reflux, and mild (e.g., C1 to C4) versus C5 to C6 disease. The limiting factor in conducting the subgroup analyses these researchers planned was lack of reporting of these key characteristics in the studies. This review examined outcomes over a lengthy duration following treatment (more than 12 months in many studies), which provided insight into short, intermediate, and longer-term effects (e.g., durability of vein closure and effects on patient-important outcomes). They did not include compression stockings or garments as a comparator, recognizing that while they could be effective in controlling symptoms of chronic venous insufficiency (e.g., edema, pain) or for preventing venous ulcer recurrence, they do not resolve the underlying incompetent veins. In some jurisdictions, including Ontario, compression stockings may be tried or required as 1rst-line treatment. Invasive treatment may be pursued because compression stockings are not practicable (e.g., donning and doffing may be challenging for an elderly person), affordable, or adequately effective.

Moreover, the authors are aware of several ongoing comparative studies that have potential relevance to the research question, including a RCT of MOCA versus CAC for the treatment of varicose veins (NCT03392753; anticipated completion of data collection: December 2019); and 2 on VenaSeal -- "Randomized controlled trial comparing the clinical outcomes after cyanoacrylate closure with VenaSeal Closure System and surgical stripping for incompetent saphenous veins" (KCT0003203; anticipated completion of data collection: February 2021) and "Global, post-market, prospective, multi-center, randomized controlled trial of the VenaSeal™ Closure System vs. surgical stripping or endothermal ablation (ETA) for the treatment of early & advanced stage superficial venous disease" (NCT03820947; anticipated completion of data collection: September 2023).

Whing and colleagues (2021) noted that GSV incompetence, causing varicose veins and venous insufficiency, makes up the majority of lower-limb superficial venous diseases. Therapeutic options for GSV incompetence include surgery (also known as HL/S), laser and RFA, and UGFS. Newer treatments include cyanoacrylate glue, MOCA, and endovenous steam ablation (EVSA). These techniques avoid the need for a general anesthetic; and may result in fewer complications and improved QOL. These treatments should be compared to inform decisions on treatment for varicosities in the GSV. This is an update of a Cochrane Review first published in 2011; these investigators examined the effects of EVLA, RFA, EVSA, UGFS, cyanoacrylate glue, MOCA and HL/S for the treatment of varicosities of the GSV. The authors concluded that their findings were limited due to the relatively small number of studies for each comparison and differences in outcome definitions and time-points reported. Technical success was comparable between most modalities. EVLA may offer improved technical success compared to UGFS or HL/S. HL/S may have improved technical success compared to UGFS. No evidence of a difference was detected in recurrence, except for a possible long-term benefit for RFA compared to EVLA or HL/S. These researchers stated that studies that provide more evidence on the breadth of treatments are needed; future trials should seek to standardize clinical terminology of outcome measures and the time-points at which they are measured.

Polidocanol for the Treatment of Varicose Veins of the Lower Extremities

Li and colleagues (2021) noted that varicose veins of the lower extremities showed earthworm-like dilatation and venous protrusion of the lower extremities. Polidocanol foam sclerotherapy, as a minimally invasive treatment with rapid recovery, less trauma and not easy to relapse, has achieved good results; however, it lacks evidence-based support. In a meta-analysis, these researchers will examine the safety and effectiveness of polidocanol in the treatment of varicose veins of the lower extremities. Chinese National Knowledge Infrastructure, Wanfang Database, Chinese Scientific Journals Database, China Biology Medicine disc, PubMed, Embase database, Web of Science, and Cochrane Library will be employed as data sources to search for RCTs of polidocanol in the treatment of varicose veins of lower extremities. The search time is set

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from the establishment of the database in December 2020 in this study. Two researchers will independently extract, delete files, extract data and evaluate the quality. Revman software version 5.3 will be used for statistical analysis of data. In this study, the safety and effectiveness of polidocanol in the treatment of varicose veins of the lower extremities will be examined in terms of total effective rate, incidence of complications and recurrence rate. The authors concluded that the findings of this study will provide reliable evidence-based support for the clinical application of polidocanol in the treatment of varicose veins of lower extremities.

Ambulatory Selective Variceal Ablation Under Local Anesthetic (the ASVAL Procedure) for the Treatment of Symptomatic Great Saphenous Vein

In a prospective study, Zolotukhin and associates (2017) examined the effect of isolated phlebectomy in patients with incompetent GSV (ambulatory selective varices ablation under local anesthesia (ASVAL) procedure) on the reflux and diameter of the trunk and evaluated recurrence rate of varicose veins at 1 year. This trial included 67 patients (51 women and 16 men; 75 limbs) with primary varicose veins and with C2 or C2,3 or C2,3,4 or C2,4 classes of chronic venous disease and GSV incompetence; age range of 17 to 71 years; mean age of 46.8 years (SD 13.9). These researchers recorded the presence or absence of reflux in the GSV with duplex US before and after surgery. The recurrence of varicose veins was examined at 12 months. All subjects underwent the ASVAL procedure. At 1 year after removing of tributaries of the incompetent trunk, 66 % of them were competent. Reflux persisted in 17 % of GSV with reflux above mid-thigh and in 61 % of trunks with reflux extended below the mid-thigh ($p = 0.0004$). The diameter of all the veins decreased significantly no matter reflux disappeared or not. Varicose veins recurred in 13.5 % cases. In 6.5 % of limbs with a reflux above the mid-thigh, the recurrence was registered at 1 year, while in the limbs with the reflux below the mid-thigh at a baseline, the recurrence rate was 25 % ($p = 0.036$). The authors concluded that isolated phlebectomy with a preservation of incompetent GSV led to disappearance of reflux in a majority of cases and to significant decrease of vein diameter in all the cases. These researchers stated that the ASVAL procedure could be considered as a less aggressive and less expensive approach in selected cases; however, clear indications for isolated phlebectomy need to be established.

Richards and colleagues (2021) stated that the ASVAL technique subscribes to the "ascending" theory of varicose vein etiology, which recommends primary ambulatory phlebectomy as a treatment for tributary varicosities and truncal vein incompetence. In a systematic review, these investigators examined the safety and effectiveness of the ASVAL technique for the treatment of symptomatic varicose veins. They carried out a comprehensive search of the Medline and Embase databases and the Cochrane Register of Controlled Trials in May 2019 revealed 11 original articles that were qualitatively reviewed. The primary outcome was the absence from recurrent varicose veins at 1-year follow-up. Secondary outcomes were resolution of GSV reflux on duplex US, change in GSV diameter, objective and subjective clinical improvement in chronic venous disease, and patient-reported outcome measures (PROMs). A total of 2,106 limbs underwent intervention in 1,734 patients reported in 2 RCTs, 1 case-control study, 3 cohort studies, and 5 case-series studies. Varicosity recurrence at 1 year ranged from 0.5 % to 13.5 % in patients. Of 1,622 limbs with diagnosed GSV incompetence before intervention, 1,114 were competent at 1 year (mean of 68.2 % [± 12.62 %]). All studies measuring GSV diameter reported statistically significant reductions in the size of the vein. The authors concluded that ASVAL may be considered as a minimally invasive treatment for early stages of chronic venous disease in the presence of truncal reflux. The evidence base should be strengthened by prospective RCTs that follow standardized procedures and report according to recognized measures of QOL alongside clinical and hemodynamic data.

External Valvuloplasty for the Treatment of Symptomatic Great Saphenous Vein

Muhlberger and colleagues (2021) noted that external valvuloplasty (eVP) is a reconstructive surgical approach for repairing the function of the terminal and pre-terminal valves. In an observational study, these researchers examined the 6-month outcomes of eVP regarding the diameter of the GSV. This trial included patients from 5 vein centers; follow-up entailed duplex US of the GSV. The VCSS as well as the CEAP classification were recorded. These investigators enrolled 210 patients, with a follow-up rate of 58 %; eVP was sufficient in 95.24 % of the patients. The GSV diameters decreased significantly from 4.4 mm (standard

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deviation (SD): 1.39) to 3.9 (SD: 1.12), 4 cm distal to the sapheno-femoral junction (SFJ); from 3.7 mm (SD: 1.10) to 3.5 mm (SD: 1.02) at the mid-thigh; from 3.6 mm (SD: 1.14) to 3.3 mm (SD: 0.94) at the knee; and from 3.1 mm (SD: 0.99) to 2.9 mm (SD: 0.78) at the mid-calf. VCSS decreased significantly from 4.76 (SD: 2.13) pre-operatively to 1.77 (SD: 1.57) 6 months post-operatively. The authors concluded that GSV function could be restored by eVP; diameters over the total length of the GSV decreased significantly. The findings of this observational study need to be validated by well-designed studies.

The authors stated that this study had several drawbacks. First, this trial had a high lost-to-follow-up rate, and these researchers re-examined only 210 of 359 patients, which represented a follow-up rate of only 58. 5 %. They attempted to motivate patients to participate in the follow-up examination using multiple phone calls. However, as re-examinations mainly took place in the afternoon, the majority of patients did not respond for occupational reasons. Nevertheless, the population's general data were similar to the pre-operative and 6-week data, indicating a comparable study population. Second, eVP was limited to approximately 15 % of all patients and patients with reflux at the SFJ and continuing reflux in the ASV benefited the most from eVP; thus, these findings represented a highly selective group of patients. Third, this study had a high number of missing data for the reflux measurements.

Pharmacological Thromboprophylaxis for Endo-Venous Varicose Vein Interventions

In a systematic review on the incidence and management of endo-venous heat-induced thrombosis (EHIT) following endo-venous thermal ablation of the great saphenous vein, Healy et al (2021) found inconsistency regarding the use of peri-procedural anti-coagulation: 13 studies reported the routine use of peri-procedural anticoagulation; 7 reported selective use; 8 studies reported that no anti-coagulation was used; and 47 study reports did not describe the use of anti-coagulation. Practice guidelines of the Society for Vascular Surgery (SVS) and the American Venous Forum (AVF) (Gloviczk et al, 2011) stated that selective prophylaxis after risk assessment is needed in patients who undergo venous surgery but that, for patients who do not have additional risk factors, surgeons should not routinely use specific thromboprophylaxis. The European Society Guidelines stated that the role of routine use of pharmacologic prophylaxis remains uncertain and more studies are needed (Wittens et al, 2015); however, the risk factors should be assessed for each individual patient using a specific risk assessment score such as the Caprini score. The AVF/SVS guidelines on EHIT also recommend an individualized approach to the use of prophylactic anti-coagulation.

Moreira et al (2022) noted that venous thrombo-embolism (VTE) is a possible complication following varicose vein surgery, reported after both open and endovascular interventions. However, there are no internationally accepted recommendations regarding post-operative VTE prevention strategies, with some investigators advocating for its use, while others recommend against it. In a systematic review, these investigators examined the safety and effectiveness of chemo-thrombo-prophylactic strategies following varicose vein surgery. They carried out a literature search using Medline, Scopus, SciELO and Web of Science databases, which returned 532 studies. A total of 10 studies were included; data were extracted using piloted forms. A total of 6,929 patients were included for analysis, out of which 70.4 % were treated by open surgery (n = 4,878) and 29.6 % by endo-venous procedures (n = 2,051; 79.1 % EVLA; 20.9 % RFA). VTE chemo-thrombo-prophylaxis was carried out in 76.3 % of the patients (n = 5,284), from which 62.5 % were treated by open surgery (n = 3,301) and 37.5 % by endo-venous interventions (n = 1,983). Among those treated by open surgery, reported DVT rates ranged between 0 % to 6.25 %, while pulmonary embolism (PE) was reported in 0 % to 0.07 % of the cases. Regarding endo-venous interventions, EHIT and DVT rates ranged between 0 % to 2.5 % and 0 % to 0.9 %, respectively, with no cases of PE described. The remaining 23.7 % of the patients did not underwent VTE chemo-thrombo-prophylaxis (n = 1,645), with DVT and PE rates after open surgery ranging between 0 % to 5.17 % and 0 % to 1.48 %, respectively. Only 1 study reported thrombotic complications after endo-venous interventions in this subgroup of patients, with post-operative EHIT rates of 7.3 %, and no information regarding PE or DVT. Bleeding complications were higher in patients undergoing chemo-thrombo-prophylaxis (0 % to 10.2 %) when compared to those who did not (0 % to 0.18 %); and were more frequent after endo-venous interventions (0 % to 10.2 % versus 0 % to 0.75 % after open surgery). The authors concluded that VTE is a possible complication following both open and endovascular varicose vein procedures, although overall VTE complications occurred less frequently after endovascular interventions. There was a clear heterogeneity regarding peri- as well as post-operative

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chemoprophylaxis regimens used. These researchers stated that further studies are needed to stratify risk factors and indications for chemo-thrombo-prophylaxis following varicose vein surgery.

The authors stated that this review had several drawbacks. First most of the studies were observational in nature, with the inherent risk of selection bias due to the lack of randomization. Second, the majority of studies were classified as low quality, with unstandardized methodologies and treatment regimens. Third, the absence of uniformed pre-operative VTE risk assessment influenced the capability to compare data among studies. These researchers stated that for more conclusions to be drawn, it is necessary to limit confounding bias that could occur at patient selection level based on pre-intervention risk for thrombo-embolism, on surgical scenarios that might have several discrepancies between different centers and by the use of other thrombo-prophylactic strategies such as compression or rapid mobilization.

In a systematic review and meta-analysis, Alameer et al (2022) examined the role of thromboprophylaxis in the prevention of venous thromboembolism in patients undergoing varicose vein interventions. PubMed, Embase and Web of Science were searched for comparative studies of patients undergoing varicose vein interventions and received either thromboprophylaxis or no thromboprophylaxis. Data were collected on the number of thrombotic events including DVT, PE and EHIT as well as bleeding events. The primary outcomes for the meta-analysis were the risk of all thrombotic events, risk of DVT and risk of bleeding. Pooled risk ratios were calculated using random effects modelling. A total of 8 studies (6,479 subjects) were included. The use of thromboprophylaxis reduces the risk of all thrombotic events (pooled risk ratio [RR] = 0.63, 95 % CI: 0.04 to 10.43) and the risk of DVT (pooled RR = 0.59, 95 % CI: 0.08 to 4.60) with no increased risk of bleeding (pooled RR = 0.66, 95 % CI: 0.06 to 7.21]. Rivaroxaban has similar effectiveness in the prevention of DVT compared to fondaparinux in patients undergoing endo-venous ablation of varicose veins (pooled RR = 0.68, 95 % CI: 0.06 to 7.41). An extended course of thromboprophylaxis reduced the risk of developing DVT compared to a short course (pooled RR = 1.40, 95 % CI: 0.44 to 4.46). However, the 2 studies reporting on the duration of thromboprophylaxis did not stratify patients according to their risk of developing VTE. The authors concluded that the use of thromboprophylaxis in patients undergoing varicose vein interventions reduced the risk of VTE with no significant increase in the risk of bleeding. However, the included studies were under-powered with high-to-moderate risk of bias. These researchers stated that more RCTs with a large sample size are needed to provide high quality evidence for clinical practice.

Turner et al (2023) stated that the VTE rate following endo-venous procedures is higher than other day-case procedures and could be reduced with pharmacological thromboprophylaxis. In a systematic review and meta-analysis, these researchers examined the rate of VTE following endo-venous varicose vein intervention with mechanical thromboprophylaxis and additional pharmacological thromboprophylaxis, versus mechanical thromboprophylaxis alone. This systematic review followed the PRISMA guidelines. Studies of endo-venous intervention for superficial venous incompetence reporting the pre-defined outcomes with at least 30 patients were eligible. Data were pooled with a fixed effects model. A total of 221 studies were included in the review (47 randomized trial arms, 105 prospective cohort studies, and 69 retrospective studies). In randomized trial arms, the rate of DVT with additional pharmacological thromboprophylaxis was 0.52 % (95 % CI: 0.23 % to 1.19 %) (9 studies; 1,095 patients; 2 events), versus 2.26 % (95 % CI: 1.81 % to 2.82 %) (38 studies; 6,951 patients; 69 events) with mechanical thromboprophylaxis alone. The rate of PE in randomized trial arms with additional pharmacological thromboprophylaxis was 0.45 % (95 % CI: 0.09 % to 2.35 %) (5 studies, 460 participants, 1 event) versus 0.23 % (95 % CI: 0.1 % to 0.52 %) (28 studies, 4,834 participants, 3 events) for mechanical measures alone. The rate of EHIT grade III to IV was 0.35 % (95 % CI: 0.09 % to 1.40 %) versus 0.88 % (9 5% CI: 0.28 % to 2.70 %). There was 1 VTE-related mortality and 1 instance of major bleeding, with low rates of minor bleeding. The authors concluded that there was a significant reduction in the rate of DVT with additional pharmacological thromboprophylaxis and routine prescription of anti-coagulation following endo-venous varicose vein intervention should be considered. VTE risk for individual study subjects was heterogeneous and risk stratification in future randomized interventional studies is critical to establish the safety and effectiveness of additional pharmacological thromboprophylaxis.

Endovenous Laser Ablation versus Conventional Surgery (Ligation and Stripping) for the Treatment of Primary Great Saphenous Varicose Vein

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Shrestha et al (2023) noted that the great saphenous varicose vein was managed with high ligation and stripping conventionally, but with the development of minimally invasive surgical techniques like EVLA, they have become popular. In a systematic review and meta-analysis, these investigators compared the short-term and long-term outcomes of these 2 therapeutic modalities on procedural time, technical success, recovery time, recurrences, cost-effectiveness, and complications. Electronic databases were searched with appropriate search terms for relevant studies, and after their screening, data was extracted. The OR was used for dichotomous data, and the MD or standardized MD (SMD) was used for continuous variables. This study identified 18 publications (10 RCTs) with a total of 1,936 patients. There was no difference in procedural time, recovery time, recurrences at 1, 2, and 5 years, or clinical severity score. The surgery group had 4.35 times higher statistically significant odds of being technically successful at 2 years, while pooling data on bruising, hematoma, sensory disturbance, infection, and phlebitis showed that the EVLA group was less likely to develop post-operative complications. The authors concluded that technical failures were more common in the EVLA, whereas post-operative complications were more common in the surgery group. Both approaches exhibited comparable clinical effectiveness, and neither modality had clear superiority over the other. Parameters like cost-effectiveness must be examined at the hospital level before choosing the right procedure for the patients.

Treatment of Varicose Tributaries either Concomitantly with or Separately from Endovenous Laser Ablation of the Incompetent Saphenous Trunk

Viljamaa et al (2023) noted that opinions on the treatment of varicose tributaries in relation to saphenous ablation in varicose disease vary. Moreover, the possible role of the tributaries regarding the recurrence of varicose disease remains unclear. The aim of the FinnTrunk Trial is to compare 2 different treatment strategies for varicose disease in a randomized setting. In group 1, the initial treatment will entail EVLA of the incompetent saphenous trunk without tributary treatment. In group 2, the varicose tributaries will be treated with UGFS concomitantly with truncal ablation. The primary outcome measure is the need for additional procedures during the follow-up. The secondary outcome measures are the cost of treatment and recurrence of varicose disease. Consecutive patients with symptomatic varicose disease (CEAP clinical class C2-C3) will be screened for the study. Patients who meet the selection criteria and give their informed consent will be scheduled for the procedure and randomized to either study group. Participants will be followed-up at 3 months, 1 year, 3 years, and 5 years. The post-procedure pain score based on a numeric rating scale (NRS) and also the use of analgesics, as well as possible procedure-related complications will be recorded at 3 months; PROMs will be recorded at 1 year. Data pertaining to the additional treatment of varicose tributaries, the AVVQ, the VCSS, and the health-related quality of life (EQ-5D-5L) will be collected at each follow-up visit. A duplex US (DUS) examination will be carried out at each visit, and data on varicose tributaries and the need for additional treatment will be recorded.

Moreover, these researchers stated that despite the clinical practice guidelines' preference for a combined approach, the opinions regarding the treatment of varicose tributaries in relation to saphenous ablation vary, and these differences may not necessarily be attributable solely to differing interpretations of the available evidence. Neither should these varying opinions come as a surprise, considering the generally benign course of varicose disease and the objectives of its treatment in addition to patients' preferences and expectations regarding their treatment. That said, these investigators hope that the FinnTrunk Trial, for its part, will aid in elucidating some aspects of this clinically important question.

Routine Post-Operative Scanning After Radiofrequency Ablation for the Treatment of Varicose Veins

Butt et al (2024) noted that RFA is a minimally invasive treatment for lower limb varicose veins. Studies indicated that RFA resulted in immediate occlusion of 90 % to 100 % of treated long saphenous veins. Evidence suggested that post-operative scans rarely change patient management or outcomes. In a retrospective study, these investigators examined the potential necessity of routine post-operative scanning in the treatment of varicose veins. Data were collected for the patients who had RFA under a single consultant from November 2015 to June 2018. Descriptive statistics were calculated to summarize patient demographics, procedural details, and outcome measures. A total of 124 patients underwent RFA. Most of

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the patients (n = 114, 92 %) showed complete ablation, indicating a high success rate for the procedure. The authors concluded that the findings of this study suggested that routine post-operative scanning should be discontinued as this did not change patient management in over 99 % of cases. These researchers stated that the resources currently used for post-operative scans could be re-directed towards other critical areas.

Glossary of Terms

Table: Glossary of Terms

Term	Definition
Sclerotherapy	Endovenous chemical ablation
Phlebectomy	Procedure that uses a small scalpel or needle to remove varicose veins

Appendix

List: Clinical, Etiological, Anatomical and Pathophysiological classification (CEAP) Classification

Clinical Classification

- C0: No visible or palpable signs of venous disease
- C1: Telangiectasias or reticular veins
- C2: Varicose veins
- C2r: Recurrent varicose veins
- C3: Edema
- C4: Changes in skin and subcutaneous tissue secondary to chronic venous disease
- C4a: Pigmentation or eczema
- C4b: Lipodermatosclerosis or atrophie blanche
- C4c: Corona phlebectatica
- C5: Healed
- C6: Active venous ulcer
- C6r: Recurrent active venous ulcer

Source: Lurie et al, 2020

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